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ENGINEERING COLLEGE
(AN AUTONOMOUS INSTITUTION)

Department of Electronics and Communication Engineering

Lecture Notes

Course Code: U23ECT304
Course Name: ENGINEERING ELECTROMAGNETICS

UNIT-I VECTOR ANALYSIS

Scalar and Vectors- Vector Algebra- Vector components and unit vector- vector field - Different Co-ordinate Systems: Cylindrical and Spherical-Relationship between Co-ordinate systems- Gradient, Divergence, and Curl – Divergence theorem - Stoke's theorem

UNIT - I

1.1 INTRODUCTION

<u>Electromagnetics:</u>

- Study of electrical and magnetic characteristics or phenomena. The electric and magnetic fields are closely related to each other. Because electric charge produced field around it.
- Moving charge or time varying charge produce a current.
- When current is going through a conductor it will be induce magnetic field around the conductor such a field is called electro-magnetic field

Some special cases of electromagnetics:

- Electrostatics: charges at rest
- Magnetostatics: charges in steady motion (DC)
- Electromagnetic waves: waves excited by charges in time-varying motion

- When an event in one place has an effect on something at a different location, we talk about the events as being connected by a "field".

A *field* is a spatial distribution of a quantity; in general, it can be either *scalar* or *vector* in nature

- Electric and magnetic fields:
  - Are vector fields with three spatial components.
  - Vary as a function of position in 3D space as well as time.

Are governed by partial differential equations derived from Maxwell's equations

- A *scalar* is a quantity having only an amplitude (and possibly phase).
- A *vector* is a quantity having direction in addition to amplitude (and possibly phase).

  Examples : velocity , acceleration, force

Various quantities involved in study of EMW

1. Scalar quantity
2. Vector quantity

1. Scalar Quantity

The scalar is a quantity whose value may be represented by a single real number which may be positive or negative.

A scalar is a quantity which is wholly characterized by magnitude

Example

Temperature, mass, volume, density, speed, electric charge.

2. Vector Quantity

A quantity which has both magnitude and also direction is called vector quantity.

Example

Force, velocity, displacement, Electric field intensity, magnetic field intensity.

Unit vector

Unit vector has a function to indicate the direction. Its magnitude is always unit (one).

Unit

$$O \xrightarrow{\vec{R}} P$$

$$\text{Unit } \vec{R} = \overrightarrow{OP} = \frac{\vec{R}}{|\vec{R}|}$$

$$\text{Magnitude } |\vec{R}| = 1$$

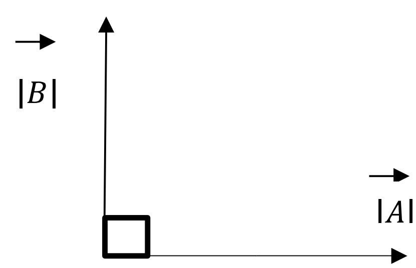
Dot Product of two vector

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta$$

θ= angle between A and B

Where $\theta = 90^o$

$$\vec{A}\ \vec{B} = 0\nabla$$



Cross Product

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \, \text{Sin}\theta \hat{a}_n$$

$\vec{A}$ and $\vec{B}$ are perpendicular to each other and $\hat{a}_n$ perpendicular to both $\vec{A}$ and $\vec{B}$.

$$\vec{A} \times \vec{B} = - \vec{B} \times \vec{A} \qquad (\times \rightarrow \text{rotation})$$

Operator Del (∇)

The del is a vector three dimensional partial operator.

$$\nabla = \frac{\partial}{\partial x}\vec{\imath} + \frac{\partial}{\partial y}\vec{\jmath} + \frac{\partial}{\partial z}\vec{k}$$

Gradient

- ✓ When Del is operated with a scalar (θ), this is called gradient of trhe scalar.
- ✓ Gradient gives maximum space rate of change of the given function in the given direction.

ϕ - scalar

ϕ - xyz

gradient of ϕ $\Rightarrow \nabla \phi = (\frac{\partial}{\partial x}\vec{\imath} + \frac{\partial}{\partial y}\vec{\jmath} + \frac{\partial}{\partial z}\vec{k})$.

$$\nabla . \phi = \frac{\partial}{\partial x}(xyz)\,\vec{\imath} + \frac{\partial}{\partial y}(xyz)\,\vec{\jmath} + \frac{\partial}{\partial z}(xyz)\vec{k}$$

$$\boxed{\nabla . \phi = yz\vec{\imath} + xz\vec{\jmath} + xy\vec{k}}$$

DIVERGENCE

- Divergence gives net outflow or inflow per unit volume
- When ∇ is operated with a vector by dot product it gives divergence of vector.

Types of Divergence

1. Positive Divergence

   Net outflow per unit volume is called positive divergence.
2. Negative Divergence

   Net inflow per unit volume is called negative divergence.

Divergence of vector (or) Solenoidal field

Divergence of vector field is zero such a field is called solenoidal field.

Consider a vector $\vec{D}$ = Dx $\vec{\imath}$ +Dy $\vec{\jmath}$+Dz$\vec{k}$

$$\text{Div } \vec{D} = \nabla . \vec{D} = [\frac{\partial}{\partial x}\vec{\imath}+\frac{\partial}{\partial y}\vec{\jmath}+\frac{\partial}{\partial z}\vec{k}]\ [\text{Dx } \vec{\imath} + \text{Dy } \vec{\jmath} + \text{Dz}\vec{k}]$$

$$\nabla . \vec{D} = \frac{\partial}{\partial x}\text{Dx} + \frac{\partial}{\partial y}\text{Dy} + \frac{\partial}{\partial z}\text{Dz}$$

$$\text{Div } \vec{A} = \text{Lt}_{\Delta v=0} \int_S \frac{\vec{A}.ds}{\Delta v} \Delta v \quad = \text{small volume}$$

$$\text{Div} = \frac{Net\ flow}{unit\ volume}$$

Curl

A curl representing number of rotation or circulation per unit area.

$$\text{Curl } \vec{E} = \int_l \frac{\vec{E}dl}{\Delta v}$$

Where the del operator cross product with a vector (or) field is called curl of the vector.

Curl = circulation or rotation per area

i) Positive curl

When the rotation or circulation is anti-clockwise direction that is called positive curl.

ii) Negative curl

When the rotation or circulation is clockwise direction that is called negative curl.

Irrotational Field

The curl of a vector field is equal to zero such as field is called Irrotational Field.

Curl= workdone/Area

Consider a vector $\vec{V}$ = Vx $\vec{\imath}$ +Vy $\vec{\jmath}$+Vz$\vec{k}$

$$\text{Curl } \vec{V} = \nabla \times \vec{V} = \begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ Vx & Vy & Vz \end{vmatrix}$$

$$\text{Curl } \vec{V} = \vec{\imath}\,(\frac{\partial}{\partial x} Vz - \frac{\partial}{\partial z} Vy) - \vec{\jmath}(\frac{\partial}{\partial x} Vz - \frac{\partial}{\partial z} Vx) + \vec{k}(\frac{\partial}{\partial x} \text{Vy} - \frac{\partial}{\partial y} Vx)$$

<u>Introduction of types of integrals related to electro magnetic theory</u>

- In electro magnetic theory the charges can be exixt or present in point form, line form, surface form, volume form
- Hence dealing with the analysis of such charge distribution, the various types of integrals are required.
  1. Line integral
  2. Surface integral
  3. volume integral

1. **<u>Line Integral</u>**

When the charges distributed in a straight line or curved path there we can use line integral.

$$\rho l = \frac{dQ}{dl}$$
$$dQ = \rho l dl$$
$$Q = \int_{l} \rho l dl$$

2. **<u>Surface Integral</u>**

When the charges are present in a perpendicular surface there we can count the charges by using surface integral

$$\rho s = \frac{dQ}{ds}$$
$$dQ = \rho s ds$$
$$Q = \int_{s} \rho s ds$$

3. **<u>Volume Integral</u>**

When the charges are distributed in uncoverage of surface while we can count the charges by using volume integral.

$$\rho s = \frac{dQ}{ds}$$
$$dQ = \rho v dv$$
$$Q = \int_{v} \rho v dv$$

ORTHOGONAL CO-ORDINATE SYSTEMS

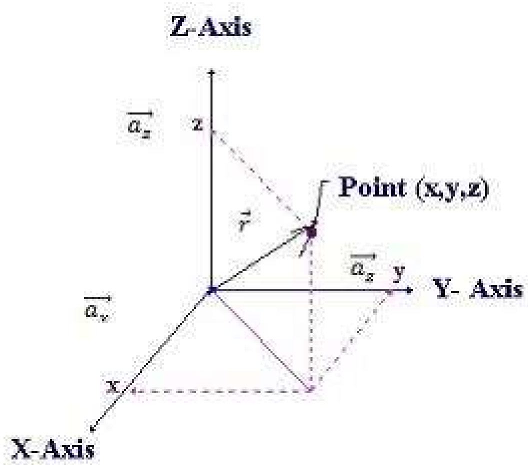
Co-ordinate system is used to describe a vector accurately in some specific direction, length, angles, and projection.

Types of orthogonal co-ordinated system

1. Cartesion (or rectangular) co-ordinate system
2. Cylindrical co-ordinate system
3. Spherical co-ordinate system

1. <u>The Cartesian co-ordinate system</u>

- In cartersion co=ordinate system three co-ordinate area x, y and z are mutually right angles to each other.
- Consider a point p(x,y,z) in space at a distance 'r' from the origin.
- The vector 'r' can be represented as
  $r = x\vec{a}_x + y\vec{a}_y + z\vec{a}_z$
  Where $\vec{a}_x, \vec{a}_y, \vec{a}_z$ are unit vectors.
  x, y and z are the components vectors. Components vector here a magnitude and direction. Unit vector have unit magnitude and direction along the co-ordinate axis.
- A unit vector in a given direction is a vector in that direction divided by its magnitude. i.e
  $\vec{a}_r = \frac{\vec{r}}{|r|}$
  $a_r = \frac{x\vec{a}x + y\vec{a}y + z\vec{a}z}{\sqrt{x^2+y^2+z^2}}$



- Consider the point p(x,y,z) & Q(x+dx, y+dy, z+dz) in rectangular co-ordinate system.
- Figure show the six planes define a rectangular parallel piped. The differential length dl from P to Q is the diagonal of the parallel piped is given by

$$dl = \sqrt{(dx^2) + (dy^2) + (dz^2)}$$

The differential area $ds = dxdy$

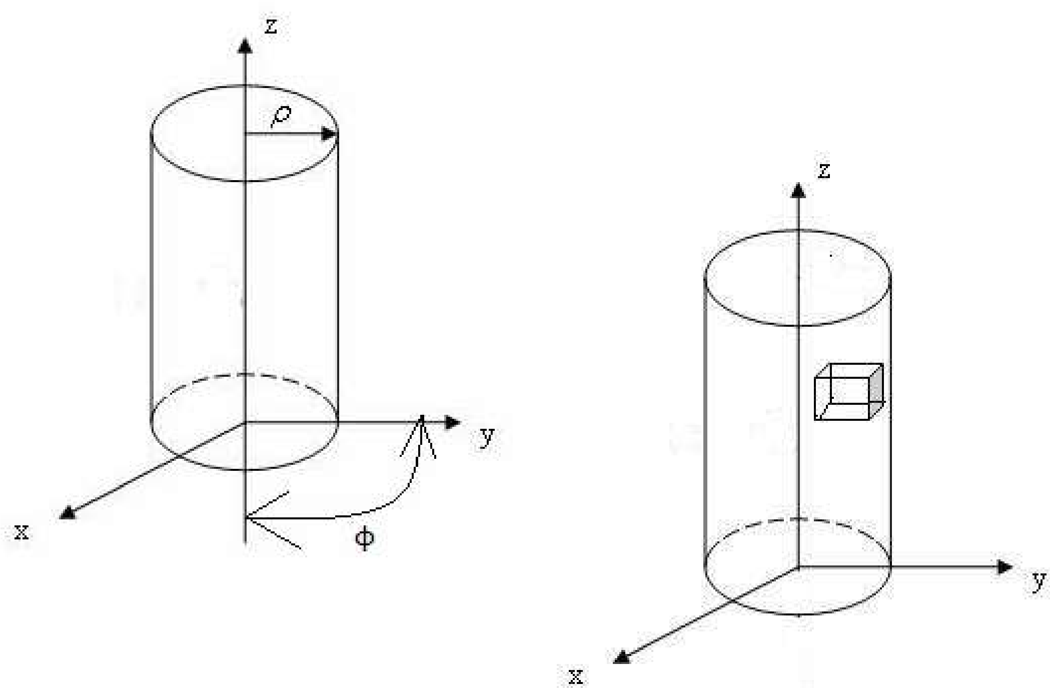
$= dydz$

$= dzdx$

The differential volume $dv = dxdydz$

2. **Cylindrical co-ordinate system**

- The cylindrical co-ordinate system is the three dimensional version of the polar co-ordinates of analytic geometry.
- In this system, consider any point as the intersection of three mutually perpendicular surfaces. They are a circular cylinder(=constant), a plane(ϕ=constant) and another plane (z=constant). The co-ordinates are ρ, ϕ and z.



- A differential volume element in cylindrical co-ordinate may be obtained by increasing ρ, ϕ and z by the differential increments dρ, dϕ and dz as shown in fig.

- It has sides of the length dρ, ρdϕ and dz

  The differential length dl = $\sqrt{(d\rho^2) + (\rho d\phi^2) + (dz^2)}$

  The differential ares $ds = \rho d\rho d\phi$

  $= d\rho dz$

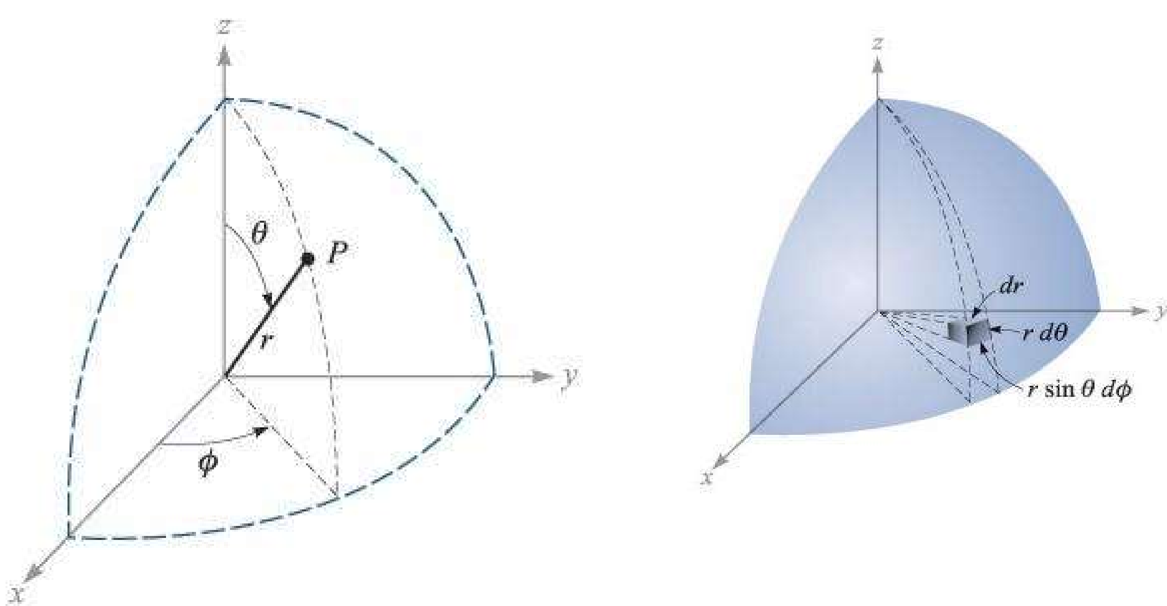
  $= \rho d\phi dz$

  The differential volume $dv = d\rho d\rho \phi dz$

  $= \rho d\rho \phi dz$

3. <u>The Spherical co-ordinate system</u>

- In this system consider any point as the point of intersection of the spherical surface (radius r=constant) conical surface (θ, angle between r and z = constant) and plane surface (ϕ= constant)
- The co-ordinates of this system are r,θ & ϕ spherical co-ordinate by r,θ & ϕ.
- A differential volume element may be obtained in spherical co-ordinate by increasing rdϕ,θ & ϕ by dr, dθ and dϕ as shown in fig.



- The sides of this volume element are dr, rdθ and rsinθdϕ.

  The differential length dl=$\sqrt{(dr^2) + (rd\theta y^2) + (rsin\theta d\phi^2)}$

  The differential are $ds = dr.rd\phi = rdrd\theta$

  $= dr\ rsin\theta d\phi = r\ sin\theta d\phi dr$

  $= rd\theta.\ rsin\theta d\phi$

$= r^2 sin\theta d\theta d\phi$

The differential volume $dv= dr.rd\theta.rsin\theta d\phi$

$dv = r^2\, sin\theta d\theta d\phi dr$

Co-ordinate	Co-ordinates
Cartesian	(x,y,z)
Cylindrical	(ρ,ϕ,z)
Spherical	(r,θ,ϕ)

Transformation between cartesian and cylindrical system

A vector in cartesion co-ordinate system can be transformed into cylindrical co-ordinate system and vice versa.

I. Conversion of cartesion to cylindrical system

The cartesion co-ordinates (x, y, z) can be converted into cylindrical co-ordinates (ρ, ϕ, z)

Given	Transform
x	$\rho=\sqrt{x^2 + y^2}$
y	$\Phi=\tan^{-1}(y/x)$
z	Z=z

II. Conversion of cylindrical to Cartesian system

The cylindrical co-ordinates (ρ,ϕ,z) can be converted into Cartesian co-ordinates(x,y,z)

Given	Transform
ρ	$x= \rho\cos\phi$
ϕ	$Y=\rho\sin\phi$
z	Z=z

Transformation between cartesian and spherical system

A vector in Cartesian co-ordinate system can be transformed into spherical co-ordinate system and vice versa.

i) Conversion of cartesian to spherical system

The cartesian coordinates (x,y,z) can be transformed into spherical co-ordinate (r,θ,ϕ)

Given	Transform
x	$r=\sqrt{x^2+y^2+z^2}$
y	$\theta=\cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)=\cos^{-1}\left(\frac{z}{r}\right)$
z	$\phi=\tan^{-1}(y/x)$

ii) Conversion of spherical to cartesian co-ordinate system

The spherical co-ordinate (r,θ,ϕ) can be transformed into cartesian coordinates (x,y,z)

Given	Transform
r	$x=r\,sin\theta cos\phi$
θ	$y=r\,sin\theta sin\phi$
ϕ	$z=r\,cos\theta$

Problem: 1

1. Transform the cartesian co-ordinates x=2, y=1, z=3 into spherical co-ordinates

Given	Transform
x=2	$r=\sqrt{x^2+y^2+z^2}=\sqrt{4+1+9}=3.74$
y=1	$\theta=\cos^{-1}\left(\frac{z}{r}\right)=\cos^{-1}\left(\frac{3}{\sqrt{14}}\right)=36.7°$
z	$\phi=\tan^{-1}(y/x)=\tan^{-1}(1/2)=26.56°$

2. Given the Cartesian co-ordinates of a point whose cylindrical co-ordinates are $\rho=1,\phi=45°,z=2$.

Given

Cylindrical co-ordinates are ρ=1,ϕ=45°,z=2.

Cartesian co-ordinates are *x,y,z*

x= $\rho cos\theta$= 1cos 45=0.707

y = $\rho sin\theta$=1sin45=0.707

Z=z=2

Cartesian co-ordinates are (0.707,0707,2)

3. Represent point p(0,1,1) m given in Cartesian co-ordinates in spherical co-ordinates.

Cartesian co-ordinates x=0, y=1m, z=1m

r^2=x^2+y^2+z^2=0+1+!

r=$\sqrt{2}$m

$cos\theta = \frac{z}{R} = \frac{1}{\sqrt{2}}$

θ=45°

tanϕ=$\frac{y}{x}$=$\frac{1}{0}$ =∞

$$\phi = 90°$$

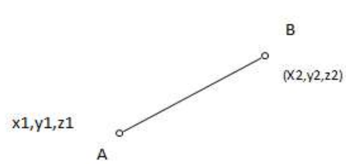
Spherical co-ordinates r=$\sqrt{2}$m,$\theta = 45°$,ϕ=90°

4. Find the distance from A(1,2,3) to B(2,0,-1) in rectangular co-ordinates.

r = $\sqrt{(x2 - x1)^2 + (y2 - y1)^2 + (z2 - z1)^2}$

= $\sqrt{(2 - 1)^2 + (0 - 2)^2 + (-1 - 3)^2}$ = $\sqrt{1 + 4 + 16}$ = $\sqrt{21}$

5. Find the distance from A(r =5, θ=20°,ϕ =120°) and B(r =2, θ=80°,ϕ =30°) in spherical co-ordinates.



x_1=rsinθcosϕ

= 5 sin20cos120

= -0.855

y1= rsinθsinϕ

= sin20sin120

=1.48

$z1 = r\cos\theta$

$= 5\cos 20$

$=4.7$

$x2= a \sin80\cos30 =1.7$

$y2 = 2\sin80\sin30=0.985$

$z2=2\cos80=0.347$

$d =\sqrt{(x2-x1)^2+(y2-y1)^2+(z2-z1)^2}$

$=\sqrt{(1.7+0.855)^2+(0.985-1.48)^2+(0.347-4.7)^2}$

$= \sqrt{6.528+0.245+18.95}$

$d = 5.07$

6. Transform the vector $\vec{A}= 10\overrightarrow{ax}- 8\overrightarrow{ay}+ 6\overrightarrow{az}$ to cylindrical co-ordinate at point (10,-8,6)
Sol.

(i) $A_r= A_x\cos\phi+ A_y\sin\phi$
$\underline{A_r=10\cos\phi -8\sin\phi}$

(ii) $\phi = \tan^{-1}(\frac{y}{x}) = \tan^{-1}(\frac{-8}{10}) = -38.6$
$A_r= 10 \cos(-38.6)-8\sin(-38.6)= 10*0.78-8*0.624$
$A_r =7.8+4.992$
$\underline{A_r=12.79}$

(iii) $A\phi = - Ax\sin\phi+Ay\cos\phi$
$A\phi=-10\sin\phi-8\cos\phi$
$\phi = tan^{-1}(\frac{y}{x})$
$\phi =-38.6$

$A\phi=-10\sin\phi - 8\cos\phi$
$= -10\sin(-38.6)-8\cos(-38.6)$
$= 10*0.623-8*0.7815$
$= 6.23-6.25$

$A\phi =0.02$

(iv) $Az=Az$ $Az=\overrightarrow{az}$

Az=6 az=6

7. A particular scalar field i) α =20e^{-x}sin($\frac{\pi y}{6}$), ii) α=25rsin ϕ. Find iys gradient at point P(0,1,1) for Cartesian and point P($\sqrt{2}$,$\frac{\pi}{2}$,5) for cylindrical co-ordinate system?

(i) <u>Cartesian system</u>

α =20e^{-x}sin($\frac{\pi y}{6}$)

$\nabla\alpha = \frac{\partial}{\partial x}[20e^{-x}\sin(\frac{\pi y}{6})]+\frac{\partial}{\partial y}[20e^{-x}\sin(\frac{\pi y}{6}]+\frac{\partial}{\partial z}[\ 20e^{-x}\sin(\frac{\pi y}{6}]$

$\frac{\partial}{\partial x}$ =-20e^{-x}sin($\frac{\pi y}{6}$) , $\frac{\partial}{\partial y}$ =20e^{-x}cos($\frac{\pi y}{6}$)($\frac{\pi}{6}$)

$\frac{\partial}{\partial z}$=0

$\nabla\alpha$ =-20e^{-x}sin($\frac{\pi y}{6}$)$\overrightarrow{ax}$+20e^{-x}($\frac{\pi}{6}$)cos($\frac{\pi y}{6}$)$\overrightarrow{ay}$at point(0,1,1)

X=0,y=1,z=1

$\nabla\alpha$ =-20e^{-0}sin($\frac{\pi}{6}$)$\overrightarrow{ax}$+20e^{-0}($\frac{\pi}{6}$)cos($\frac{\pi}{6}$)$\overrightarrow{ay}$

$\nabla\alpha$ = -20sin($\frac{\pi}{6}$)$\overrightarrow{ax}$+20 ($\frac{\pi}{6}$)cos($\frac{\pi}{6}$)$\overrightarrow{ay}$

= -20*(1/2)$\overrightarrow{ax}$+(10.46)(0.866)$\overrightarrow{ay}$

$$\nabla\alpha = -10\overrightarrow{ax} + 9.064\overrightarrow{ay}$$

(ii) <u>Cylindrical system</u>

$$\alpha = 25rsin\phi$$

$\nabla\alpha = \frac{\partial\alpha}{\partial r}\overrightarrow{ar}+\frac{1}{r}\frac{\partial\alpha}{\partial\phi}\overrightarrow{a\phi}+\frac{\partial\alpha}{\partial z}\overrightarrow{az}$

$$\nabla\alpha = [\frac{\partial}{\partial r}(25rsin\phi)\overrightarrow{ar} + \frac{1}{r}\frac{\partial}{\partial\phi}(25rsin\phi)\overrightarrow{a\phi} + \frac{\partial}{\partial z}(25rsin\phi)\overrightarrow{az}\ ]$$

$\frac{\partial}{\partial r} = 25sin\phi$,$\frac{\partial}{\partial\phi}$=$25rcos\phi$, $\frac{\partial}{\partial z}$=0

$\nabla\alpha = 25sin\phi\overrightarrow{ar}$+$\frac{1}{r}$25cos$\phi\overrightarrow{a\phi}$

$\nabla\alpha = 25sin\phi\overrightarrow{ar}$+25cos$\phi\overrightarrow{a\phi}$ at point P($\sqrt{2}$,$\frac{\pi}{2}$,5)

$$\nabla\alpha = 25\sin\left(\frac{\pi}{2}\right)\overrightarrow{ar}$$

$$\nabla\alpha = 25\overrightarrow{ar}$$

1.3 DIVERGENCE THEOREM

The surface integral of a normal component of any vector field is equal to volume of divergence of that the particular vector field.

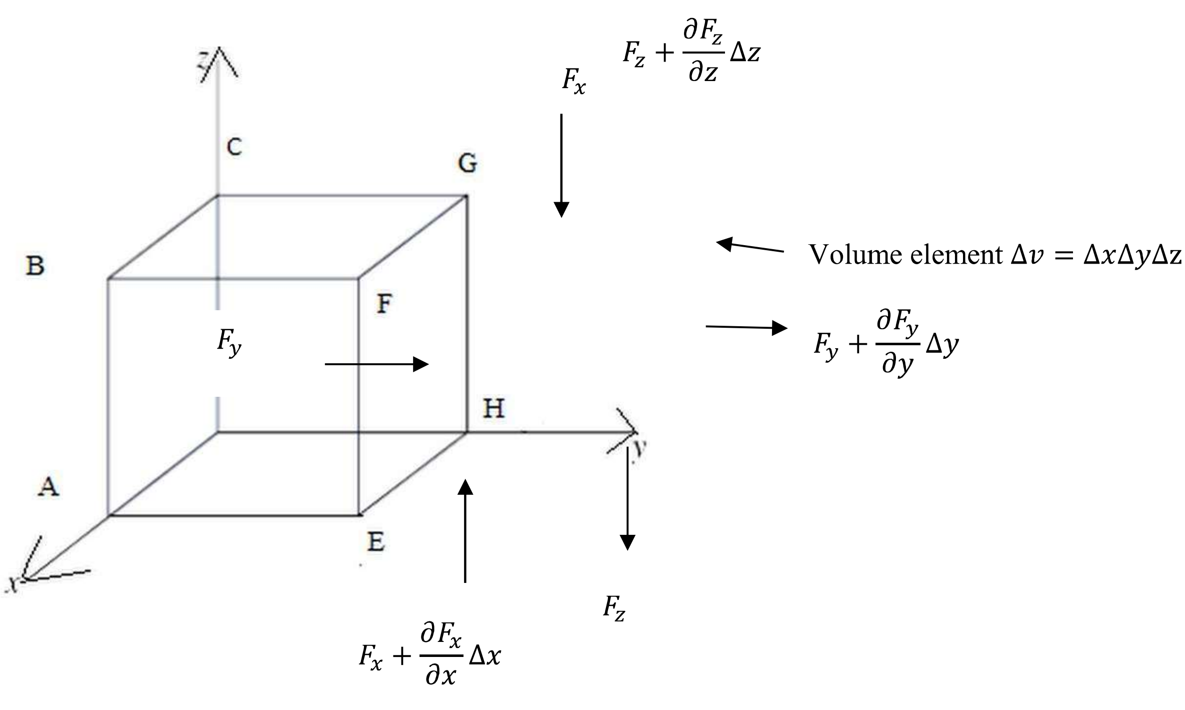
Consider $\vec{F}$ is vector

$$\oint_S \vec{F} ds = \int_V \nabla.\vec{F} dv$$

$$\oint\!\!\oint_S \vec{F} ds = \iiint_V \nabla.\vec{F}\, dv$$

<u>Proof</u>

- Consider an element volume ∇v=$\nabla x \nabla y \nabla z$ of a parallelepiped as shown in fig.
- Here $\vec{F}$ is vector field. The flux of any vector F through a surface is given by the surface integral of the vector over that surface.



- The flux passing out of the volume is treated as positive and that passing inward as negative.

- Let Fx,Fy,Fz be the components of F along the coordinate axes, so that

  $F=F_x\vec{i}+F_y\vec{j}+F_z\vec{k}$

  By definition of divergence i.e Div $\vec{F}$ = $\nabla.\vec{F}$

$$\nabla.\vec{F} = \left(\frac{\partial Fx}{\partial x} + \frac{\partial Fy}{\partial y} + \frac{\partial Fz}{\partial z}\right)$$

$$\iiint_V \nabla.\vec{F}\, dv = \iiint_V \left(\frac{\partial Fx}{\partial x} + \frac{\partial Fy}{\partial y} + \frac{\partial Fz}{\partial z}\right) dxdydz$$

$$= \iiint_V \frac{\partial Fx}{\partial x} dxdydz + \iiint_V \frac{\partial Fy}{\partial y} dxdydz + \iiint_V \frac{\partial Fz}{\partial z} dxdydz$$

Where dv=dxdydz

Consider an element volume as shown in fig.

$\iiint_V \frac{\partial Fy}{\partial y} dxdydz = \iint_S (\int \frac{\partial Fy}{\partial y} dy) dxdz$

$\int_{y1}^{y2} \frac{\partial Fy}{\partial y} dy = Fy_2 - Fy_1$

Where Fy_2-Fy_1 are the y components of the vector on the left & right hand side of the origin along the y axis. Thus

$\iiint_V \frac{\partial Fy}{\partial y} dxdydz = \iint_S (Fy_2 - Fy_1)$dxdzAdd

$= \iint_S Fy\, ds_y$

where dxdz is the y component of the surface area (element ds). The above equation gives the surface integral of Fy with the y component of ds over the whole surface.

Similarly

$\iiint_V \frac{\partial Fy}{\partial y} dxdydz = \iint_S (Fx_2 - Fx_1)dydz$

$= \iint_S Fx ds_x$

The above equation gives the surface area integral of Fy with the y components of ds over the whole surface.

$\iiint_V \frac{\partial Fz}{\partial z} dxdydz = \iint_S (Fz_2 - Fz_1)dzdy$

$= \iint_S Fz ds_z$

The above equation gives the surface area integral of Fz with the y components of ds over the whole surface.

- Add the results 1,2,3, the result is the surface integral of the vector $\vec{F}$ over the whole surface bounding the volume

From equation 1, 2, 3

$$\iiint_V \left(\frac{\partial Fx}{\partial x} + \frac{\partial Fy}{\partial y} + \frac{\partial Fz}{\partial z}\right) dxdydz = \iint_S F_x ds_x + \iint_S F_y ds_y + \iint_S F_z ds_z$$

$$= \oiint_S \vec{F} ds$$

$$\iiint_V \nabla.\vec{F} dv = \oiint_S \vec{F} ds$$

Problems:

1. Using divergence theorem, evaluate $\oiint_S \vec{F}n\,ds$ where $\vec{F}$=2xy$\vec{\imath}$+y$^2\vec{\jmath}$+4yz$\vec{k}$ and S is the surface of the cube bounded by x=0, x=1, y=0, y=1 and z=0, z=1.

   By divergence theorem,

$$\oiint_S \vec{F} ds = \iiint_V \nabla.\vec{F}\, dv$$

$$\nabla.\vec{F} = \left(\vec{\imath}\frac{\partial}{\partial x} + \vec{\jmath}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right).\left(2xy\vec{\imath} + y^2\vec{\jmath} + 4yz\vec{k}\right)$$

$$= \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(4yz)$$

$$= 2y+2y+4y = 8y$$

$$\iiint_V \nabla.\vec{F}\, dv = \int_0^1\int_0^1\int_0^1 8y\,dxdydz$$

$$= \int_0^1\int_0^1 8y[z]_0^1 dxdy$$

$$= \int_0^1\int_0^1 8y\,dxdy$$

$$= \int_0^1\int_0^1 8\left[\frac{y^2}{2}\right]_0^1 dx$$

$$= 4\int_0^1 dx$$

$$= 4\,[x]_0^1 \quad = 4$$

$$\iiint_V \nabla.\vec{F}\, dv = 4.$$

2. If F= x$^2\vec{a}_x$+y$^2\vec{a}_y$+z$^2\vec{a}_z$, then find divergence of F and curl of F.

$$\text{Div } F = \nabla . F = [\vec{a}_x\frac{\partial}{\partial x} + \vec{a}_y\frac{\partial}{\partial y} + \vec{a}_z\ \frac{\partial}{\partial z}](x^2\vec{a}_x + y^2\vec{a}_y + z^2\vec{a}_z)$$

$$= \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(z^2)$$

$= 2x+2y+2z$

$\nabla.F\ = 2(x+y+z)$

$$\text{Curl} = \nabla \times F = \begin{vmatrix} \vec{a}x & \vec{a}y & \vec{a}z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= \vec{a}_x[\frac{\partial}{\partial x}(z^2) - \frac{\partial}{\partial z}(y^2)] - \vec{a}_y[\frac{\partial}{\partial x}(z^2) - \frac{\partial}{\partial z}(x^2)] + \vec{a}_z[\frac{\partial}{\partial x}(y^2) - \frac{\partial}{\partial y}(x^2)]$$

$$= \vec{a}_x.0 + \vec{a}_y.0 + \vec{a}_z.0\ = 0$$

1.4 STOKE'S THEOREM

Line integral of any vector field is equal to the surface integral of curl of the vector field.

$\vec{F}$ is a vector

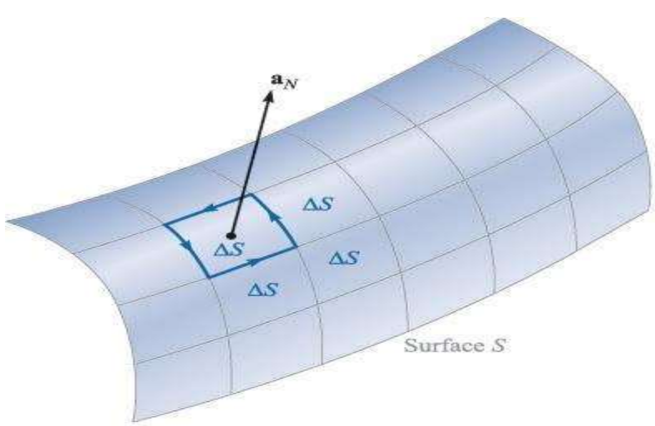
$$\oint_l \vec{F} dl = \int_S \nabla \times \vec{F} ds$$

Or

$$\oint_l \vec{F} dl = \iint_S \nabla \times \vec{F}. ds$$

Proof:

- Consider an arbitrary surface this Is broken up into incremental surface of area Δs as shown in fig



- If F is any field vector, then by definition of the curl to one of these incremental surfaces.

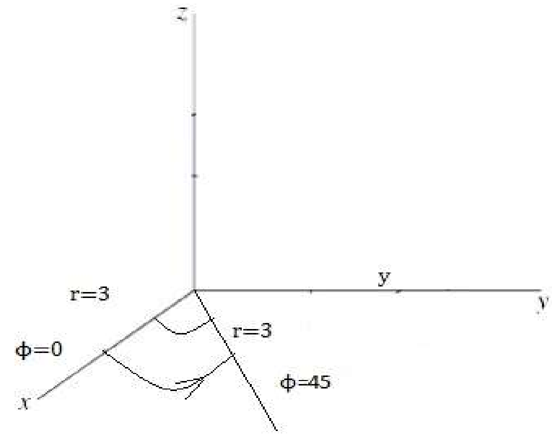
$$\frac{\oint F.dl\Delta s}{\Delta s} = (\Delta \times F)_N$$

Where,

N -> normal to the surface

$dl\Delta s$ -> closed path of an incremental area Δs

The curl of F normal to the surface can be written as



$$\frac{\oint F.dl\Delta s}{\Delta s} = (\Delta \times F).a_N$$

or

$\oint F.dl\Delta s = (\Delta \times F).a_N\Delta s$ $a_N \rightarrow$ unit vector

$= (\Delta \times F).\Delta s$

The closed integral for whole surface s is given by the surface integral of the moment of the normal component of curl F.

$$\oint_l \vec{F}dl = \iint_S \nabla \times \vec{F}.ds$$

Problem 1

1. Verify stoke's thorem for a vactor field $\vec{F} = r^2\cos\phi\vec{a}_r + z\sin\phi\vec{a}_z$ around the path L defined by $0 \le r \le 3$, $0 \le r \le 45°$, and z=0

   Sol.

   From Stoke's theorem

$$\oint_l \vec{F}dl = \int_S \nabla \times \vec{F}ds$$

To evaluate RHS, find $\nabla \times \vec{F}$

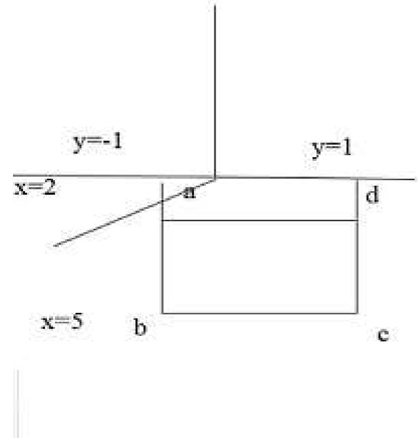
$$\nabla \times \vec{F} = [\frac{1}{r}\frac{\partial Fz}{\partial \phi} - \frac{\partial F\phi}{\partial z}]\vec{a}_r + \frac{\partial Fr}{\partial z} - \frac{\partial Fz}{\partial r}\vec{a}_\phi + [\frac{1}{r}\frac{\partial (rF\phi)}{\partial \phi} - \frac{1}{r}\frac{\partial Fr}{\partial \phi}]\vec{a}_z$$

$Fr = r^2\cos\phi$, $F\phi = 0$, $Fz = Z\sin\phi$

$\nabla \times \vec{F}$ =[$\frac{1}{r} * 0 - 0$] $\vec{a}_r$+[0-0] $\vec{a}_\phi$+[$\frac{1}{r}(0) - \frac{1}{r}$ (r^2(-sin ϕ)]$\vec{a}_z$

= rsinϕ$\vec{a}_z$

$\overrightarrow{ds}$=$rdrd\phi\vec{a}_z$



$\int_S \nabla \times \vec{F} ds$=$\int_S$(rsinϕ $\vec{a}_z$) . ($rdrd\phi$)$\vec{a}_z$

= $\int_{\phi=0}^{45°}\int_{r=0}^{3} r^2 sin\phi dr d\phi$=$[\frac{r^3}{3}]_0^3$ $[-cos\phi]_0^{45°}$

=[9][=0.707-(-1)] =9*0.2928 =2.636

Thus stoke's theorem is verified.

2. Evaluate both sides of stoke's theorem for the feet $\vec{H}$=6xy$\vec{a}_x$-3y^2$\vec{a}_y$a/m. The rectangular path around the region is given by 2≤ $x \leq$ 5, =1≤ $y \leq$ 1, z=0 let the positive direction of ds be $\vec{a}_z$

<u>Sol</u>

a to b = x=2 to 5	y=-1
b to c = x=5	y=-1 to 1
c to d = x=5 to 2	y=1
d to a x=2	y=1 to -1

$\oint_{ab} \vec{H}.dl$=$\int_2^5$($6xy$ $\vec{a}_x$-3y^2$\vec{a}_y$). $a_x\vec{a}_x$

= $\int_2^5 6xy(a_x\vec{a}_x)$dx

=6$[\frac{yx^2}{2}]_2^5$

=6[$\frac{y}{2}\left(\frac{25}{2}\right) - y(\frac{4}{2})$]

=6($\frac{21}{2}$)y

$\oint_{ab} \vec{H}.dl$=63y

$\oint_{bc} \vec{H}.dl = \int_2^5$($6xy$ $\vec{a}_x$-3y^2$\vec{a}_y$)dx$\vec{a}_x$ y=1

= $\int_2^5 6xy(\vec{a}_x\vec{a}_x)$dx

= 6 $\int_2^5[xy$dx]

$=6y[\frac{x^2}{2}]\begin{matrix}2\\5\end{matrix}$

$=6y[\frac{4}{2} - \frac{25}{2}]$

$=6y[\frac{-21}{2}]$

$\oint_{bc}\vec{A}.dl$=-63y

$\oint_{cd}\vec{H}.dl$=$\int_{-1}^{1}$(6xy$\vec{a}_x$-3y$^2\vec{a}_y$)dy$\vec{a}_y$

=$\int_{-1}^{1}-3y^2$dy

=-3$[\frac{y^3}{3}]\begin{matrix}1\\-1\end{matrix}$

=-3$[\frac{1}{3}]$+3$[\frac{-1}{3}]$=-2

$\oint_{da}\vec{H}.dl = \int_{1}^{1}$(6xy$\vec{a}_x$-3y$^2\vec{a}_y$)dy$\vec{a}_y$

= $\int_{-1}^{1}-3y^2$dy = 2

$\int_L\vec{H}$.dL = 63y-63y+2-2=-63-63 =-126

$$\nabla \times \vec{F}=\begin{bmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy & -3y^2 & 0 \end{bmatrix}$$

$$\nabla \times \vec{F}=\overrightarrow{a_x}[\frac{\partial(0)}{\partial x}-\frac{\partial}{\partial z}(-3y^2)]-\overrightarrow{a_y}[\frac{\partial(0)}{\partial x}-\frac{\partial}{\partial z}(6xy)]+\overrightarrow{a_z}[\frac{\partial}{\partial x}(-3y^2)-\frac{\partial}{\partial y}(6xy)]$$

$\nabla \times \vec{F}$=-6x$\overrightarrow{a_z}$

$\int_S\nabla \times \vec{F}ds\,\overrightarrow{a_z}$=$\int_S(-6xy\,\overrightarrow{a_z}).dxdy\overrightarrow{a_z}$

= $\int_{-1}^{1}\int_{2}^{5}-6xdydx$

=$\int_{-1}^{1}\left[\frac{-6x^2}{2}\right]_2^5 dy$=$\int_{-1}^{1}\frac{-126}{2}dy$

$$\int_S \nabla \times \vec{F}ds = -126$$

NIT-II ELECTROSTATICS

Coulomb's Law-Electric field intensity (E) - Field due to point and continuous charges distribution-Field of a line charge Field of a sheet charge - Electric flux density (D) - Gauss's law - Application of Gauss's Law - Electrical potential- Point charge and system of charge- Electric dipole –Capacitance.

COULOMB'S LAW

According to coulomb's law, force between any two point charges is

1. Directly proportional to the product of magnitudes of the two charges
2. Inversely proportional to the square of the distance between them.

Consider the two point charges Q1 and Q2 shown in fig.

As per coulomb's law

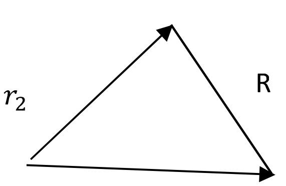
$$F \,\alpha\, \frac{Q_1 Q_2}{r^2}$$

$$F = k\frac{Q_1 Q_2}{r^2}$$

$$K = \frac{1}{4\pi\varepsilon_o}$$

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon r^2}$$

<u>Vector form of coulomb's law</u>



$$\overrightarrow{a_r} = \frac{r_1 - r_2}{|r_1 - r_2|}$$

- Consider two charges Q1 and Q2 at distance vector r1 and r2 from the origin respectively. R represents the distant vector from Q1 and Q2.
  $[\vec{F} = \frac{Q_1 Q_2}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}]$

$$[\vec{F} = \frac{Q_1 Q_2}{4\pi\varepsilon_o R^2} \frac{r_1 - r_2}{|r_1 - r_2|}]$$

Problem: 1

Find the factor of interaction between two charges spaced 10cm apart in a vacuum. The charges are 4×10^{-8} and 6×10^{-5} coulomb respectively. If the same charges are aeparated by the same distance in kerosene (ε_r=2) what is the force of interaction?

Given

$Q_1 = 4\times10^{-8}$ c

$Q_2 = 6\times10^{-5}$ c

For volume $F = \frac{Q_1 Q_2}{4\pi\varepsilon_o r^2}$

$$= \frac{4\times10^{-8} \times 6\times10^{-5}}{4\pi \frac{1}{36\pi\times10^{9}}} = 0.0216\text{N}$$

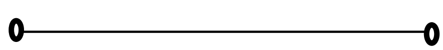
For kerosene ε_r=2

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_o r^2} = 0.0108\text{N}$$

Problem:2

Find the force on a charge Q_2 = 10μc at point(2,0,0) by a charge Q_1=20μc at point(1,0,0) in free space. Dimensions are in meters.

Q1(1,0,0) Q2 (2,0,0)



Q_1=20μc , Q_2 = 10μc

Distance between two chares $r = \sqrt{(1-2)^2 + (0-0)^2 + (0-0)^2} = 1\text{m}$

By coulomb's law, the force between two charges Q_1 and Q_2 is

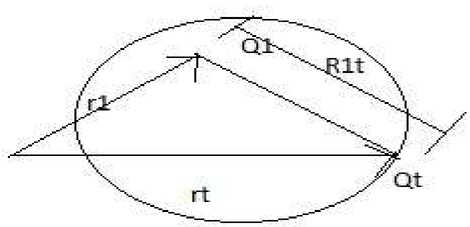
$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_o r^2}$$

$$= \frac{10\times10^{-6}\times20\times10^{-6}}{4\pi\times\frac{10^{-9}}{36\pi}\times(1)^2} = 1.8 \text{ Newtons.}$$

ELECTRIC FIELD INTENSITY (E)

- Electric field intensity (E) is exists of field around a charge in which it exerts a force on a test charge.
- This region where a particular charge exerts a force on test charge located in that region.
- Electric field intensity equal to force exerted per unit charge

$$\vec{E} = \frac{\vec{F}}{Q}$$



$$\overrightarrow{a_R} = \frac{r_t - r}{|r_t - r|}$$

$$\vec{E} = \frac{\vec{F}}{Q}$$

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

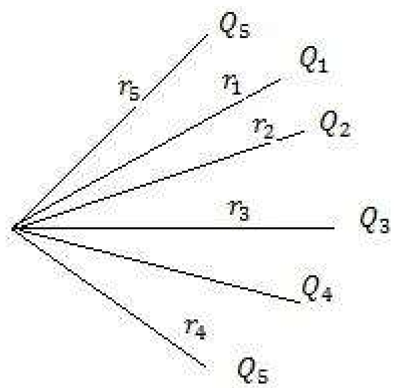
$$\vec{E} = \frac{\vec{F}}{Q_1} = \frac{Q_t}{4\pi\varepsilon_o R_1 t^2}\overrightarrow{a_R}$$

$$\vec{E} = \frac{Q_1}{4\pi\varepsilon_o R_1 t^2} \cdot \frac{r_t - r}{|r_t - r|}$$

- Unit of electric field intensity N/c or v/m

<u>Superposition principle</u>

- If there are more than two point charge when each exerts force on the test charge then the net force on the test charge can be obtained by using superposition principle or principle of superposition
- Consider a point charge $Q_t$surrounded by number of charges Q_1,Q_2,Q_3,....Q_n that the total force on Q_t is vector sum of all the forces exerted on Q_t due to each of the other point charges.
- Consider force exerted on $Q_t$due to Q_1 according to principle of superposition effects of other charges supressed



$$F_{Q_1Q_t} = \frac{Q_1Q_t}{4\pi\varepsilon_o R_1t^2}\frac{r_t-r}{|r_t-r|}$$

$$F_{Q_1Q_t} = \frac{Q_2Q_t}{4\pi\varepsilon_o R_2t^2}\frac{r_t-r_2}{|r_t-r_2|}$$

$$F_{Q_n} = \frac{Q_nQ_t}{4\pi\varepsilon_o R_nt^2}\frac{r_t-r_n}{|r_t-r_n|}$$

$$F_{total} = F_{Q_1Q_t}+F_{Q_1Q_t}+\ldots\ldots+F_{Q_nQ_t}$$

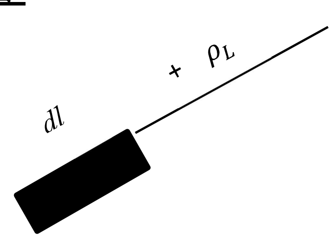
$$F = \frac{Q_t}{4\pi\varepsilon_o}\sum_{i=1}^{n}\frac{Q_i}{R_it^2}\frac{r_t-r_n}{|r_t-r_n|}\frac{r_t-r_i}{|r_t-r_i|}$$

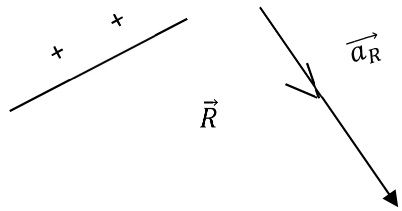
ELECTRIC FIELD INTENSITY DUE TO VARIOUS CHARGE DISTRIBUTION

- The electric field intensity due to a point charge Q is given by

$$\vec{E} = \frac{Q}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

1. **$\vec{\mathbf{E}}$ due to line charge**





- Consider a line charge distribution having a charge density ρ_L.
- The charge dQ on the differential length dl is,

$$dQ = \rho_L dl$$

- Hence the differential electric field $d\vec{E}$ at point P due to dQ is given by

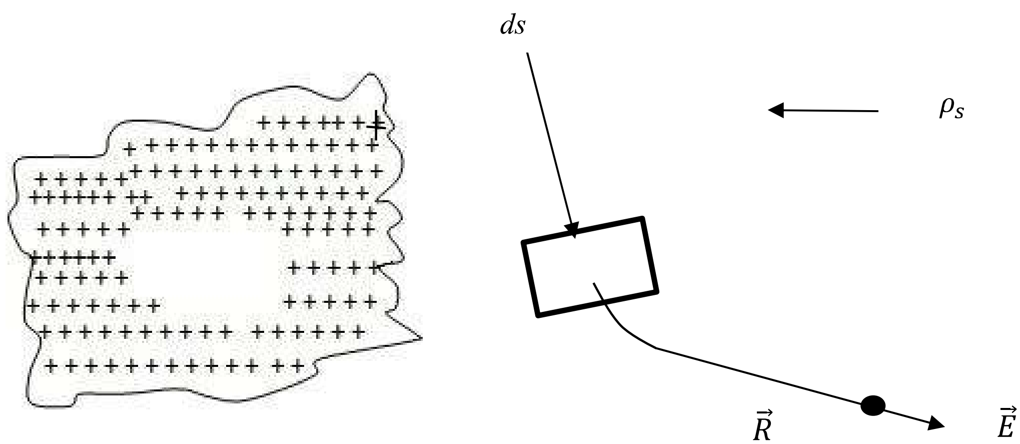
$$d\vec{E} = \frac{dQ}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

$$= \frac{\rho_L dl}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

- Hence the total $\vec{E}$ at a point P due to line charge can be obtained by integrating $d\vec{E}$ over the length of the charge.

$$\vec{E} = \int_L \frac{\rho_L dl}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

2. $\vec{E}$ due to surface charge



- Consider a surface charge distribution having a charge densityρ_s
- The charge dQ on the differential surface area ds is

$$dQ = \rho_s ds$$

- Hence the differential electric field $d\vec{E}$ at a point P due to dQ is given by,

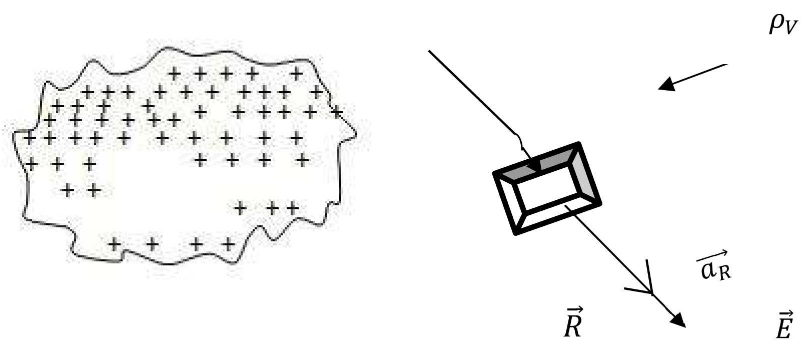
$$d\vec{E} = \frac{dQ}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

$$= \frac{\rho_s ds}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

- The total $\vec{E}$ at a point P is to be obtained by integrating d$\vec{E}$over the surface area on which charge is distributed.

$$\vec{E} = \int_s \frac{\rho_s ds}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

3. <u>$\vec{E}$ due to volume charge</u>



- Consider a volume charge distribution having a charge densityρ_v
- The charge dQ on the differential surface area ds is

  dQ =$\rho_v dv$
- Hence the differential electric field d$\vec{E}$ at a point P due to dQ is given by,

$$d\vec{E} = \frac{dQ}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

$$= \frac{\rho_v dv}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

- The total $\vec{E}$ at a point P is to be obtained by integrating d$\vec{E}$over the surface area on which charge is distributed.

$$\vec{E} = \int_s \frac{\rho_v dv}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

- Thus if there are all possible types of charge distributions, then the total $\vec{E}$ at appoint is the vector sum of individual electric field intensity produced by each of the charges at a point under consideration

$\vec{E}_{total} = \vec{E}_p + \vec{E}_l + \vec{E}_s + \vec{E}_v$

Where $\vec{E}_p$, $\vec{E}_l$, $\vec{E}_s$ *and* $\vec{E}_v$ are the field intensity due to point, line, surface and volume charge distributions respectively.

Electric field due to discrete charge

[Charge are not uniform]

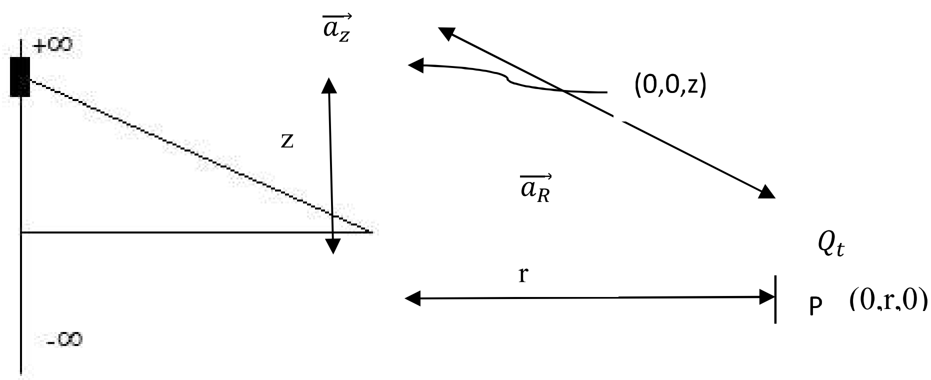
$$E_1=\frac{F_{Q_1Q_t}}{Q_1}=\frac{Q_t}{4\pi\varepsilon_o R_1t^2}\frac{r_t-r_1}{|r_t-r_1|}$$

$$E_2=\frac{Q_t}{4\pi\varepsilon_o R_2t^2}\frac{r_t-r_2}{|r_t-r_2|}$$

$$E_n=\frac{Q_t}{4\pi\varepsilon_o R_nt^2}\frac{r_t-r_n}{|r_t-r_n|}$$

$$E=\frac{Q_t}{4\pi\varepsilon_o}\sum_{i=1}^{n}\frac{1}{R_it^2}\frac{r_t-r_n}{|r_t-r_n|}\frac{r_t-r_i}{|r_t-r_i|}$$

ELECTRIC FIELD DUE TO INFINITE LINE CHARGE



- Consider an infinite long straight line carrying uniform line charge having density of ρ_l. Let this line lies in z-axis from -∞ to ∞. Hence it is called infinite line charge.
- Consider a point P is on y-axis at which electric field intensity to be calculated. The distance of the point P from origin is 'r'.

$$\vec{E}=\frac{Q}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

Charges are line form Q = $\int \rho_l\, dl$

$$\vec{E}=\int\frac{\rho_l}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}dl \qquad \text{(for a small infinite line charge)}$$

$$\vec{E}=\int_{-\infty}^{\infty}\frac{\rho_l}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}dz$$

$\vec{E}=\int_{-\infty}^{\infty}\frac{\rho_l}{4\pi\varepsilon_o R^2}\frac{r\overrightarrow{a_R}-Z\overrightarrow{a_z}}{|r\overrightarrow{a_R}-Z\overrightarrow{a_z}|}dz$ Note

$$R=|r\vec{p}-r\vec{l}|=\sqrt{r^2+z^2}\overrightarrow{r_p} \;=\; r\overrightarrow{a_y}$$

$$\vec{E} = \int_{-\infty}^{\infty} \frac{\rho_l}{4\pi\varepsilon_o(r^2+z^2)} \frac{r\overrightarrow{a_R} - Z\overrightarrow{a_z}}{|r\overrightarrow{a_R} - Z\overrightarrow{a_z}|}\ dz\vec{r} \text{ dl= } z\overrightarrow{a_z}$$

$$\vec{E} = \int_{-\infty}^{\infty} \frac{\rho_l}{4\pi\varepsilon_o(r^2+z^2)} \frac{r\overrightarrow{a_R} - Z\overrightarrow{a_z}}{\sqrt{r^2+z^2}}\ dz\overrightarrow{a_R} = \frac{\overrightarrow{r_p} - \text{r}\overrightarrow{dl}}{|\overrightarrow{r_p} - \text{r}\overrightarrow{dl}|}$$

$$\vec{E} = \int_{-\infty}^{\infty} \frac{\rho_l(r\overrightarrow{a_R} - Z\overrightarrow{a_z})}{4\pi\varepsilon_o(r^2+z^2)^{3/2}}\ dz\overrightarrow{a_R} = \frac{\text{r}\overrightarrow{a_y} - \text{z}\overrightarrow{az}}{|\text{r}\overrightarrow{a_y} - \text{z}\overrightarrow{az}|}$$

- The charges present in the z-axis induced electric field will be cancelled with each other within the axis, so z component at the equal to zero.

$$\therefore Z\overrightarrow{a_z} = 0$$

$$Z = r\tan\theta$$

$$dz = rsec^2\theta d\theta$$

$$\theta = tan^{-1}(\frac{Z}{r})$$

$z = -\infty$; $\theta = \frac{-\pi}{2}$

$z = \infty$; $\theta = \frac{\pi}{2}$

$$\vec{E} = \int_{-\pi/2}^{\pi/2} \frac{\rho_l r\overrightarrow{a_y} rsec^2\theta d\theta}{4\pi\varepsilon_o(r^2 + r^2tan^2\theta)^{3/2}}$$

$$\vec{E} = \int_{-\pi/2}^{\pi/2} \frac{\rho_l}{4\pi\varepsilon_o} \frac{r^2\overrightarrow{a_y}sec^2\theta d\theta}{(r^2)^{3/2}(1 + tan^2\theta)^{3/2}}$$

$$= \frac{\rho_l}{4\pi\varepsilon_o r} \int_{-\pi/2}^{\pi/2} \frac{sec^2\theta d\theta}{(sec^2\theta)^{3/2}} \overrightarrow{a_y}$$

$$= \frac{\rho_l}{4\pi\varepsilon_o r} \int_{-\pi/2}^{\pi/2} \frac{1}{sec\theta} \overrightarrow{a_y} d\theta$$

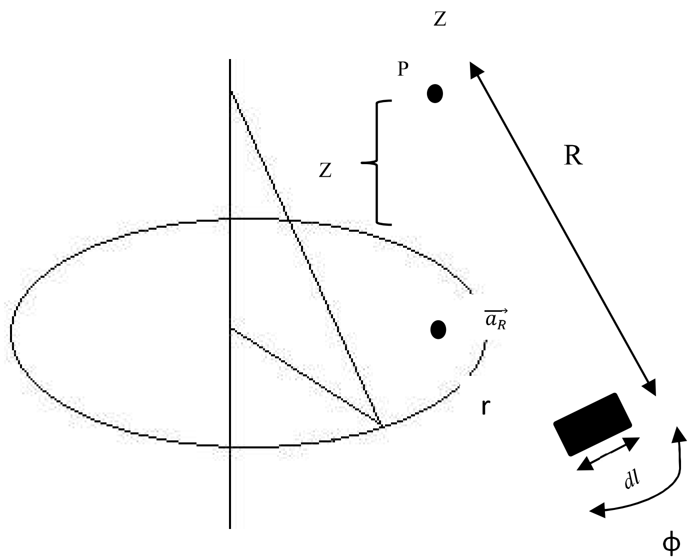
$$\vec{E} = \frac{\rho_l}{4\pi\varepsilon_o r} \int_{-\pi/2}^{\pi/2} cos\theta\, \overrightarrow{a_y} d\theta$$

$$= \frac{\rho_l}{4\pi\varepsilon_o r} \overrightarrow{a_y} [\sin\theta]_{-\pi/2}^{\pi/2}$$

$$= \frac{\rho_l}{4\pi\varepsilon_o r} \overrightarrow{a_y} [2]$$

$$\vec{E} = \frac{\rho_l}{2\pi\varepsilon_o r} \overrightarrow{a_y}$$

ELECTRIC FIELD DUE TO CHARGE CIRCULAR RING



- Consider a circular ring of radius r, placed in xy plane with centre at origin carrying a uniformly along its circumference.
- The charge density is ρ_l. Let point P is at a perpendicular distance from the ring.
- Consider a small differential length dl on this ring. The charge if dl is dQ
- The electric field intensity due to a charge is written as

$$\vec{E} = \frac{\vec{F}}{Q} = \frac{Q}{4\pi\varepsilon_o R^2}\overrightarrow{a_R}$$

$$\vec{E} = \int \frac{\rho_l}{4\pi\varepsilon_o r}\overrightarrow{a_R}$$

$\overrightarrow{a_R}$ --- unit vector

$Positive\ vector\ of\ dl = \mathrm{r}\overrightarrow{a_R}$

$$Positive\ vector\ of\ p = z\overrightarrow{a_z}$$

Limit is 0 to 2π

R $=\sqrt{r^2 + z^2}$

$\overrightarrow{a_R} = \frac{z\overrightarrow{a_z} - \mathrm{r}\overrightarrow{a_r}}{|z\overrightarrow{a_z} - \mathrm{r}\overrightarrow{a_r}|}$

$$\overrightarrow{a_R} = \frac{z\overrightarrow{a_z} - \mathrm{r}\overrightarrow{a_r}}{\left|\sqrt{r^2 + z^2}\right|}$$

$$dl = rd\phi$$

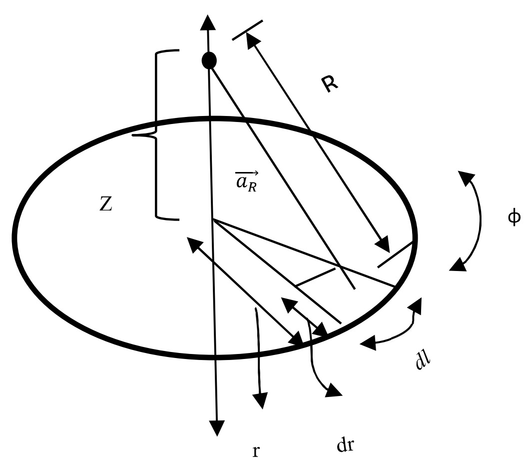
$$\vec{E} = \int_0^{2\pi} \frac{\rho_l r d\phi}{4\pi\varepsilon_o} \frac{z\overrightarrow{a_z}}{|z^2 + r^2|^{3/2}}$$

$$= \frac{\rho_l r z d\phi \overrightarrow{a_z}}{4\pi\varepsilon_o [z^2 + r^2]^{3/2}} \int_0^{2\pi} d\phi$$

$$= \frac{\rho_l r z d\phi \overrightarrow{a_z}}{4\pi\varepsilon_o [z^2 + r^2]^{3/2}} [2\pi]$$

$$\vec{E} = \frac{\rho_l r z d\phi \overrightarrow{a_z}}{2\pi\varepsilon_o [z^2 + r^2]^{3/2}}$$

ELECTRIC FIELD INTENSITY DUE TO INFINITE DISC (OR) SHEET CHARGES:



- Consider an infinite sheet or disc of charges having uniform cnarges density (ρ_s) placed in xy plane.
- The point P at which electric field intensity ($\vec{E}$) to be calculated on z-axis.
- Consider the differential surface are ds carrying charge dQ.
- The normal direction to ds is z-direction.

  The electric field intensity written as

$$\vec{E} = \frac{Q}{4\pi\varepsilon_o R^2} \overrightarrow{a_R}$$

$$= \frac{\int \rho_s ds}{4\pi\varepsilon_o R^2} \overrightarrow{a_R}$$

$$Limit\ of\ radius \rightarrow 0\ to\ \infty$$
$$Limit\ of\ radius \rightarrow 0\ to\ 2\pi$$

ds =*dr d*ϕ

ds=*rdrd*ϕ

$$\overrightarrow{a_R} = \frac{z\overrightarrow{a_z} - r\overrightarrow{a_r}}{|z\overrightarrow{a_z} - r\overrightarrow{a_r}|}$$

$$\overrightarrow{a_R} = \frac{z\overrightarrow{a_z} - r\overrightarrow{a_r}}{\sqrt{r^2 + z^2}}$$

$$\vec{E} = \int_0^\infty \int_0^{2\pi} \frac{\rho_l r d\phi dr}{4\pi\varepsilon_o(r^2+z^2)} \cdot \frac{z\overrightarrow{a_z} - r\overrightarrow{a_r}}{\sqrt{r^2+z^2}}$$

$$Radial\ components\ r\ \overrightarrow{a_r} = 0$$

$$\vec{E} = \int_0^\infty \int_0^{2\pi} \frac{\rho_l r d\phi dr}{4\pi\varepsilon_o(r^2+z^2)^{3/2}} z\overrightarrow{a_z}$$

$u^2 = z^2 + r^2$

r=0 , u=z

r=∞ , u=∞

(1) $=> 2ydu=2rdr$

$$\vec{E} = \int_z^\infty \int_0^{2\pi} \frac{\rho_s r d\phi dr z\overrightarrow{a_z}}{4\pi\varepsilon_o(u^2)^{3/2}}$$

$$= \int_z^\infty \int_0^{2\pi} \frac{\rho_s r d\phi dr\ z\overrightarrow{a_z}}{4\pi\varepsilon_o u^3}$$

$u^2 = z^2 + Q^2$

$2udu=2rdr$

$udu=rdr$

$$\vec{E} = \int_0^{2\pi} \int_z^\infty \frac{\rho_s d\phi\ udu\ z\overrightarrow{a_z}}{4\pi\varepsilon_o u^3}$$

$= \frac{z\overrightarrow{a_z}\rho_s}{4\pi\varepsilon_o} \int_0^{2\pi} \int_z^\infty \frac{1}{u^2} dud\phi$

$= \frac{z\overrightarrow{a_z}\rho_s}{4\pi\varepsilon_o} \int_0^{2\pi} d\phi [\frac{-1}{u}]_Z^\infty$

$= \frac{z\overrightarrow{a_z}\rho_s}{4\pi\varepsilon_o} \frac{1}{z} \int_0^{2\pi} d\phi$

$\vec{E} = \frac{\overrightarrow{a_z}\rho_s}{4\pi\varepsilon_0} 2\pi$

$$\vec{E} = \frac{\overrightarrow{a_z}\rho_s}{2\varepsilon_0}$$

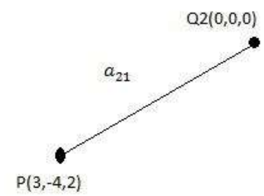
Problem: 1

Calculate electric field intensity at P(3,-4,2) in free space called by

a) $Q_1=2\mu c$ at (0,0,0)
b) $Q_2=3\mu c$ at (-1,2,3)
c) $Q_1=2\mu c$ at (0,0,0) and $Q_2=3\mu c$ at (-1,2,3)

<u>Sol</u>

a)



Unit vector $\overrightarrow{a_{21}} = \frac{r_1-r_2}{|r_1-r_2|} = \frac{3\overrightarrow{a_x}-4\overrightarrow{a_y}+2\overrightarrow{a_z}}{\sqrt{9+16+4}}$

$$= \frac{3\overrightarrow{a_x}-4\overrightarrow{a_y}+2\overrightarrow{a_z}}{\sqrt{29}}$$

Electric field intensity at P due to Q_1

$E = \frac{Q1}{4\pi\varepsilon_0 r^2}\overrightarrow{a_{21}}$

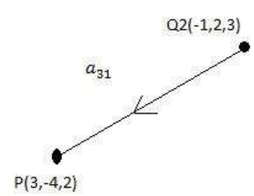
$$= \frac{2\times10^{-6}}{4\pi\times\frac{1}{36\pi\times10^{9}\times(\sqrt{29})^2}} \cdot \frac{3\,\overrightarrow{a_x}+4\overrightarrow{a_y}+2\overrightarrow{a_z}}{\sqrt{29}}$$

$$= \frac{2\times9\times10^{3}}{29\sqrt{29}} \quad (3\overrightarrow{a_x} + 4\overrightarrow{a_y} + 2\overrightarrow{a_z})$$

$E = 345\overrightarrow{a_x}-460\overrightarrow{a_y}+230\overrightarrow{a_z}$ v/m

b)

$$\vec{a}_{31} = \frac{\overrightarrow{a_x}(3-1)+\overrightarrow{a_y}(-4-(2))+\overrightarrow{a_z}(2-3)}{(3+1)^2+(-4-2)^2+(2-3)^2}$$



$$= \frac{4\overrightarrow{a_x}-6\overrightarrow{a_y}-\overrightarrow{a_z}}{\sqrt{16+36+1}}$$

$$= \frac{4\overrightarrow{a_x}-6\overrightarrow{a_y}-\overrightarrow{a_z}}{\sqrt{53}}$$

Electric intensity at P due to Q_2

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}\overrightarrow{a_{31}}$$

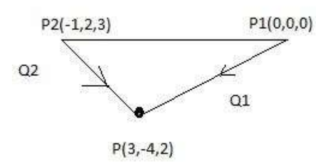
$$= \frac{3\times10^{-6}}{4\pi\times\frac{1}{36\pi\times10^{-6}}\times(\sqrt{53})^2}\frac{4\overrightarrow{a_x}-6\overrightarrow{a_y}-\overrightarrow{a_z}}{\sqrt{53}}$$

$$E = 280\,\overrightarrow{a_x} - 420\overrightarrow{a_y} - 70\,\overrightarrow{a_z}\ \text{V/m}$$

C)

$$\vec{a}_{31}=\frac{3\overrightarrow{a_x}-4\overrightarrow{a_y}+2\overrightarrow{a_z}}{\sqrt{9+16+4}}$$

$$=\frac{3\overrightarrow{a_x}-4\overrightarrow{a_y}+2\overrightarrow{a_z}}{\sqrt{29}}$$



Electric field intensity at P due to Q_1

$$E_1=345\overrightarrow{a_x}-460\overrightarrow{a_y}+230\overrightarrow{a_z}\ \text{v/m}$$

Electric field intensity at P due to Q_2

$$E_2=280\,\overrightarrow{a_x} - 420\overrightarrow{a_y} - 70\,\overrightarrow{a_z}\ \text{V/m}$$

Total electric field intensity at P due to Q_1 and Q_2

$$E=E_1+E_2$$

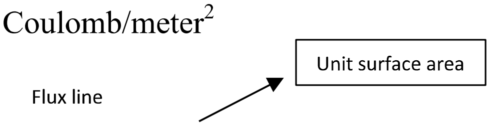
$$= 345\overrightarrow{a_x}-460\overrightarrow{a_y}+230\overrightarrow{a_z}+280\,\overrightarrow{a_x} - 420\overrightarrow{a_y} - 70\,\overrightarrow{a_z}$$

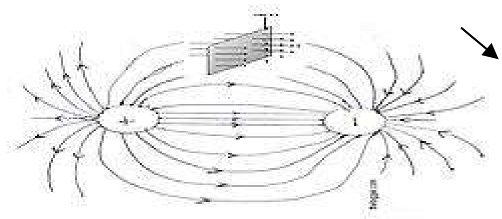
$$E=65\overrightarrow{a_x}-880\overrightarrow{a_y}+160\overrightarrow{a_y}\ \text{V/m}$$

ELECTRIC FLUX DENSITY (D)

Electric flux density or displacement density is defined as the electric flux per unit area.

$$D = \frac{Q}{A}\ \text{Coulomb/meter}^2$$





Consider the two point charges as shown in fig. The flux lines originating from positive charge and terminating at negative charge.

Consider a unit surface area as shown in fig. The number of flux lines are passing through this surface area.

The net flux passing normal through the unit surface area is called the electric flux density(D)

GAUSS'S LAW

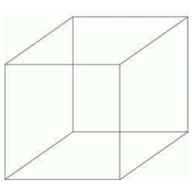
The electric flux passing through any closed surface is equal to the total charge enclosed by the surface.

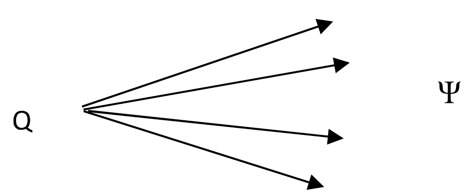
$$\Psi=Q$$

Where

$\Psi \rightarrow$ Total electric flux leaving normally from a closed surface

$Q \rightarrow$ Net charge enclosed by the closed surface





Proof:

Consider a charge Q at the origin of a spherical co-ordinate system, Whose co-ordinates are r, θ, ϕ as shown in fig.

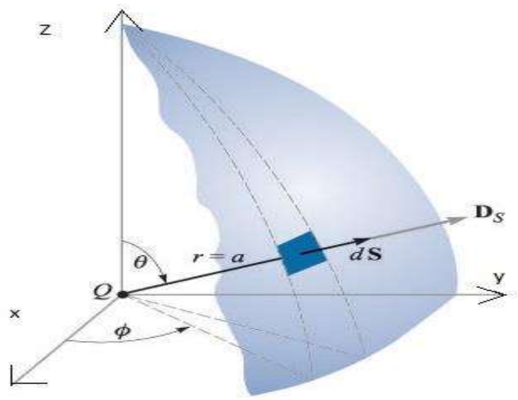
The electric field intensity due to the charge Q is

$E = \frac{Q}{4\pi\varepsilon_0 r^2}$

The electric flux density D=εE

$D = \frac{Q}{4\pi r^2}$

Consider a small element of area ds on the surface the sphere at a distance r from the origin as shown in fig.



$$ds = r.d\theta r sin\theta d\phi$$

$= r^2.sin\theta d\theta d\phi$

The electric flux Ψ= ∫ dΨ

$= \oint_s D ds$

$\Psi = \int_s \frac{Q}{4\pi r^2} r^2.sin\theta d\theta d\phi$

$= \frac{Q}{4\pi} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} sin\theta d\theta d\phi$

$= \frac{Q}{4\pi} \int_{\phi=0}^{2\pi} [-cos\theta]_0^{\pi} d\phi$

$= \frac{Q}{4\pi} \int_{\phi=0}^{2\pi} 2\ d\phi = \frac{Q}{2\pi} [\phi]_0^{2\pi} = \frac{Q}{2\pi} [2\pi]$

$\Psi = Q$

Hence proved.

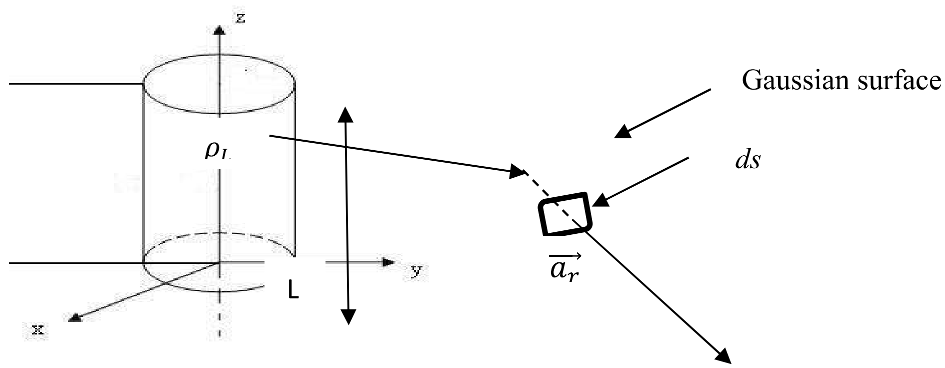
APPLICATIONS OF GAUSS'S LAW

- The Gauss's law can be used to find $\vec{E}$ or $\vec{D}$ for symmetrical charge distributions, such as point charge an infinite line charge, an infinite sheet of charge and a spherical distribution of charge.
- Gauss's law is also used to find the charge enclosed or flux passing through the closed surface.

While selecting the closed Gaussian surface to apply the Gauss's law, following conditions must be satisfied,

1. $\vec{D}$ is everywhere either normal or tangential to the closed surface i.e θ=$\frac{\pi}{2}$ or π. So that $\vec{D}.\vec{ds}$ becomes Dds or zero respectively
2. $\vec{D}$ is constant over the portion of the closed surface for which $\vec{D}.\vec{ds}$ is not zero.

1. <u>Infinite Line charge</u>



- Consider an infinite line charge of density ρ_Lc/m lying [illegible] z-axis from $-\infty$ to ∞ .
- Consider the Gaussian surface as the right circular cylinder with z-axis as its axis and radius 'r' as shown in fig. The length of the cylinder is L.
- The flux density at any point on the surface is direct radially outwards i.e in the $\vec{a_r}$ direction according to cylindrical co-ordinate system.
- Consider differential surface area ds as shown which is at a radial distance r from the line charge. The direction normal to ds is $\vec{a_r}$.
- As the line charge is along z-axis, there can nit be any component of $\vec{D}$ in z-direction. So $\vec{D}$ has only radial component.

$$Q=\oint_S \vec{D}.\vec{ds}$$

The integration is to be evaluated for side surface, top surface and bottom surface.

$$Q = \oint_{side} \vec{D}.\vec{ds}+\oint_{top} \vec{D}.\vec{ds} +\oint_{bottom} \vec{D}.\vec{ds}$$

$\vec{D} = D_r \overrightarrow{a_r}$ as has only radial component

$\overrightarrow{ds} = r d\phi dz \overrightarrow{ds}$ normal to $\overrightarrow{a_r}$ direction

$\vec{D}.\overrightarrow{ds} = D_r r d\phi dz(\overrightarrow{a_r}.\overrightarrow{a_r})$ $\overrightarrow{a_r}.\overrightarrow{a_r} = 1$

$= D_r r d\phi dz$

Now D_r is constant over the side surface.

- As $\vec{D}$ has only radial component and no component along $\overrightarrow{a_z}$ and -$\overrightarrow{a_r}$. Hence integrations over top and bottom surface is zero.

$$\oint_{top} \vec{D}.\overrightarrow{ds} = \oint_{bottom} \vec{D}.\overrightarrow{ds} = 0$$

$$Q = \oint_{side} \vec{D}.\overrightarrow{ds}$$

$$= \oint_{side} D_r r d\phi dz$$

$$= \int_{z=0}^{L} \int_{\phi=0}^{2\pi} D_r r d\phi dz$$

$$= r\, D_r [z]_0^L [\phi]_0^{2\pi}$$

$$Q = 2\pi D_r L$$

$$\vec{D} = D_r \overrightarrow{a_r}$$

$$= \frac{Q}{2\pi r L} \overrightarrow{a_r}$$

But $\frac{Q}{L} = \rho_L$ C/m

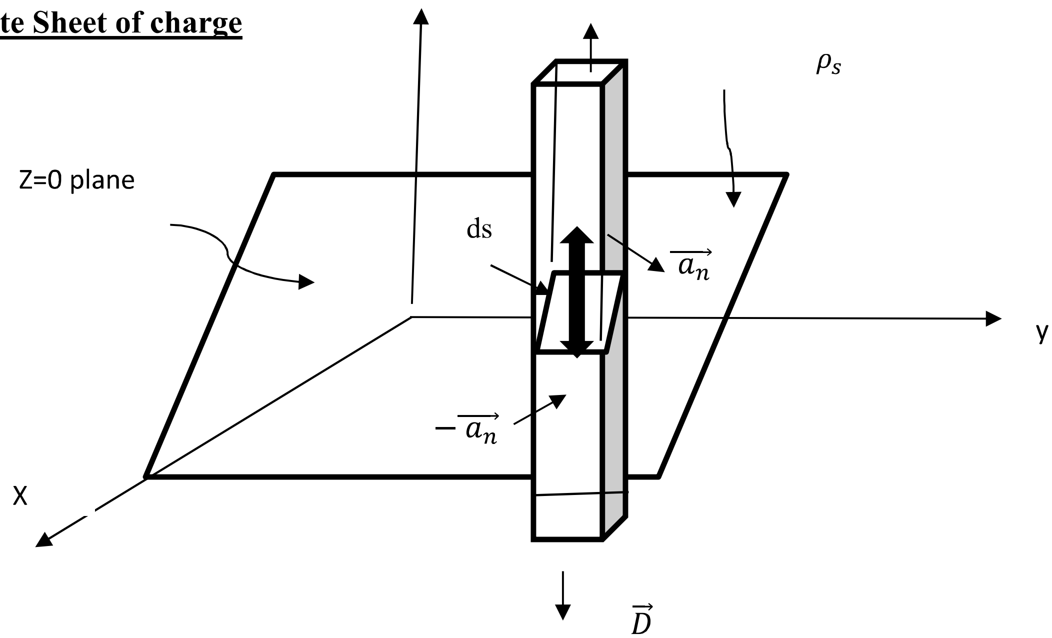
$\vec{D} = \frac{\rho_L}{2\pi r} \overrightarrow{a_r}$ C/m --- Due to infinite line charge

$\vec{E} = \frac{\vec{D}}{\varepsilon_0} = \frac{\rho_L}{2\pi\varepsilon_0 r} \overrightarrow{a_r}$ V/m

∴ The result are same as obtained from the coulomb's law

z $\vec{D}$

2. **Infinite Sheet of charge**



- Consider the infinite sheet of charge of uniform charge density ρ_sc/m^2, lying in the z=0 plane. i.e xy palne.
- Consider a rectangular box as a Gaussian surface which is cut in the sheet of charge to give *ds=dxdy*.

$\vec{D}$ acts normal to the plane i.e $\overrightarrow{a_n} = \overrightarrow{a_z}$

$\overrightarrow{a_n} = -\overrightarrow{a_z}$ direction

Hence $\vec{D}$=0 in x and y directions.

Hence the charge enclosed can be written as

$Q=\oint_s \vec{D}.\overrightarrow{ds}$

$Q = \oint_{side} \vec{D}.\overrightarrow{ds} + \oint_{top} \vec{D}.\overrightarrow{ds} + \oint_{bottom} \vec{D}.\overrightarrow{ds}$

But $\oint_s \vec{D}.\overrightarrow{ds}$=0 as $\vec{D}$has no component in x and y direction

Now, $\vec{D} = D_z \overrightarrow{a_z}$for top surface

And *ds=dxdy.*

$\vec{D}.\overrightarrow{ds} = D_z\, dxdy(\overrightarrow{a_z}.\overrightarrow{a_z})$

$= D_z\, dxdy$

$\vec{D} = D_z(-\overrightarrow{a_z})$for bottom surface.

$$\overrightarrow{ds} = dxdy(-\overrightarrow{a_z})$$

$$\overrightarrow{D}.\overrightarrow{ds} = D_z\, dxdy(\overrightarrow{a_z}.\overrightarrow{a_z})$$

$$= D_z\, dxdy$$

$$Q = \oint_{top} \overrightarrow{D}.\overrightarrow{ds} + \oint_{bottom} \overrightarrow{D}.\overrightarrow{ds}$$

Now , $\oint_{top} \overrightarrow{D}.\overrightarrow{ds} = \oint_{bottom} \overrightarrow{D}.\overrightarrow{ds}$= A= Area of surface

$$Q=2D_zA$$

But, Q $=\rho_s \times A \rho_s$= surface charge density

$$\rho_s=2D_z$$

$$D_z=\frac{\rho_s}{2}$$

$$\overrightarrow{D}=D_z\overrightarrow{a_z}$$

$$\overrightarrow{D} = \frac{\rho_s}{2}\overrightarrow{a_z} \text{ C/m}^2$$

$$\vec{E}= \frac{\vec{D}}{\varepsilon_0} = \frac{\rho_s}{2\varepsilon_0}\overrightarrow{a_z} \text{ V/m}$$

The result are same as obtained by the coulomb's law for the infinite sheet of charge.

Problem:1

The flux density $\vec{D}=\frac{r}{3}\overrightarrow{a_r}$nc/m^2 is in free space.

a) Find $\vec{E}$ at r= 0.2 m.
b) Find the total electric flux leaving the space of r=0.2m
c) Find the total charge within the sphere of r=0.3m

Sol

a) $\vec{E}= \frac{\vec{D}}{\varepsilon_0} = \frac{r}{3\varepsilon_0}\overrightarrow{a_r}$

$\vec{E}= \frac{0.2\times10^{-9}}{3\times8.854\times10^{-12}} = 7.5295\, \overrightarrow{a_r}$ V/m

b) $Q=\Psi=\oint_S \overrightarrow{D}.\overrightarrow{ds}$

Consider a differential area Ds normal to $\overrightarrow{a_r}$ which is $r^2.sin\theta d\theta d\phi$.

$$\overrightarrow{ds} = r^2.sin\theta d\theta d\phi\overrightarrow{a_r}$$

$\vec{D}=\frac{r}{3}\vec{a_r}$

$\vec{D}.\vec{ds}=\frac{r^3}{3}\ sin\theta d\theta d\phi$ $\qquad\qquad\qquad\qquad$ $\vec{a_r}.\vec{a_r}=1$

$Q=\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi}\frac{r^3}{3}\ sin\theta d\theta d\phi=\frac{r^3}{3}[-cos\theta]_0^{\pi}[\phi]_0^{2\pi}$

$=\frac{4}{3}\pi r^3$ nc

∴ At r = 0.2m,

$Q=\frac{4}{3}\pi(0.2)^3$=0.0335 nc=33.51 Pc

c) At r = 0.3m,

$Q=\frac{4}{3}\pi(0.3)^3$=0.113 nc=113.097 pc

ELECTRIC POTENTIAL

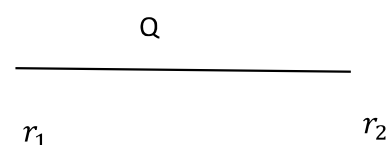
An electric charge produces an electric field around it ad if a test charge is brought in to this region, it experience a force. The force is given by

F=QE

Since there is a movement of charge in the electric field from one point $r_1$to another point r_2, there will be workdone against the force.

W=$-\int_{r_1}^{r_2}q.E\ dr$

$=-q\int_{r_1}^{r_2}E\ dr$



Potential difference(v) is defined as the work done in moving a unit positive charge from one point to another in an electric field.

Work done on unit positive chrge per change is

$$V=\frac{w}{q}$$

$$V=\int_{r_1}^{r_2}E\ .dr \quad \text{Joules/coulomb}$$

But $\qquad$ E = $\frac{Q}{4\pi\varepsilon r^2}$

$$V=-\frac{Q}{4\pi\varepsilon}\int_{r_1}^{r_2}\frac{1}{r^2}dr$$

$$=-\frac{Q}{4\pi\varepsilon}\left[\frac{1}{-r}\right]_{r_1}^{r_2}$$

$$= -\frac{Q}{4\pi\varepsilon}\left[\frac{1}{r_1} - \frac{1}{r_2}\right] \text{ volts}$$

This is the potential difference between two point r_1 and r_2

$$V = \frac{Q}{4\pi\varepsilon r_1} - \frac{Q}{4\pi\varepsilon r_2}$$

$$V = V_1 - V_2$$

If the test charge is moved from infinity to a given point in the electric field

$V_2 = 0$ $\qquad\qquad\qquad [\therefore \frac{1}{r^2} = 0]$

Then $V = V_1$

Absolute potential or potential at a point is defined as the work done is moving a unit positive charge from infinity to a given point in an electric field.

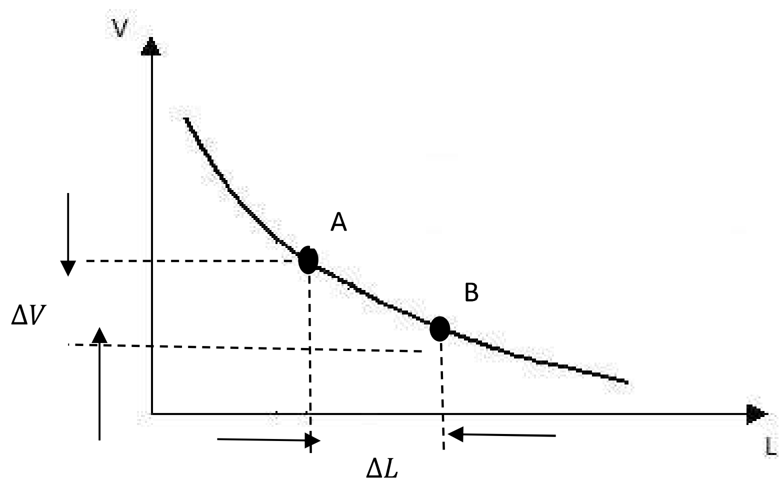
$$V = \frac{Q}{4\pi\varepsilon r}\ volts$$

POTENTIAL GRADIENT

- Consider an electric field $\vec{E}$ due to positive charge placed at the origin of a sphere. Then,

$$V = \int \vec{E}.\overrightarrow{dL}$$

$$= \frac{Q}{4\pi\varepsilon r}$$



- The potential decreases as distance of point from the charge increases. This is shown in fig.
- It is known that the line integral of $\vec{E}$ between the two points gives a potential differential between the two points.
- For an elementary length ΔL, We can write

$$\therefore V_{AB} = \Delta V$$
$$= -\vec{E}.\Delta\vec{L}$$

- The rate of changr of potential with respect to the distance is called the potential gradient.

Grad V = $Lt_{\Delta l \to 0}\left(\frac{\Delta V}{\Delta l}\right) = \frac{dV}{dl}$

$$dV = -Edl$$

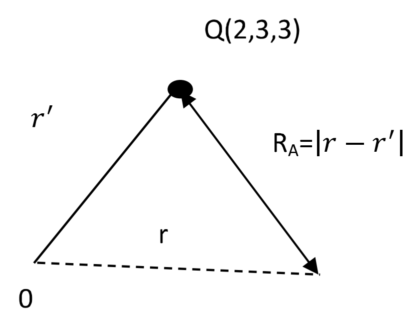
$$= -Ecos\theta dl$$

$$\frac{dV}{dl} = -Ecos\theta$$

If $\theta = 0$, $\frac{dV}{dl}$ is maximum

$$|\text{Grad V}| = \frac{dV}{dl} = -E$$

$$\Delta V = -E$$



$$\Delta V = \frac{\partial V}{\partial x}i + \frac{\partial V}{\partial y}j + \frac{\partial V}{\partial z}k = -E$$

The gradiant of a scalar function V can be written as

$$\Delta V = \frac{\partial V}{\partial l}i + \frac{\partial V}{\partial l_2}j + \frac{\partial V}{\partial l_3}k$$

Where

$\partial l_1 = h_1 \partial P_1$

$\partial l_2 = h_2 \partial P_2$

$\partial l_3 = h_3 \partial P_3$

$$\Delta V = \left(\frac{1}{h_1}\frac{\partial V}{\partial P_1}\right) i + \left(\frac{1}{h_2}\frac{\partial V}{\partial P_2}\right) j + \left(\frac{1}{h_3}\frac{\partial V}{\partial P_3}\right) k$$

In Cartesian system

$$h_1 = h_2 = h_3 = 1$$

$$P_1 = x, \qquad P_2 = y, \qquad P_3 = z$$

$$\Delta V = \frac{\partial V}{\partial x} i + \frac{\partial V}{\partial y} j + \frac{\partial V}{\partial Z} k$$

In Cylindrical co-ordinate system

$$h_1 = 1, \quad h_2 = r, \qquad h_3 = 1$$

$$P_1 = \rho, \qquad P_2 = 0, \qquad P_3 = z$$

$$\Delta V = \frac{\partial V}{\partial \rho} i + \frac{\partial V}{\partial \theta} j + \frac{\partial V}{\partial Z} k$$

Spherical co-ordinate system

$$h_1 = 1, \quad h_2 = r, \qquad h_3 = 1$$

$$P_1 = R, \qquad P_2 = \theta, \qquad P_3 = \phi$$

$$\Delta V = \frac{\partial V}{\partial R} i + \frac{1}{R}\frac{\partial V}{\partial \theta} j + \frac{1}{Rsin\theta}\frac{\partial V}{\partial \phi} k$$

Problem 1

Find the electric potential at a point (4,3)m due to a charge of 10^{-9}C located at the origin in free space.

Sol.

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

$r = \sqrt{4^2 + 3^2}$ =5m

$$V = \frac{10^{-9}}{4\pi\frac{1}{36\pi\times10^{-9}}(5)} = \frac{9}{5} = 1.8\text{V}$$

Problem 2

IF same charge Q=0.4 nc in above example is located at (2,3,3) then obtain the absolute potential of point A (2,2,3)

Sol.

Now the Q is located at (2,2,3)

The potential at A is given by

$$V_A = \frac{Q}{4\pi\varepsilon_0 RA}$$

$$RA = |r - r'| = \sqrt{(2-2)^2 + (2-3)^2 + (3-3)^2} = 1$$

$$V_A = \frac{0.4\times 10^{-9}}{4\pi\times 8.854\times 10^{-12}\times 1} = 3.595V$$

POSSION'S AND LAPLACE EQUATIONS

- According to Gauss's law in point form, the divergence of electric flux is equal to the charge density.

$$\nabla D = \rho v$$

w.k.t $\quad D = \varepsilon E$

$$\nabla(\varepsilon E) = \rho v$$

$$\varepsilon\nabla . E = \rho v$$

$$\nabla . E = \frac{\rho v}{\varepsilon}$$

w.k.t $\quad E = -\nabla V$

$$\nabla . (-\nabla V) = \frac{\rho v}{\varepsilon}$$

$$\nabla . (\nabla V) = \frac{-\rho v}{\varepsilon}$$

$$\nabla^2 V = \frac{-\rho v}{\varepsilon}$$

The above equation is the poission's equation.

$$\nabla.\nabla \text{ V} = \frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial V}{\partial y}\right) + \frac{\partial}{\partial Z}\left(\frac{\partial V}{\partial Z}\right) k$$

$= \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$

Poisson's equation for Cartesian co-ordinate system is written as

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = \frac{-\rho v}{\varepsilon}$$

For cylindrical co-ordinate system, the poisson's equation is

$$\nabla^2 V = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho^2}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^{\,2}} = \frac{-\rho v}{\varepsilon}$$

For spherical co-ordinate system, the poisson's equation is

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 sin\theta}\frac{\partial}{\partial \theta}\left(sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 sin\theta}\frac{\partial^2 V}{\partial \phi^{\,2}} = \frac{-\rho v}{\varepsilon}$$

If the volume charge density (ρv) is zero, then

$$\nabla^2 V = 0 \qquad \rightarrow \textit{This is called laplace equation}$$

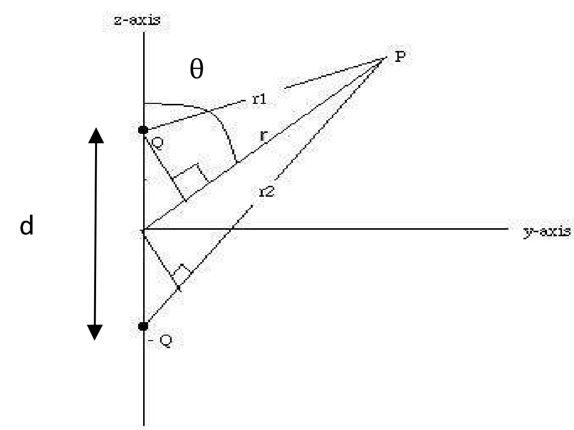
$$\nabla^2 \rightarrow \textit{Laplarian operator}$$

DIPOLE (OR) ELECTRIC DIPOLE (OR) POTENTIAL DUE TO DIPOLE

- An electric dipole or simply dipole is nothing but two equal and opposite charge separated by a very small distance. The product of charge and spacing is called electric dipole moment.
- Let Q an –Q be the two charge separated by a small distance d. The product of charge Q and spacing d is called dipole moment.

m=Qd

- Let P be any point at distance of r_1, r_2 &r_3 from +Q and =Q and mid point of dipole respectively as shown in fig.



Potential at P dye to +Q is

$$V_1 = \frac{Q}{4\pi\varepsilon r_1}$$

Potential at P due to –Q is

$$V_2 = \frac{-Q}{4\pi\varepsilon r_2}$$

The resultant potential at P

$$V = V_1 + V_2 = \frac{Q}{4\pi\varepsilon}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

If the point is too far away from the diploe, the distance r_1 and r_2 are written as

$$r_1 = r - \frac{d}{2}\cos\theta$$

$$r_2 = r + \frac{d}{2}\cos\theta$$

$$V = \frac{Q}{4\pi\varepsilon}\left(\frac{1}{r - \frac{d}{2}\cos\theta} - \frac{1}{r + \frac{d}{2}\cos\theta}\right)$$

$$= \frac{Q}{4\pi\varepsilon}\left(\frac{r + \frac{d}{2}\cos\theta - r - \frac{d}{2}\cos\theta}{r^2 - \left(\frac{d}{2}\cos\theta\right)^2}\right)$$

$$= \frac{Q}{4\pi\varepsilon}\left(\frac{r + \frac{d}{2}\cos\theta - r - \frac{d}{2}\cos\theta}{r^2 - \left(\frac{d}{2}\cos\theta\right)^2}\right)$$

$= \frac{Q}{4\pi\varepsilon}\left(\frac{d\cos\theta}{r^2}\right)$ $\therefore \frac{d}{2} << r^2$

$V = \frac{m\cos\theta}{4\pi\varepsilon r^2}$ $m = Qd$

This shows that the potential is directly proportional to the dipole moment and inversely proportional to the square of the distance.

DIPOLE MOMENT

Dipole may be described by its dipole moment m. If Q is the charge and r is the vector (distance) from the negative to the positive charge, the dipole moment is given by

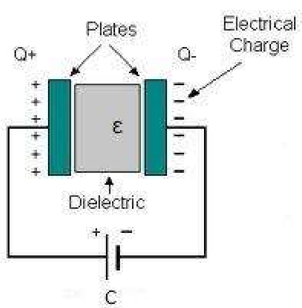
$$m = Qr$$

IF there are n dipole per unit volume ΔV, then there are $n\Delta V$ dipole and the total diploe moment is given by

$$m_{total} = \sum_{i=1}^{n\Delta V} m_i$$

CAPACITOR

- A capacitor is an electric device which consists of two conductor separated by a dielectric medium.



- Consider a capacitor composed of two conductor plates of area 'A' separated by a dielectric medium whose permittivity is ε.
- The space separation between the plates is d. If potential 'V' is applied across the plates, the positive charge Q is deposited on one plate and the negative charge –Q is deposited on other plate as shown in fig. The net charge is Zero.
- The capacitance of two conducting planes is defined as the ratio of magnitude of charge on either of the conductors. It is given by

$$C = \frac{Q}{V}$$

- The unit of capacitance is coulombs/volt or Farad.
- Assume that there is a uniform charge density D over the plates and dielectric medium

$$D = \frac{Q}{A} \text{ C/m}^2$$

- It is also written as interms of electric field E

$$D = \varepsilon E$$

$$\frac{Q}{A} = \varepsilon E$$

$$Q = A\varepsilon E$$

But electric field is given by

$$E = \frac{V}{d} \quad V/m$$

Sub. the value of E in above equation

$$Q = A\varepsilon \frac{V}{d}$$

$$\frac{Q}{V} = \frac{A\varepsilon}{d}$$

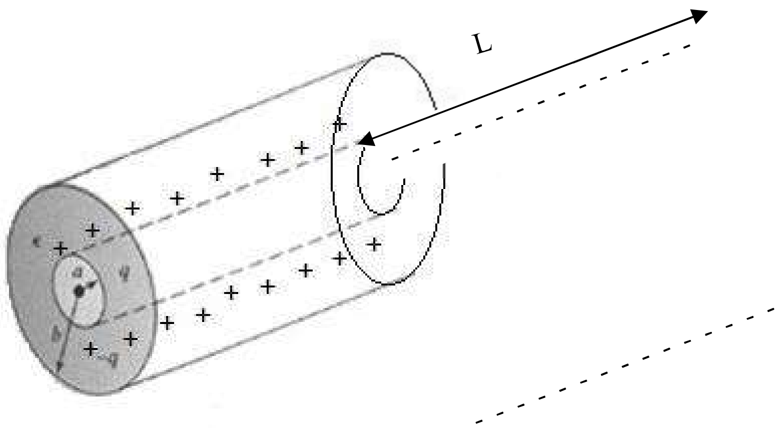
Capacitance is given by

$$C = \frac{Q}{V}$$

$$C = \frac{A\varepsilon}{d} \qquad \varepsilon = \varepsilon_0 \varepsilon_r$$

$$C = \frac{A\varepsilon_0\varepsilon_r}{d} \text{ Farad}$$

CAPACITANCE OF COAXIAL CABLE



$+\rho_L$

- Consider a co-axial cable or co-axial capacitor, let inner radius is ‘a’ and outer radius is ‘b’.
- Two concentric conductors are separated by dielectric of permittivity of ε_0
- The length of cable l, the iner conductor carries charge density ($+\rho_l$). Then the equal charge density ($-\rho_l$) exists on the outer conductor.

$$\rho_l = \frac{Q}{L}$$
$$Q = \rho_l L$$

- Assuming cylindrical co-ordinated system electric field intensity ($\vec{E}$) will be radial from inner o outer conductor and for infinite line of charge the electric intensity is given by

$$\vec{E} = \frac{\rho_l}{2\pi\varepsilon_0 r}\overrightarrow{a_r}$$

- $\vec{E}$is directed from inner conductor to outer conductor

$$C = \frac{Q}{V}$$

$$Q = \rho_l . L$$

$$V = -\int_b^a \vec{E}.dl$$

$$\vec{E} = \frac{\rho_l}{2\pi\varepsilon r}\overrightarrow{a_r}$$

$$V = -\int_b^a \frac{\rho_l}{2\pi\varepsilon r}\overrightarrow{a_r}.dl$$

$$=-\int_b^a \frac{\rho_l}{2\pi\varepsilon r}\overrightarrow{a_r}.dr\overrightarrow{a_r}$$

$$= -\frac{\rho_l}{2\pi\varepsilon}\int_b^a \frac{1}{r}\,dr$$

$$= -\frac{\rho_l}{2\pi\varepsilon}[\ln r]_b^a$$

$$= -\frac{\rho_l}{2\pi\varepsilon}[\ln(a) - \ln(b)]$$

$$= \frac{\rho_l}{2\pi\varepsilon}[\ln(b) - \ln(a)]$$

$$V = \frac{\rho_l}{2\pi\varepsilon}[\ln\frac{b}{a}]$$

$$C = \frac{\rho_l . L}{\frac{\rho_l}{2\pi\varepsilon}\ [\ln\frac{b}{a}]}$$

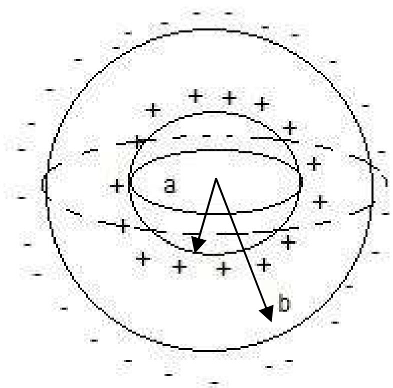
$$C = \frac{2\pi\varepsilon L}{\ln\frac{b}{a}} Farad$$

CAPACITANCE OF SPHERICAL SHEL (OR) SPHERICAL SHAPE TWO CONDUCTOR

- Consider a spherical capacitor formed by two concentric spherical conductor shells of radius ‘a’ and ‘b’.
- The capacitor radius of outer sphere is ‘b’ while that of inner sphere is ‘a’.
- The permitivity of dielectric between two conductor is ε_0 . The inner conductor carries positive charge(+Q) which constituting negative charge on outer conductor.
- Consider Gaussian surface as sphere of radius ‘r’ it can be obtained that electric field intensity ($\vec{E}$)is in radial direction.

$$C = \frac{Q}{V}$$

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\overrightarrow{a_r} \rightarrow \vec{E}\ due\ to\ point\ charge$$



$$V = -\int_b^a \vec{E}.dl$$

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\overrightarrow{a_r}$$

$$= -\int_b^a \frac{Q}{4\pi\varepsilon_0 r^2}\overrightarrow{a_r}.dr\overrightarrow{a_r}$$

$$= -\frac{Q}{4\pi\varepsilon_0}\int_b^a \frac{1}{r^2}\ dr$$

$$= -\frac{Q}{4\pi\varepsilon}\left[-\frac{1}{r}\right]_b^a$$

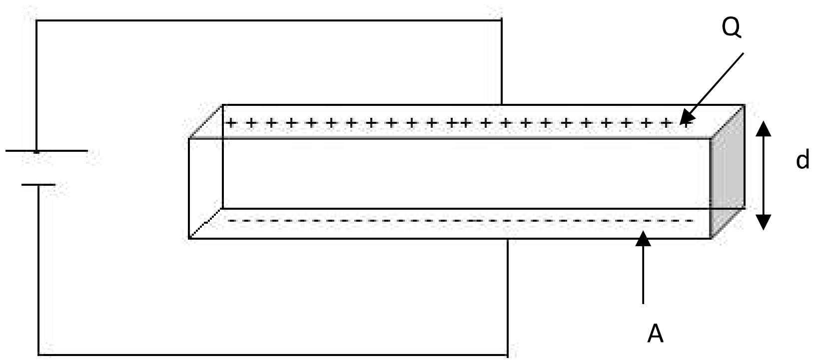
$$= -\frac{Q}{4\pi\varepsilon_0}[-\frac{1}{a}+\frac{1}{b}]$$

$$V = \frac{Q}{4\pi\varepsilon_0}[\frac{1}{a}-\frac{1}{b}]$$

$$C = \frac{Q}{\frac{Q}{4\pi\varepsilon_0}[\frac{1}{a}-\frac{1}{b}]]}$$

$$C = \frac{4\pi\varepsilon_0}{[\frac{1}{a}-\frac{1}{b}]}$$ Farad

ENERGY STORED IN A CAPACITOR: [Energy stored in a electrostatic field]



Energy stored in a capacitor is given by

$$W_E = \frac{1}{2}\int_V \overrightarrow{D}.\overrightarrow{E}.dv$$

$$\vec{E} = \frac{V}{d}, \quad \vec{D} = \varepsilon\vec{E}$$

$$W_E = \frac{1}{2}\int_V \varepsilon\overrightarrow{E}.\overrightarrow{E}.dv$$

$$= \frac{1}{2}\varepsilon\int_V \vec{E}^2 dv$$

$$= \frac{1}{2}\varepsilon\int_V \frac{V^2}{d^2}dv$$

$$= \frac{1}{2}\varepsilon\frac{V^2}{d^2}\int_V dv$$

$$W_E = \frac{1}{2}\varepsilon\frac{V^2}{d^2}(volume)$$

$$= \frac{1}{2}\varepsilon\frac{V^2}{d^2} \times Ad$$

$$= \frac{1}{2}\varepsilon V^2\frac{A}{d} \qquad C = \frac{\varepsilon A}{d}$$

$$W_E = \frac{1}{2}\,CV^2$$

If the dielectric is free space then there is increasing stored energy.

Problem:1

Determined the capacitance of a parallel plate capacitance with two metal plates of size 30cm×30cm separated by 5mm in air medium.

Given

A=0.3×0.3 = 0.09m^2

d=5$\times 10^{-3}$m

$$\varepsilon_0 = \frac{1}{36\pi \times 10^9}$$

$$C = \frac{\varepsilon_0 A}{d}$$

$$= \frac{0.09}{36\times 5\pi} \times 10^{-6}$$

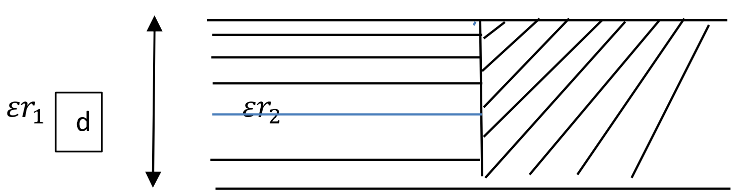
$$= 1.59\times 10^{-10}$$

= 15.9 nF

Problem:2

Find the capacitance of a parallel plate capacitor containing two dielectrics$\varepsilon_{r1} = 2$ and $\varepsilon_{r1} = 3$, each comprising one half the volume as shown in fig. Assume A= 2m^2 and d=1mm.

Sol.



The capacitance of a capacitor consisting of dielectric 1 ($\varepsilon_{r1} = 2$) alone is C_1

$$C_1 = \frac{\varepsilon_0 \varepsilon_{r1} {}^{A}\!/_{2}}{d}$$

$$= \frac{8.854\times 10^{-12}\times 2\times 1}{10^{-13}}$$

= 17.708 nF

The capacitance of a capacitor consisting of dielectric 2 ($\varepsilon_{r2} = 3$) alone is C_2

$$C_2 = \frac{\varepsilon_0 \varepsilon_{r2} {}^{A}\!/_{2}}{d}$$

$$= \frac{8.854\times10^{-12}\times3\times1}{10^{-13}}$$

= 26.562 nF

These two capacitances C_1 and C_2 are in parallel. The effective capacitance is

$C = C_1 + C_2$

= 17.708+26.562

= 44.270 nF

Problem:3

Determine the capacitance of a parallel plate capacitor composed of thin foil sheet, 20 cm square for plate separated through a glass dielectric 0.4cm thick with relative permittivity 6.

Given.

Area of plates, A= 20cm^2= 20×10^{-4} m^2

Distance of separation, d=0.4cm = $0.4\times 10^{-2} m$

Relative Permittivity, ε_r=6

$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$ Farad

$$= \frac{8.854\times10^{-12}\times6\times20\times10^{-4}}{0.4\times10^{-2}}$$

$$= \frac{1.06248\times10^{-13}}{0.4\times10^{-2}}$$

$C = 26.562\times 10^{-12}$Farads

Problem:4

Find the energy stored in a parallel plate capacitor of 0.5m by 1m has a separation of 2cm and a voltage difference of 10v

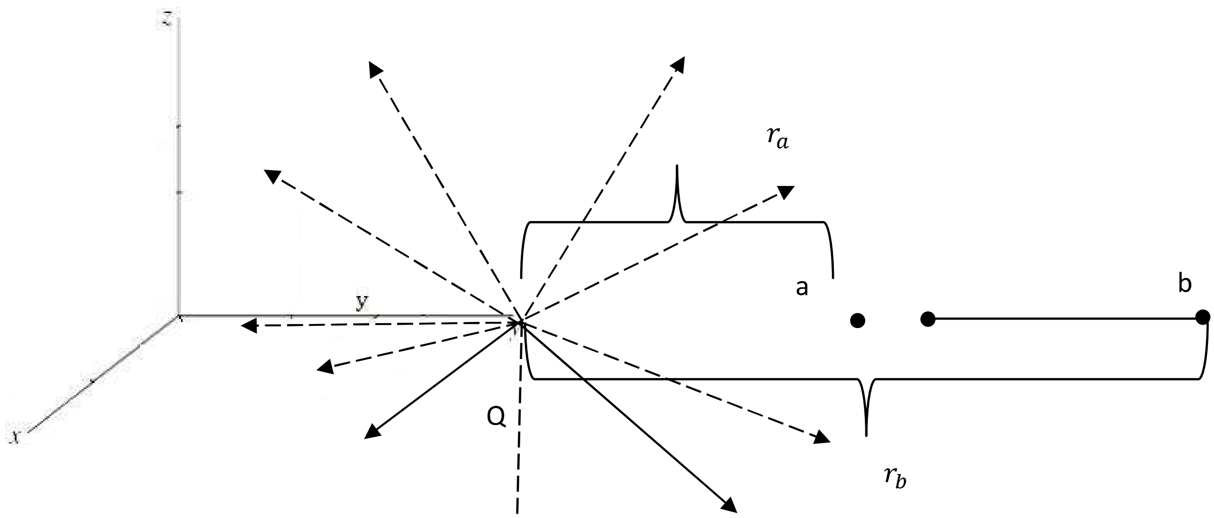
Sol

$$C = \frac{\varepsilon_0 A}{d} = \frac{8.854 \times 10^{-12} \times 0.5 \times 1}{02 \times 10^{-2}} = 2.2135 \times 10^{-10} F$$

Energy stored $E = \frac{1}{2}\, CV^2 = \frac{1}{2} \times 2.2135 \times 10^{-12} \times 10^2$

$=1.10675 \times 10^{-8}\, Joules$

POTENTIAL DUE TO POINT CHARGE



- Consider a point charge, located at the origin of a spherical co-ordinate system, producing $\vec{E}$ radially in all directions as shown in the fig.
- Assuming free space, the field $\overrightarrow{E}$ due to a point charge Q at a point having radial distance r from origin is given by

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\overrightarrow{a_r}$$

- Consider a unit charge which is placed at a point B which is at a radial distance of r_B, from the origin.
- It is moved against the direction of $\vec{E}$ from point B to point A. The point A is at a radial direction of r_A from the origin.
- The potential difference V_{AB} between point A and B is given by

$$V = -\int_a^b \vec{E}.dl \qquad dl = \overrightarrow{a_r}.dr$$

$$V = -\int_{r_a}^{r_b} \frac{Q}{4\pi\varepsilon_0 r^2}\overrightarrow{a_r}dr\overrightarrow{a_r}$$

$$= \frac{-Q}{4\pi\varepsilon_0}\int_{r_a}^{r_b} \frac{1}{r^2}\overrightarrow{a_r}dr\overrightarrow{a_r}$$

$$= \frac{-Q}{4\pi\varepsilon_0}\left[\frac{-1}{r}\right]_{r_a}^{r_b}$$

$$= \frac{-Q}{4\pi\varepsilon_0}[\frac{-1}{r_b} + \frac{-1}{r_a}]$$

$$V = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_b} - \frac{1}{r_a}\right]$$

$$r_b = \infty$$

$\rightarrow Absolute\ ptential\ at\ 'a'$

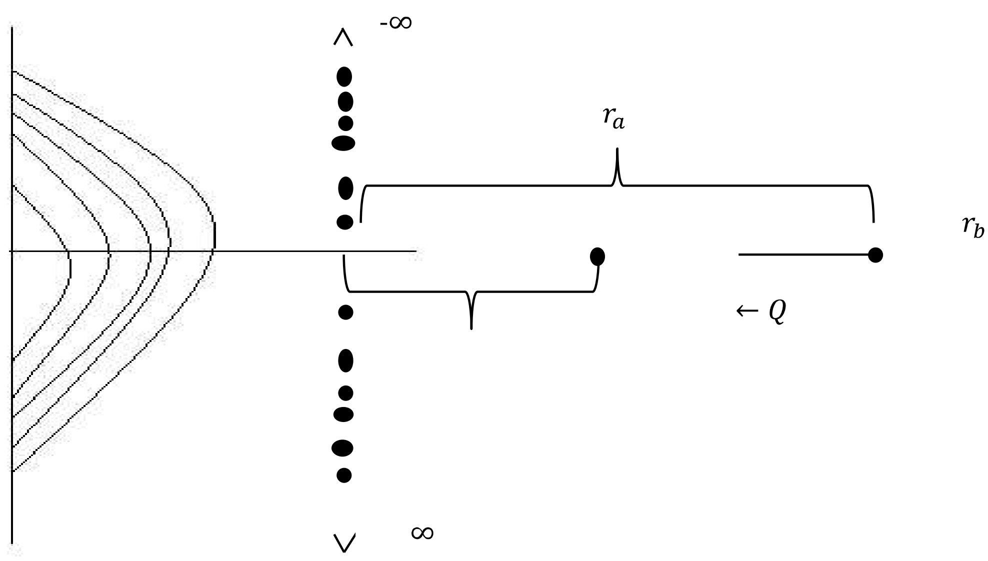
$$V_{AB} = \frac{Q}{4\pi\varepsilon_0}\left[-\frac{1}{r_a}\right]$$

- Similarly absolute potential at 'b'

$$r_a = \infty$$

$$V_{AB} = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{r_b}\right]$$

POTENTIAL DUE TO INFINITE LINE CHARGE



- The $\vec{E}$ due to infinite line charge along z-axis is

$$\vec{E} = \frac{\rho_L}{2\pi\varepsilon_0 r}\overrightarrow{a_r}$$

$$V = -\int_a^b \vec{E}.dl$$

$\vec{E}$ due to infinite line charge

$$V = -\int_{r_a}^{r_b} \frac{\rho_L}{2\pi\varepsilon_0 r}\overrightarrow{a_r}dr\overrightarrow{a_r}$$

$$= -\frac{\rho_L}{2\pi\varepsilon_0}\int_{r_a}^{r_b}\frac{1}{r}dr$$

$$= -\frac{\rho_L}{2\pi\varepsilon_0}[\ln r]_{r_a}^{r_b}$$

$$= -\frac{\rho_L}{2\pi\varepsilon_0}[\ln r_b - \ln r_b]$$

$$= -\frac{\rho_L}{2\pi\varepsilon_0}[\ln\left(\frac{r_b}{r_a}\right)]$$

Problem:1

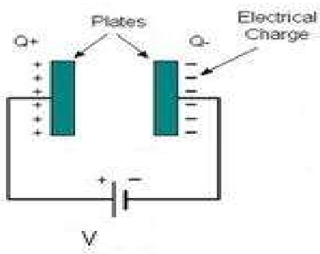
Find the field intensity and potential at a distance 100mm from a positive point charge of 10nc.

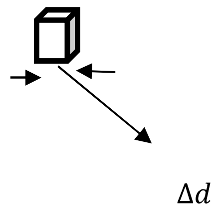
$$E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{10 \times 10^{-9}}{4\pi\frac{1}{36\pi \times 10^{-9}} \times (0.1)^2} = 9KV/m$$

$$E = \frac{Q}{4\pi\varepsilon_0 r} = \frac{10 \times 10^{-9}}{4\pi\frac{1}{36\pi \times 10^{-9}} \times (0.1)} = 900V$$

ENERGY DENSITY

- Consider a elementary cube of side Δd parallel to the plates of a capacitor as shown in fig.





The capacitance of elementary capacitor is

$$\Delta C = \frac{\varepsilon A}{\Delta d}$$

$$= \frac{\varepsilon (\Delta d)^2}{\Delta d}$$

$$\Delta C = \varepsilon \Delta d \qquad \rightarrow (1)$$

Energy stored in the element capacitor is

$$\Delta W = \frac{1}{2} \Delta C (\Delta V)^2 \rightarrow (2)$$

But potential difference across the elementary cube is

$$\Delta V = E.\Delta d \qquad \rightarrow (3)$$

Where E → electric field exist in the cube.

Sub (1) &(3) in (2)

$$\Delta W = \frac{1}{2} \varepsilon \Delta d (E.\Delta d)^2$$

$$= \frac{1}{2}\ \varepsilon E^2 (\Delta d)^3$$

$$= \frac{1}{2}\ \varepsilon E^2 \Delta V \Delta V = \Delta d^3\ \ \textit{is elementary volume}$$

The energy density is given by

$$\frac{\Delta W}{\Delta V} = \frac{1}{2} \varepsilon E^2$$

$\frac{\Delta W}{\Delta V} = \frac{1}{2}\, DE\ Joules/m^3$

Two marks

1. **What is scalar quantity?**

   A quantity which has magnitude only is called scalr quantity. It is represented by length.

   Ex. Temperature, Mass, Volume and energy.

2. **What is vector quantity?**

   A quantity which has both magnitude and also direction is called vector quantity. It is graphically represented by a line with an arrow to show magnitude and direction.

   Ex. Force, velocity and acceleration.

3. **Define unit vector?**

   Unit vector has a function to indicate the direction. Its magnitude is always unity (one).

   Magnitude $|\vec{R}| = 1$

4. **Define cross product or vector product.**

   $$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| sin\theta \widehat{a_n}$$

   $\vec{A}$&$\vec{B}$ are perpendicular to each other and $\widehat{a_n}$ perpendicular to both $\vec{A}$&$\vec{B}$.

   $$\vec{A} \times \vec{B} = \overrightarrow{-B} \times \vec{A}$$

5. **Define gradient.**

   When Del is operated with a scalar (ϕ). This is called gradient of scalar.

   Gradient of ϕ $=> \nabla\phi = \left(\frac{\partial}{\partial x}\vec{\imath} + \frac{\partial}{\partial y}\vec{\jmath} + \frac{\partial}{\partial z}\vec{k}\right). Xyz$

   $$\nabla.\phi = \frac{\partial}{\partial x}(xyz)\vec{\imath} + \frac{\partial}{\partial y}(xyz)\vec{\jmath} + \frac{\partial}{\partial z}(xyz)\vec{k}$$

   $$\nabla.\phi = \mathrm{yz}\vec{\imath} + xz\vec{\jmath} + xy\vec{k}$$

6. **Define divergence.**
   - Divergence gives net outflow or inflow per unit volume.
   - When ∇ is operated with a vector by dot product it gives divergence of vector.

   Types of divergence

   1. Positive divergence

      Net outflow per unit volume is called positive divergence.

   2. Negative divergence

      Net inflow per unit volume is called negative divergence.

7. **Define divergence of vector (or) solenoidal field?**

   Divergence of vector field is zero such a field is called solenoidal field.

   $$\text{Div.}\,\vec{A} = \mathrm{Lt}_{\Delta V=0} \int_S \frac{\vec{A}.ds}{\Delta V} \qquad \Delta V = small\ volume$$

8. **Define curl.**

When the del operator cross product with a vector (or) field is called curl of the vector.

$$Curl = \frac{Circulation\ or\ rotation}{Area}$$

9. **Define coordinate system and gives its type.**

Co-ordinate system is used to describe a vector accurately in some specific direction, length, angles and projection.

Types of co-ordinate system

1. Cartesian (rectangular) co-ordinate system.
2. Cylindrical co-ordinate system.
3. Spherical co-ordinate system.

10. **Give the conversion of Cartesian to cylindrical.**

Given	Transform
X	$\rho=\sqrt{x^2+y^2}$
Y	$\Phi=\tan^{-1}(y/x)$
Z	Z=z

11. **Give the conversion of cylindrical to Cartesian.**

Given	Transform
P	$x=\rho\cos\phi$
Φ	$Y=\rho\sin\phi$
Z	Z=z

12. **Give the conversion of cylindrical to spherical.**

Given	Transform
X	$r=\sqrt{x^2+y^2+z^2}$
Y	$\theta=\cos^{-1}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)=\cos^{-1}\left(\frac{z}{r}\right)$
Z	$\phi=\tan^{-1}(y/x)$

13. Give the conversion of spherical to Cartesian system.

Given	Transform
R	x= r $sin\theta cos\phi$
Θ	y=r $sin\theta sin\phi$
Φ	z= r $cos\theta$

14. **Define divergence theorem**.

The surface integral of a normal component of any vector field is equal to volume integral of divergence of that the particular vector field.

$$\oiint_S \vec{F}\ \mathrm{ds} = \iiint_V \nabla.\vec{F}\mathrm{dv}$$

15. Define stoke's thorem.

The line integral of any vector field is equal to the surface integral of curl of the vector field.

$$\oint_l \vec{F}dl = \iint_S \nabla \cdot \vec{F}dv$$

16. State coulomb's law.

According to coulomb's law, force between any two point charge is

1. Directly proportional to the product of magnitude of the two charges
2. Inversely proportional to the square of the distance between them.

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

17. Define electric field intensity.

It is define as the force exerted per unit charge.

$$\vec{E} = \frac{\vec{F}}{Q}$$

18. Define the formula for electric field intensity at a point due to 'n' no. of point charge.

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\overrightarrow{a_r}$$

19. Write the formula for electric field intensity due to line charge

$$\vec{E} = \int_L \frac{\rho_L dl}{4\pi\varepsilon_0 R^2} \overrightarrow{a_r}$$

20. Give the electric field intensity due to surface charge.

$$\vec{E} = \int_L \frac{\rho_s ds}{4\pi\varepsilon_0 R^2} \overrightarrow{a_r}$$

21. Write the electric field intensity due to volume charge.

$$\vec{E} = \int_L \frac{\rho_v dv}{4\pi\varepsilon_0 R^2} \overrightarrow{a_r}$$

22. What are the types of charge distributers?

There are four types of charge distribution, namely

a) Line charge
b) Point charge
c) Surface charge
d) Volume Charge

23. Define point charge.

Point charge is one whose maximum dimension is very small in comparison with any other length

24. Define line charge.

If the charge is uniformly distributed along a line, it is called line charge. The line may be finite or infinite.

25. Define volume charge.

If the charge is distributed uniformly over a volume then it is called as a volume charge

26. Define line charge density.

It is the ratio of total charge in coulomb to total length in meters and denoted as L.

$$\rho_L = {Total\ charge}/{Toatal\ length\ in\ c/m}$$

27. Define surface charge density.

It is the ratio of total charge per unit area in meters and denoted as S.

$$\rho_S = {Total\ charge}/{Toatal\ area\ in\ c/m^3}$$

28. Define volume charge density.

It is the ratio of total charge per unit volume in meters and denoted as V.

$$\rho_V = {}^{Total\ charge}\!\big/_{Toatal\ volume\ in\ c/m^3}$$

29. Give the electric field intensity due to infinite line charge

$$\vec{E} = \frac{\rho_L}{2\pi\varepsilon_0 r}\overrightarrow{a_y}$$

30. Give the electric field intensity due to infinite disc (or) sheet charge.

$$\vec{E} = \frac{\rho_S\overrightarrow{a_z}}{\varepsilon_0}$$

31. **Give the principle of superposition**.

If a system consists of n point charge namely $q_1, q_2, \ldots, q_n$ then the force on its charge is given by the vector sum of all individual forces given ny coulomb's law. This is linear superposition.

32. Define electric flux density (D)

Electric flux density or displacement density is defined as the electric flux per unit area.

$$D = \frac{Q}{A}\ coulomb/meter^2$$

33. State Gauss law.

The electric flux passing through any closed surface is equal to the total charge enclosed by the surface.

$\Psi = Q$

34. Give the application of gauss law.

To find the electric field intensity for symmetrical or uniform charge configuration.

To find the electric field intensity for unsymmetrical or non-uniform field.

35. Define potential or Potential difference.

Potential difference (v) is defined as the workdone in moving a unit positive charge from one point to another in electric field.

36. Define potential gradient.

The rate of change of potential with respect to the distance is called the potential gradient.

$$Grad\ V = \mathrm{Lt}_{\Delta l \to 0}\left(\frac{\Delta V}{\Delta l}\right) = \frac{dv}{dl}$$

37. Write the poisson's and laplace 's equation.

$$\nabla^2 V = \frac{\rho_v}{\varepsilon} \rightarrow poisson'sequation$$

$$\nabla^2 V = 0 \rightarrow LAplace'sequation$$

38. Define dipole (or) electric dipole (or) potential due to dipole.

An electric diploe or simply diploe is nothing but two equal opposite charges separated by a very small distance. The product of charge and spacing is called electric diploe moment.

39. Give the potential due to electric diploe.

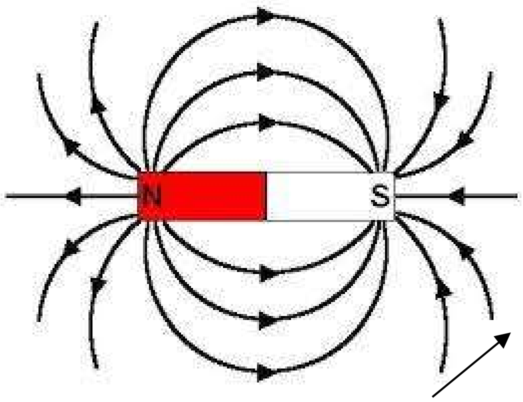
$$V = \frac{m cos\theta}{4\pi\varepsilon r^2} \qquad m = Qd$$

UNIT-III

UNIT-III	MAGNETOSTATICS

Lorentz Law of force, magnetic field intensity (H) - Biot-Savart's Law - Ampere's Law - Magnetic field intensity due to straight conductors, an infinite sheet of current, at the centre of the toroid, along the axis of the circular loop and solenoid – Magnetic Flux-Magnetic flux density (B) – Derivation of the steady magnetic field.

BASIC CONCEPTS



Closed path

Magnetic lines of force

The region around a permanent magnet or the space around a current carrying conductor which is occupied by magnetic lines of force is called magnetic field.

Magnetic Flux Lines And Their Properties

These are purely imaginary lines used to represent the magnetic field and to facilitate our understanding of magnetic field.

1. They are closed lines. They always start from north pole and end at south pole outside the magnet. Inside the magnet they start from south pole and end at north pole. The path followed by the magnetic flux line is called magnetic circuit.
2. They are non-intersecting lines.
3. The number of lines passing across unit cross sectional area is proportional to the magnitude of current in the conductor.

Scalar and vector magnetic Potential

The scalar and vector magnetic potential used to find the current density of a conductor or current density in a particular region.

Two rules of Emf

Gradients Div. Curl

Div. curl $\vec{F} = 0$

Curl .Gradient $\phi = 0$

$\vec{F}$=vector

ϕ=scalar

$$\vec{E} = -\nabla V$$

$$\vec{H} = -\nabla V_m$$

Scalar Magnetic Potential

Curl of gradient of a scalar is equal to zero.

$$V_m \to Scalar\ magnetic\ potential$$

According to the rule $\nabla \times \nabla(V_m) = 0 \quad \to \quad (1)$

$\nabla \times \vec{H} = \vec{J} \quad \to \quad (2)$

(1) → (2) $\quad \vec{H} = \nabla . V_m$

$\vec{J} = 0$

- A scalar magnetic potential can be defined for source free where current density is equal to zero.
- So scalar magnetic potential is applicable only in permanent magnet $\vec{J} = 0$ & $I = 0$

VECTOR MAGNETIC POTENTIAL

Divergence of curl of a vector =0

$$\nabla . (\nabla \times \vec{A}) = 0 \quad \to \quad (1)$$

$$\nabla . \vec{B} = 0 \quad \to \quad (2)$$

$$\vec{B} = \nabla \times \vec{A}$$

According to rule,

$$\nabla \times \vec{H} = \vec{J}$$

$$\vec{B} = \mu \vec{H}$$

$$\vec{H} = \frac{\vec{B}}{\mu}$$

$$\nabla \times \frac{\vec{B}}{\mu} = \vec{J}$$

$$\nabla \times \vec{B} = \mu \vec{J}$$

$$\nabla \times \left|\nabla \times \vec{A}\right| = \mu \vec{J}$$

$$\nabla \times \nabla \times \vec{A} = \mu \vec{J}$$

$$\nabla \times \nabla \times \vec{A} = \begin{vmatrix} \nabla & \nabla^2 \\ A & \nabla . A \end{vmatrix}$$

$$= \nabla(\nabla . \vec{A}) - \nabla^2 \vec{A}$$

$$\nabla(\nabla . \vec{A}) - \nabla^2 \vec{A} = \mu \vec{J}$$

$$\vec{J} = \frac{\nabla(\nabla . \vec{A}) - \nabla^2 \vec{A}}{\mu}$$

MAGNETIC FORCE ON A MOVING CHARGE[Force on a current element]

- A charged particle in motion in a magnetic field of flux density ‘B’ is experienced a force.
- The forced is proportional to the magnitude of the charge Q, its velocity V and flux density B and to the sine of the angle between V and B.
- The direction of the force is perpendicular to both V and B.

  F= Q $(\vec{V} \times \vec{B})$

  F=QVB sinθ
- The electric force on a charged particle in electric field of intensity E is

$$F = QE$$

- The force on a moving particle due to combined electric and magnetic field is obtained

$$F = \overset{Q_E}{\rightarrow} + \left(\overset{Q_V}{\rightarrow} \times \overset{B}{\rightarrow}\right)$$

$$F = Q[\vec{E} + (\vec{V} * \vec{B})]$$

This force is called <u>Lorentz force</u>

- The force on current element is

$$dF = dQ\ (V \times B)$$

$$= dQ\ (\frac{dl}{dt} \times B) \qquad [\therefore V = \frac{dl}{dt}\ velocity$$

$$= \frac{dQ}{dt}(dl \times B) \qquad [\therefore I = \frac{dQ}{dt}\ current]$$

$$dF = Idl \times B$$

$$F = (I \times B)l$$

This is also called Lorentz force.

- The magnitude of the force is

$$F = BIlsin\theta$$

BIOT-SAVART LAW

- Biot-Savart law states that the magnetic field intensity (H) produced at a point P due to differential current element (*Idl*) is

1. Proportional to the product of current and differential length

$$dH \propto Idl$$

2. It is proportional to angle between the element and the line joining the point P to the element.

$$dH \propto sin\theta$$

3. It is inversely proportional to square of distance between the point P and element

$$dH \propto \frac{1}{R^2}$$

$$dH\ \propto \frac{I\ dl\ sin\theta}{R^2}$$

$$dH = \frac{KI\ dl\ sin\theta}{R^2}$$

$$K = \frac{1}{4\pi}$$

$$\overrightarrow{dH} = \frac{I\ \overrightarrow{dl}\ sin\theta \overrightarrow{a\phi}}{4\pi R^2}$$

$$\overrightarrow{dH} = \frac{I\,\overrightarrow{dl}\,sin\theta\overrightarrow{a\phi}}{4\pi R^2}$$
$$\overrightarrow{dH} = \frac{I\,\overrightarrow{dl}\overrightarrow{a\phi}\,sin\theta}{4\pi R^2}$$
$$\overrightarrow{dH} = \frac{I\,\overrightarrow{dl}\overrightarrow{a\phi}}{4\pi R^2}$$
$$\vec{H} = \int \frac{I\,\overrightarrow{dl}\overrightarrow{a\phi}}{4\pi R^2}$$

AMPERE'S LAW

Ampere's law state that line integral of magnetic field intensity around a closed path is exactly equal to the current enclosed by the path.

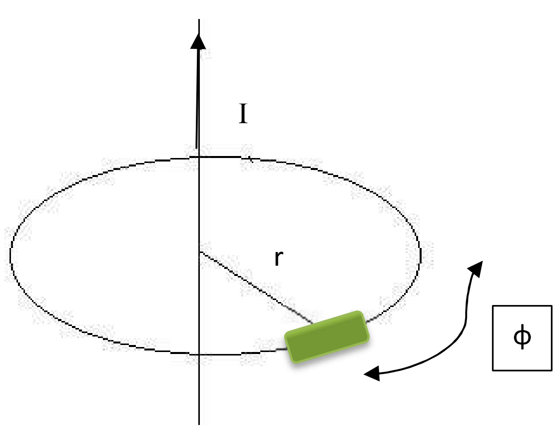
$$\oint \vec{H}dl = I$$

Another form of Ampere's law divided by Δs

$$\oint \frac{\vec{H}dl}{\Delta s} = \frac{I}{\Delta s}$$

<u>Application of Amperes law</u>

- Ampere's law used to finding the magnetic field intensity due to current carrying conductor



$$dl = rd\phi a\phi$$
$$\vec{H} = \vec{H}.a\phi$$

- Consider a direct current going through a conductor is I , which is inducing magnetic field intensity around the conductor.

$$\oint \vec{H}.dl = I$$

$$\int_0^{2\pi} \vec{H}.\overrightarrow{a\phi}.rd\phi\overrightarrow{a\phi} = I$$

$$\int_0^{2\pi} \vec{H}.r.d\phi = I$$

$$\vec{H}\int_0^{2\pi} d\phi = \frac{I}{r}$$

$$\vec{H} \times 2\pi = \frac{I}{r}$$

$$\vec{H} = \frac{\mathrm{I}}{2\pi r}$$

MODIFIED AMPERE'S LAW

- Ampere law deals with conductor
- The modified ampere law deals with both conductor and dielectric.

$\nabla \times \vec{H} = \vec{J}$

$\vec{J} = \vec{J}_{con} + \vec{J}_{dis}$

$\vec{J}_{con} = \sigma\vec{E}$, $\quad \vec{J}_{con} = \varepsilon\frac{d\vec{E}}{dt}$

$\vec{J} = \sigma\vec{E} + \varepsilon\frac{d\vec{E}}{dt}$

$\nabla \times \vec{H} = \sigma\vec{E} + \varepsilon\frac{d\vec{E}}{dt} \qquad \vec{E} = E_0e^{jwt}$

$\nabla \times \vec{H} = \sigma\vec{E} + \varepsilon\frac{dE_0e^{jwt}}{dt}$

Time varying $\vec{E}$

$\vec{E} = \overrightarrow{E_0}e^{jwt}$

$\nabla \times \vec{H} = \sigma\vec{E} + \varepsilon jwE_0e^{jwt}$

$\nabla \times \vec{H} = \sigma\vec{E} + jw\varepsilon\vec{E}$

$J_c = \sigma\vec{E}$

$J_d = w\varepsilon\vec{E}$

$\frac{J_c}{J_d} = \frac{\sigma\vec{E}}{w\varepsilon\vec{E}}$

$J_c \gg J_d \;\; => \frac{J_c}{J_d} \gg 1 \rightarrow conductor$

$$J_c \ll J_d \;\; => \frac{J_c}{J_d} \ll 1 \rightarrow dielectric$$

2.9 HELMHOLTZ'S THEOREM

A vector A is uniquely prescribed within a region by its divergence and its curl.

Let $\nabla . A = \rho_v$ --------------------(A)

$\nabla \times A = \rho_s$ ------------------(B)

Where,

$\rho_v => source\ density\ of\ A$

$\rho_s => circulation\ density\ of\ A$

Any vector A satisfying equation (A) and (B) with both ρ_v and ρ_s vanishing at infinity can be written as the sum of two vectors: one irrotational (zero curl), the other solenoid (zero divergence) . This is called Helmholtz's theorem.

$$A = -\nabla V + \nabla \times B$$

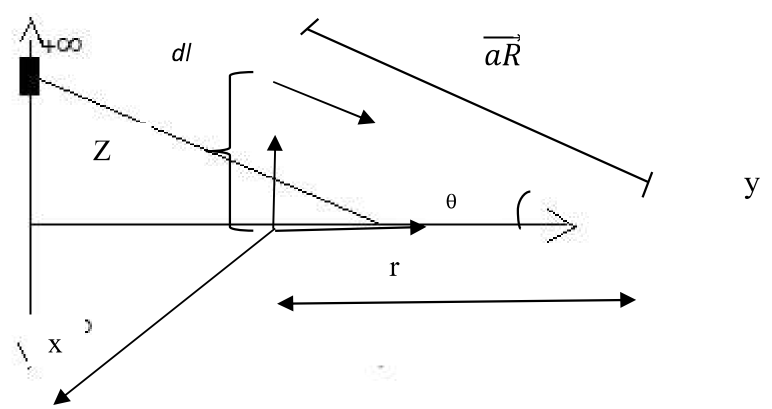
Let $A_i = -\nabla V$

$$A_s = \nabla \times B$$

From the result of $\nabla^2 A = \nabla(\nabla . A) - \nabla \times \nabla \times A$

$\nabla^2 A = \nabla \rho_v - \nabla \times \rho_s$ Gradient of divergence of A minus curl of curl A

<u>**$\vec{H}$ due to infinite long straight conductor**</u>



- Consider an infinite long straight conductor, along z-axis. The current passing through the conductor is 'I' ampere.
- The field intensity $\vec{H}$ at a point P is to be calculated, which is at a distance 'r' from the z-axis.

- ➢ Consider small differential element (dl), along the z-axis, at a distance Z from origin.

$$\vec{H} = \int \frac{I\,\vec{dl} \times \overrightarrow{a\phi}}{4\pi R^2}$$

$$= \int \frac{I\,\vec{dl} \times \overrightarrow{aR}}{4\pi R^2}$$

$$\vec{dl} = dz.\overrightarrow{az}$$

$$\overrightarrow{aR} = \frac{r\overrightarrow{ai} - z\overrightarrow{az}}{\sqrt{r^2 + z^2}}$$

Here taking circular co-ordinates

$$\vec{dl} \times \vec{ar} = \begin{vmatrix} \vec{ar} & \overrightarrow{a\phi} & \vec{az} \\ 0 & 0 & dz \\ \frac{r}{\sqrt{r^2+z^2}} & 0 & \frac{-z}{\sqrt{r^2+z^2}} \end{vmatrix}$$

$$= -\overrightarrow{a\phi}\ [0 - dz(\frac{r}{\sqrt{r^2+z^2}})]$$

$$= a\phi \left[dz(\frac{r}{\sqrt{r^2+z^2}})\right]$$

$$\vec{dl} \times \vec{ar} = \frac{rdz}{\sqrt{r^2+z^2}}\overrightarrow{a\phi}$$

$$R^2 = r^2 + z^2$$

$$Limit\ are - \infty\ to\ \infty$$

$$\vec{H} = \int_{-\infty}^{\infty} \frac{Irdz}{4\pi[r^2+z^2]\sqrt{r^2+z^2}}\overrightarrow{a\phi}$$

$$= \int_{-\infty}^{\infty} \frac{Irdz}{4\pi[r^2+z^2]^{3/2}}\overrightarrow{a\phi}$$

$$Z = rtan\theta$$

$$dz = rsec^2\theta d\theta$$

$$\theta = tan^{-1}\left(\frac{Z}{r}\right)$$

Limit $z = \infty$, $\theta = \frac{\pi}{2}$

$$z = -\infty \quad , \quad \theta = -\frac{\pi}{2}$$

$$\vec{H} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{Ir.rsec^2\theta d\theta}{4\pi[r^2 + r^2tan^2\theta]^{3/2}} \overrightarrow{a\phi}$$

$$\vec{H} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{Ir^2sec^2\theta d\theta}{4\pi r^3[1 + tan^2\theta]^{3/2}} \overrightarrow{a\phi}$$

$$\vec{H} = \frac{I}{4\pi r} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{sec^2\theta}{[sec^2\theta]^{3/2}} d\theta \overrightarrow{a\phi}$$

$$= \frac{I}{4\pi r} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{sec\theta} d\theta \overrightarrow{a\phi}$$

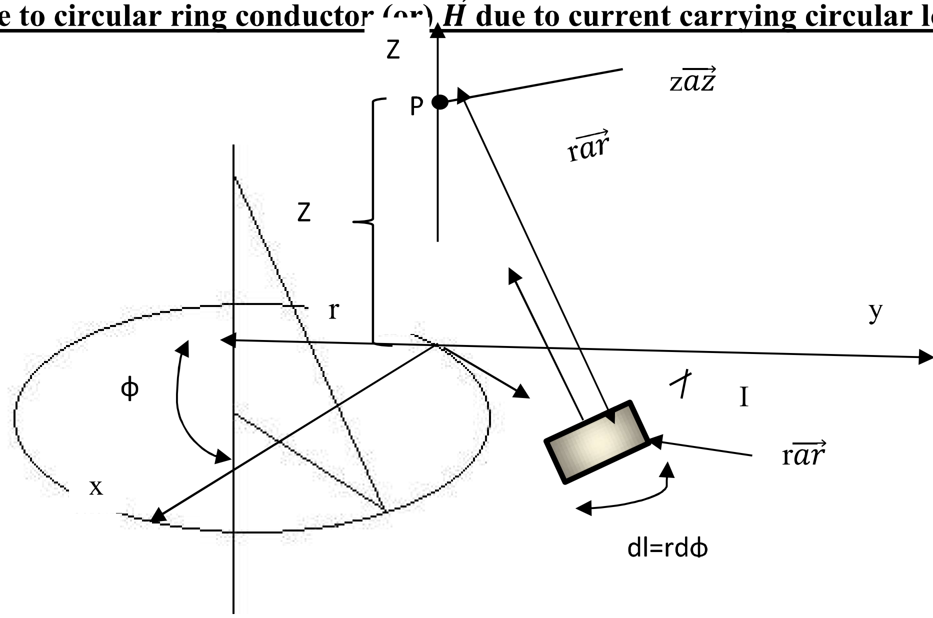
$$= \frac{I}{4\pi r} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} cos\theta \;\; d\theta \overrightarrow{a\phi}$$

$$= \frac{I}{4\pi r} [sin\theta]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \overrightarrow{a\phi}$$

$$= \frac{I}{4\pi r} [1 + 1] \overrightarrow{a\phi}$$

$$\vec{H} = \frac{2I}{4\pi r} \overrightarrow{a\phi}$$

$\vec{H}$ due to circular ring conductor (or) $\vec{H}$ due to current carrying circular loop



- Consider a circular loop carrying a current I amp, placed in xy plane, with z axis as its axis as shown in fig.
- The magnetic field intensity $\vec{H}$ at point is to obtained. The point P is at a distance Z from the plane of the circular loop, along its axis.
- The radius of the circular loop is r. Consider the differential length $\vec{dl}$ of the circular loop as shown in the fig.

$$\vec{H} = \int \frac{I\,\vec{dl}\ \times \vec{aR}}{4\pi R^2}$$

$$\vec{aR} = \frac{z\vec{az} - r\vec{ar}}{\sqrt{z^2 + r^2}}$$

$$\vec{dl} = rd\phi\vec{a\phi}$$

$$\vec{dl} \times \vec{aR} = \begin{bmatrix} \vec{ar} & \vec{a\phi} & \vec{az} \\ 0 & rd\phi & 0 \\ \frac{-r}{\sqrt{z^2 + r^2}} & 0 & \frac{z}{\sqrt{z^2 + r^2}} \end{bmatrix}$$

$$= \vec{ar}\left[(rd\phi(\frac{z}{\sqrt{z^2+r^2}}))\right] + \vec{az}\left[(\frac{r}{\sqrt{z^2+r^2}})(rd\phi)\right]$$

$$\vec{dl} \times \vec{aR} = \frac{rd\phi z.\vec{ar}}{\sqrt{z^2 + r^2}} + \frac{r^2 d\phi\vec{az}}{\sqrt{z^2 + r^2}}$$

Here radial component is zero due to magnetic field intensity in radial component is vanish.

$$\vec{dl} \times \vec{aR} = \frac{r^2 d\phi\vec{az}}{\sqrt{z^2 + r^2}}$$

Limit $are\ 0\ to\ 2\pi$

$$\vec{H} = \int_0^{2\pi} \frac{I\,r^2 d\phi\vec{az}}{4\pi\sqrt{z^2 + r^2}[z^2 + r^2]}$$

$$= \int_0^{2\pi} \frac{I\,r^2 d\phi\vec{az}}{4\pi[z^2 + r^2]^{3/2}}$$

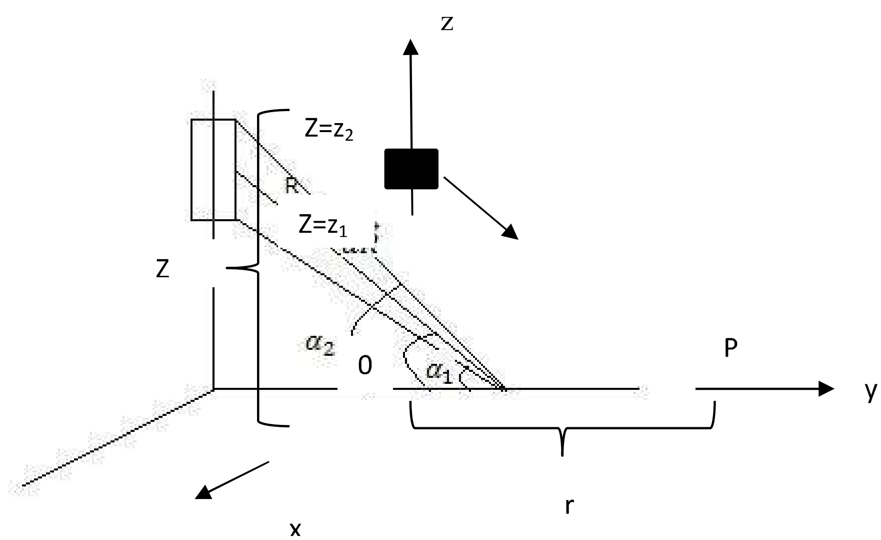
$$= \frac{I\,r^2\vec{az}}{4\pi[z^2 + r^2]^{3/2}} \int_0^{2\pi} d\phi$$

$$= \frac{I\,r^2\vec{az}}{4\pi[z^2 + r^2]^{\frac{3}{2}}}[2\pi]$$

$$\vec{H} = \frac{I\, r^2 \overrightarrow{az}}{2[z^2 + r^2]^{\frac{3}{2}}}$$

$\vec{H}$ DUE TO finite length of current carrying conductor

- Consider a conductor of finite length placed in z-axis. It carries a direct current I the perpendicular distance of point P from the z-axis is r.
- The conductor is placed such that one end is at z=z1 while the other end is z=z2.
- Consider a differential element dl along z-axis at distance z from origin.



$$\vec{H} = \int \frac{I\, \overrightarrow{dl} \times \overrightarrow{aR}}{4\pi R^2}$$

$$\overrightarrow{dl} = dz_a \vec{z}$$

$$\overrightarrow{aR} = \frac{r\overrightarrow{ar} - z\overrightarrow{az}}{\sqrt{r^2 + z^2}}$$

Component of z-axis is zero.

$$\overrightarrow{aR} = \frac{r\overrightarrow{ar}}{\sqrt{r^2 + z^2}}$$

$$\overrightarrow{dl} \times \overrightarrow{aR} = \begin{bmatrix} \overrightarrow{ar} & \overrightarrow{a\phi} & \overrightarrow{az} \\ 0 & 0 & dz \\ \frac{r}{\sqrt{z^2+r^2}} & 0 & \frac{-z}{\sqrt{z^2+r^2}} \end{bmatrix}$$

$$= \frac{rdz\overrightarrow{a\phi}}{\sqrt{r^2+z^2}}$$

Limit Z1 to Z2

$$\vec{H} = \int_{z1}^{z2} \frac{I\ rdz\overrightarrow{a\phi}}{4\pi R^2\sqrt{r^2+z^2[r^2+z^2]}}$$

$$= \frac{I}{4\pi}\int_{Z1}^{Z2} \frac{rdz\overrightarrow{a\phi}}{[r^2+z^2]^{3/2}}$$

Z=r tanθ $\qquad \theta = tan^{-1}\left(\frac{z}{r}\right)$

$$dz = r\ sec^2\theta d\theta$$

Z=Z1, $\theta = \alpha_1$

Z=Z2, $\theta = \alpha_2$

$$\vec{H} = \frac{I\overrightarrow{a\phi}}{4\pi}\int_{\alpha_1}^{\alpha_2} \frac{r^2 sec^2\theta d\theta}{r^3(sec^3\theta)}$$

$$= \frac{I\overrightarrow{a\phi}}{4\pi r}\int_{\alpha_1}^{\alpha_2} \frac{sec^2\theta d\theta}{(sec^3\theta)}$$

$$\vec{H} = \frac{I\overrightarrow{a\phi}}{4\pi r}\int_{\alpha_1}^{\alpha_2} cos\theta d\theta$$

$$= \frac{I\overrightarrow{a\phi}}{4\pi r}[sin\theta]_{\alpha_1}^{\alpha_2}$$

$$\vec{H} = \frac{I\overrightarrow{a\phi}}{4\pi r}[sin\alpha_1 - sin\alpha_2]$$

- Magnetic flux density due to circular ring(or) circular loop current carrying conductor is

$$\vec{B} = \mu\vec{H}$$

$$\vec{B} = \frac{\mu I r^2 \overrightarrow{az}}{2[r^2 + z^2]^{3/2}}$$

- Magnetic flux density due to infinity straight conductor is

$$B = \frac{\mu I a\phi}{2\pi r}$$

Problem 1

A circular coil of radius am carries a current of 4A what is the value of magnetic field intensity at the centre?

Given

I=4A , a=2m

$$H = \frac{I}{2a}$$

$H = \frac{4}{2\times 2} = 1A/m$

2.5 MAGNETIC FLUX DENSITY (B)

- Magnetic flux density is defined as the magnetic flux passing per unit area. Its unit is weber/metre^{2}or Tesla.

$$B = \frac{\phi}{A} \text{weber/metre}^2$$

Also defined as

$$B = \mu H$$

μ→ $permeability\ of\ medium$

$$\mu = \mu_0 \mu_r (H/m)$$

$$\mu_0 = 4\pi * 10^{-7} (H/m) \quad (permeability\ of\ free\ space)$$

$$\mu_r = relative\ permeability$$

$$H \rightarrow MAgnetic\ field\ intensity\ (A/m)$$

MAGNETIC FLUX(ϕ)

Magnetic flux is defined as the flux ϕ passing through any area.

$$\phi = \int_S \vec{B} ds$$

In a particular surface the incoming flux is equal to the outgoing magnetic flux

$$\phi = 0$$

$$\int_S \vec{B} ds = 0 \rightarrow integral\ form\ of\ gauss\ law$$

By applying divergence theorem

$$\iint_S B.ds = \nabla.B$$

Then, $\nabla.B = 0$

The total magnetic flux passing through any closed surface is equal to zero.

MAGNETIC FIELD INTENSITY($\vec{H}$)

- The quantitative measure of strongest or weakness of the magnetic field is given by magnetic field intensity or magnetic field strength.
- Magnetic field intensity is defined as force experienced by unit north pole.
- The magnetic field intensity H can be made independent of the medium and is defined by equation

$$H = \frac{B}{\mu}$$

$$B = \mu H$$

Problem 1

A round copper conductor is carrying a current of 250A. Determine the magnetising force flux density at a distance of 10cm from the conductor.

Sol.

*I=250A, r=10cm=10*10^{-2}=0.1m*

$$B = \frac{\mu_0 NI}{2\pi r} \qquad \text{(N=1)}$$

$$B = \frac{4\pi \times 10^{-7} \times 1 \times 250}{2 \times \pi \times 0.1}$$

$$B = 0.5 \times 10^{-3} Wb/m^2$$

$$B = 0.5m\, Wb/m^2$$

Magnetic force $H = \frac{B}{\mu_0}$

$$H = \frac{0.5 \times 10^{-3}}{4 \times \pi \times 10^{-7}}$$

$$= 397.88\, AT/m$$

Two marks

1. **Define magnetic vector potential.**

   It is defined as that quantity whose curl gives the magnetic flux density.

   $B = \nabla \times A$

   Where A is the magnetic vector potential

   $A = \frac{\mu}{4\pi} \iiint_V \frac{J}{r} dr$ web/m

2. **Define Biot-Savart law.**

   The magnetic field intensity (*H*) produce at a point P due to differential current element (*Idl*) is

   a) Proportional to the product of current and differential length.

   $$dH \propto Idl$$

   b) It is proportional to angle between the element and the line joining the point P to the element.

   $$dH \propto sin\theta$$

   c) It is inversely proportional to square of the distance between the point P and element

   $$dH \propto \frac{1}{R^2}$$

3. **Give the magnetic field intensity due to a finite and infinite wire carrying a current.**

   For infinite wire

   $$\vec{H} = \frac{I\vec{a\phi}}{2\pi r}$$

   For finite wire

   $$\vec{H} = \frac{I\vec{a\phi}}{4\pi r}[sin\alpha_2 - sin\alpha_1]$$

4. **Define magnetic flux density.**

   Magnetic flux density is defined as the magnetic flux passing per unit area. Its unit is weber/meter^2

$$B = \frac{\phi}{A} weber/meter^2$$

5. What is the difference between scalar and vector magnetic potential.

Magnetic scalar potential is a quantity whose negative gives the magnetic intensity.Magnetic vector potential is a quantity whose curl gives the magnetic flux density.

6. Give Lorentz force equation

$$F = Q[\vec{E} + \vec{V} \times \vec{B}]$$
$$where$$

F= Lorentz force
Q= Charge of the moving charge
V= Velocity of charge
B= Magnetic field
E= Electric field intensity

7. State gauss law for magnetic field.

The total magnetic flux passing through any closed surface is equal to zero

8. State Ampere's law

Ampere's law states that line integral of magnetic field intensity around a closed path is exactly equal to the current enclosed by the path.

$$\oint \vec{H} dl = I$$

9. Find the maximum torque on a 100 turn rectangular coil 0.2m by 0.3m, carrying a current of 2A in the field of flux density 5 wb/m^2?

<u>Given</u>

N= 100
A=0.2*0.3=0.06m^2
B=5 wb/m^2
T_{max}=NIAB
= 100*2*0.06*5
T_{max}=60 N-m

10. Give the force on a current element

The force on a current element Idl is given by $dF = I \times Bdl$
= $BIdlsin\theta\ Netwon$

11. Define Magnetic dipole.

A small bar magnetic with pole strength Qm and length l may be treated as magnetic dipole whose magnetic moment is $Q_m l$

12. State ampere's circuit law.

The ampere's circuited law states that the line integral of magnetic filed intensity $\vec{H}$ around a closed path is exactly equal to the direct current enclosed by tht path.

The numerical representation of ampere's circuital law is,

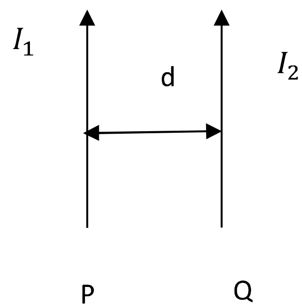
$$\oint \vec{H} dl = I$$

UNIT-IV MAGNETIC FORCES AND MATERIALS

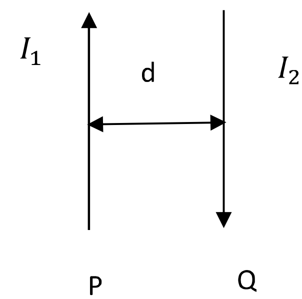
Magnetic materials - Magnetization - Boundary conditions – Magnetic Scalar and vector potential - Magnetic force – Force on a moving charge, differential current element- Force between differential current element- Torque on a closed circuit - Inductance and mutual inductance.

FORCE BETWEEN CURRENT CARRYING CONDUCTOR

- Consider two straight long parallel conductor P and Q separated by a distance ‘d’.
- Let I_1 and I_2 be the currents flowing in conductor P and Q respectively.
- Consider a conductor P produced a magnetic field whose flux density is B at conductor Q



a.Force of attraction



b.Force of repulsion

$$B = \frac{\mu_0 I_1}{2\pi d}$$

The force on conductor Q due to P

$$F = BI_2 l$$

Where l is the length of the conductor

$$F = \frac{\mu_0 I_1}{2\pi d} I_2 l$$

- If the current of conductor P and Q are flowing in same direction, there is a force of attraction
- If current are flowing in opposite direction, there is a force of repulsion. However the value of force is same

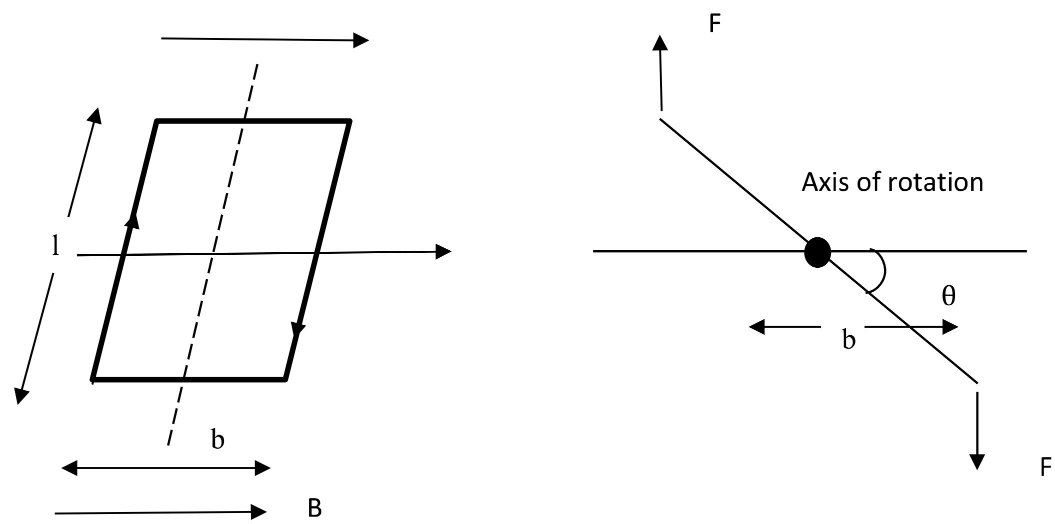
$$F = \frac{\mu_0 I_1 I_2 l}{2\pi d}\ N$$

- If the conductor are infinitely long the force per unit length is

$$F = \frac{\mu_0 I_1 I_2}{2\pi d}\ N/m$$

TORQUE ON CLOSED CONDUCTORS: (MAGNETIC TORQUE)

- When a current loop is placed parallel to a magnetic field, forces act on the loop that tend to rotate it. The tangential force multiplied by the radial distance at which it acts is called Torque or mechanical moment on the loop.



a) Axis of rotation b) Current loop

- Consider the rectangular loop of length 'l' and breadth 'b' carrying a current I in a uniform magnetic field of flux density B as shown in fig.
- The loop makes an angle θ with respect to magnetic flux density B.

The force acting on the loop

$$F = BIl sin\theta$$

- If the loop plane is parallel to the magnetic field , the total torque on the loop

$$T = 2\times torque\ on\ each\ side$$

$$T = 2 \times force \times distance$$

$$= 2\ BIl\ sin\theta\ \frac{b}{2}$$

$$= BIlb sin\theta$$

$$= BIA sin\theta \qquad (Area\ A=lb)$$

The magnetic moment of loop is IA

- The magnetic moment is a vector with the direction given by the unit normal $\hat{n}$ to the plane of the loop.

$$m = IA\hat{n}$$

$$T = mB sin\theta\hat{n}$$

- ❖ In vector form, torque can be represents as

$$\vec{T} = \vec{m} \times \vec{B}$$
$$m = \frac{T}{B}$$

- ❖ The magnetic moment is defined as the maximum torque on loop per unit magnetic induction.

Problem 1:

Find the maximum torque on an 100 turn rectangular coil, 0.2 by 0.3m, carrying a current of 2A in the field of flux density 5 web/m^3

Given

N=100

A=0.2*0.3=0.06m^2

I=2A

B=5 web/m^2

$T_{max}=NIAB$

= 100*2*0.06*5

= 60 Newton-meter

Problem 2:

Two wires carrying current in the same direction of 3A and 6A are placed with their axes 5 cm apart, free space permeability =$4\pi\times 10^{-7} H/m$. Calculate the force between them in kg/m length.

Sol:

Force between two parallel conductors is given by

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi d}$$

d = distance of separation = 5cm

= 5*10^{-2} m

$I_1=3A$

$I_2 = 6A$

l = *length of conductor*

Hence force per unit meter length is given by ,

$$\frac{F}{l} = \frac{\mu I_1 I_2 l}{2\pi d} = \frac{\mu_0 \mu_r I_1 I_2}{2\pi d}$$

$$\frac{F}{l} = \frac{4 \times \pi \times 10^{-7} \times 3 \times 6}{2\pi \times 5 \times 10^{-2}}$$

$$= 7.2 \times 10^{-5} N/m$$

$$\frac{F}{l} = 72 \mu N/m$$

The force expressed in kg/m is given by

$$\frac{F}{l} = \frac{7.2 \times 10^{-5} N/m}{9.8}$$

$$= 7.3469 \times 10^{-6}\ kg/m$$

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

The total emf induced in a circuit is equal to the time rate of decrease of the total magnetic flux linking the circuit

- Faraday's law: an electromotive force (emf) is induced in a loop if the magnetic flux through the loop changes in time
  - this emf is an induced voltage
    - recall we get a voltage by doing a path integral of an electric field
  - magnitude of induced voltage is proportional to how fast the flux through the loop changes in time
  - the sign of the induced voltage is such that it would induce a current in the loop that would produce a flux that opposes the change in the flux through the loop (Lenz's law)

Faraday's law can be stated as,

$$\boxed{e = -N\frac{d\phi}{dt} \text{ volts.}}$$

where N = Number of turns in the circuit

e = Induced e.m.f.

t- time in seconds

assume single turn circuit i.e. $N = 1$, then Faraday's law can be stated as,

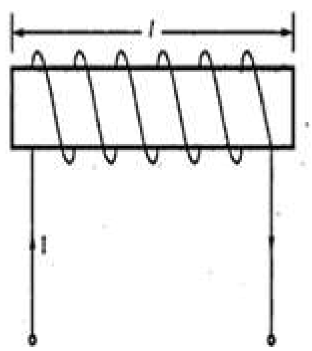
$$e = -\frac{d\phi}{dt} \text{ volts}$$

the induced e.m.f. is given by,

$$\boxed{e = \oint \bar{E} \cdot d\bar{L}}$$

INDUCTANCE OF SOLENOIDS

Consider a solenoid of N turns as shown in the fig



solenoid with N turns

Let the current flowing through the solenoid be I ampere, let the length of the solenoid be l and the cross sectional area be A

The field intensity inside the solenoid is given by

$$H = \frac{NI}{l} \text{ A/m}$$

$$\text{Total flux linkage} = N\phi = N(B)(A)$$

$$= N(\mu H)(A)$$

$$\text{Total flux likage} = \mu NHA$$

$$= \mu N\left[\frac{NI}{l}\right] A$$

$$= \frac{\mu N^2 IA}{l}$$

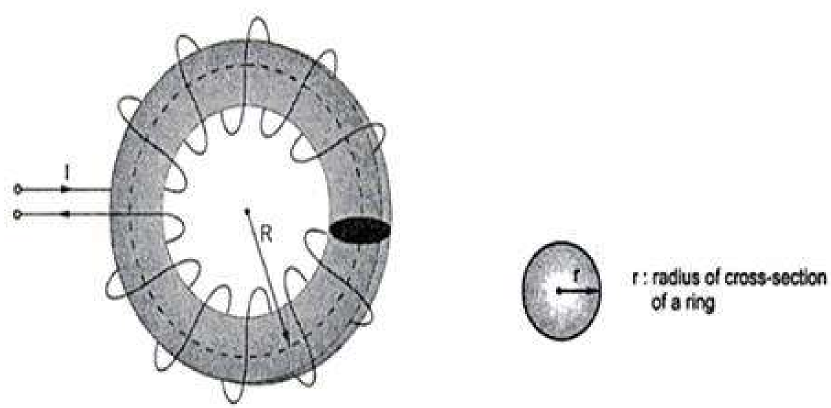
the inductance of a solenoid is given by

$$L = \frac{\text{Total flux linkage}}{\text{Total current}}$$

$$L = \frac{\mu N^2 IA}{l(I)}$$

$$\boxed{L = \frac{\mu N^2 A}{l} \quad H}$$

INDUCTANCE OF TOROID



(a) Toroidal ring

(b) Cross-sectional view of a toroidal ring

The magnetic flux density inside a toroid ring is given by

$$B = \frac{\mu NI}{2\pi R}$$

The total flux linkage of a toroidal ring having N turns is given by

Total flux linkage = N ϕ

But Φ = (B) (A) where A = Area of cross-section of a toroidal ring

∴ Total flux linkage = N (B) (A)

$$= N\left(\frac{\mu NI}{2\pi R}\right)(A)$$

$$= \frac{\mu N^2 IA}{(2\pi R)}$$

$$L = \frac{\text{Total flux linkage}}{\text{Total current}}$$

$$L = \frac{\mu N^2 IA}{(2\pi R)(I)}$$

$$\boxed{L = \frac{\mu N^2 A}{(2\pi R)} \ H}$$

For a toroid with N number of turns h as a height of toroid with r1 as inner radius and r2 as outer radius, the inductance is given by

$$L = \frac{\mu N^2 h}{2\pi} \ln\left(\frac{r_2}{r_1}\right) \ H$$

INDUCTANCE OF TRANSMISSION LINES

INDUCTANCE OF TWO TRANSMISSION LINES

Consider two conductors (wires) A and B of radius a and b respectively are separated by the distance ‘d’

The conductor A carries a current of +I and conductor B carries a current of –I

The internal flux linkage with conductor A is given by $\Phi 1 = \mu_0 \mu r I/8\pi$

The external flux linkage with conductor A is given by $\Phi 2 = \mu_0 I/2\pi \ln (d/a)$

The total flux linkage of A

$\Phi = \Phi 1 + \Phi 2$

$= \mu_0 \mu r I / 8\pi + \mu_0 I / 2\pi \ln (d/a)$

Inductance of transmission lines

$L = \Phi 1 / I$ = Total flux linkage/ total current

Similarly for conductor B

The Total flux linkage

$\Phi = = \mu_0 \mu r I / 8\pi + \mu_0 I / 2\pi \ln (d/b)$

The inductance of conductor A

$LA = \Phi / I = \mu_0 / 4\pi [\mu r/2 + 2 \ln (d/a)]$ H/m

Similarly the inductance of conductor B

$LB = \Phi / I = \mu_0 / 4\pi [\mu r/2 + 2 \ln (d/b)]$ H/m

The loop (total) inductance

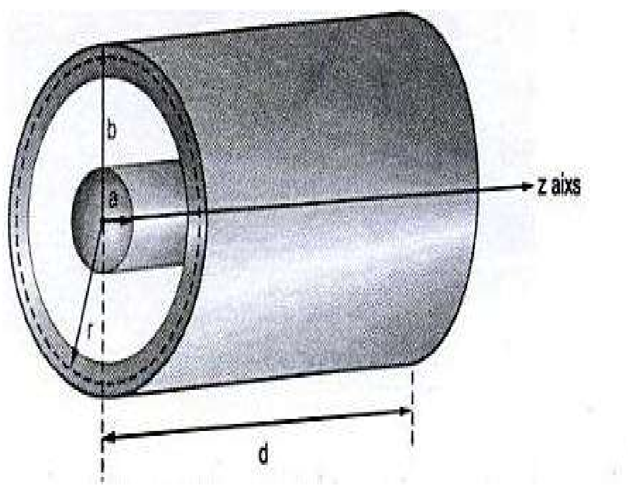
$L = LA + LB$

$= \mu_0 / 4\pi [\mu r + 2 \ln (d/a) + 2 \ln(d/b)]$

$L = \mu_0 / 4\pi [\mu r + 2 \ln(d^2/ab)]$

INDUCTANCE OF CO-AXIAL CABLE

Consider a co-axial cable with inner conductor radius a and outher conductor radius b as shown in the below fig



Let the current through the coaxial cable be I

For the co-axial cable the field intensity at any point between inner and outer conductors is given by

$$H = \frac{I}{2\pi r} \quad \text{where} \quad a < r < b$$

$$B = \mu H = \frac{\mu I}{2\pi r}$$

Now assume that the axis of the cable is along z-axis

The magnetic flux density will be in radial plane extending from r = a to r = b and z = 0 to z = d

$$\bar{B} = \frac{\mu I}{2\pi r} \bar{a}_\phi \ T$$

total magnetic flux is given by

$$\phi = \int_S \bar{B} \cdot d\bar{S}$$

$$d\bar{S} = dr\, dz\, \bar{a}_\phi$$

$$\phi = \int_{z=0}^{z=d} \int_{r=a}^{r=b} \left(\frac{\mu I}{2\pi r} \bar{a}_\phi \right) \cdot \left(dr\, dz \bar{a}_\phi \right)$$

$$\phi = \frac{\mu I}{2\pi} [z]_0^d \ [\ln r]_a^b$$

$$\phi = \frac{\mu I d}{2\pi} \ln\left(\frac{b}{a}\right)$$

The inductance of co-axial cable

L= total flux linkage / total current

L = $\frac{\mu I d}{2\pi} ln\left(\frac{b}{a}\right)$ henry

The inductance of a co-axial cable per unit length

L/d = μ / 2π ln (b / a) henry / m

INDUCTANCE

Any conductor carrying a current produces a field around it which in turn produces the magnetic flux

If the current in the coil is alternating with respect to time, flux linkage will also vary

The value of flux depends upon flux density which in turn depends upon the current in the coil at a particular instant.

The value of flux density at the centre of a solenoid increases with and is proportional to the current flowing in the coil.

If there are N turns in the coil each flux lines that passes through the whole length of solenoid link the current N times.

If all the flux lines link all the turns then the total line linkage is equal to

A = N Φ

The unit of inductance is henry

The inductance if a coil is said to be 1 henry, if the current in the coil changes at a rate of 1 amp/sec

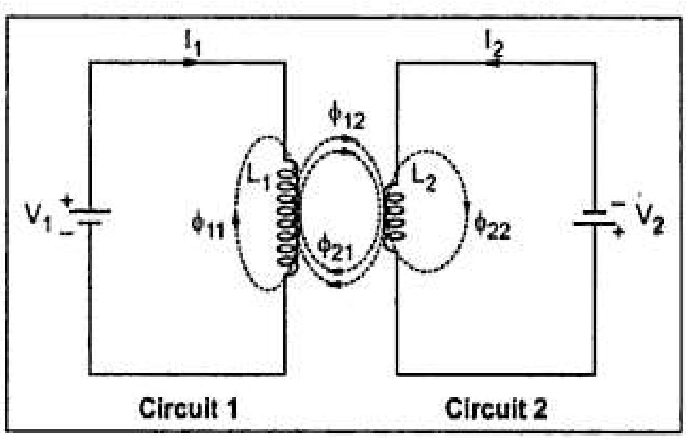
Inductance is classified as self inductance and mutual inductance

SELF INDUCTANCE

Inductance is also referred as self inductance as the flux produced by the current flowing through the coil links with the coil itself.

MUTUAL INDUCTANCE

Consider two magnetic circuits with two different coils carrying two different currents



Let coil 1 of N1 turns with inductance L1 carries current I1, then the magnetic flux produced by the current I1 flowing through coil 1 is represented by Φ 11.

Let coil 2 of N2 turns with inductance L2 carries current I2, then the magnetic flux produced by the current I2 flowing through coil 2 is represented by Φ 22.

The flux produced in coil1 also links with coil2 and produces another flux represented by Φ 12

Similarly the flux produced in coil2 also links with coil1 and produces another flux represented by Φ 21

The total flux linkage of the second coil due to the flux, produced by the current I1 in first coil is N2 Φ 12

Similarly the total flux linkage of the second coil due to the flux, produced by the current I2 in first coil is N1 Φ 21

With respect to the flux linkage the mutual inductance between the two coils can be defined as the ratio of flu linkage of one coil to the current in other coil

Thus the mutual inductance between circuit 1 and circuit 2 is given by

$M12 = N2\ \Phi\ 12 / I1$ henry

φ12 is the flux produced by I1 which links the coil 2

Similarly

$M21 = N1\ \Phi\ 21 / I2$ henry

If the medium surrounding the magnetic circuits is linear,

$\boxed{M12 = M21}$

Note: The mutual inductance depends on the magnetic interaction between the two currents.

From the above statements

$M12 \cdot M21 = L11 \cdot L22$

Let $L11=L1$, $L22 = L2$ & $M12 = M21 = M$

$M^2 = L1L2$

$M = \sqrt{L1L2}$

Suppose K1 times the flux produced by I1 links with secondary then

$M21/L11 = K1N2/N1$

Similarly

Suppose K2 times the flux produced by I2 links with primary then

$M12/L22 = K2N1/N2$

From the above statement

$M12M21/L11L22 = K1K2$

If $M12 = M21 = M$, $L11 = L22 = L$

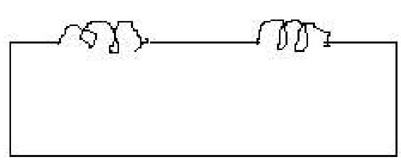
$M^2 / L_1L_2 = K_1K_2 = \text{say } K^2$

$$\boxed{M = K\sqrt{L1L2}}$$

INDUCTORS IN SERIES

SERIES CONNECTION – AIDING

V(t)



Consider two coupled coils of self inductance L1 and L2. Let M be the mutual inductance between them. The two coils are connected in series (aiding)

Since the current is entering both coils at the dotted terminal, it is called series aiding connection.

By applying kirchoff's law

$L1\ di/dt + M\ di/dt + L2\ di/dt + M\ di/dt = V(t)$

$(L1+L2+2M)\ di/dt = V(t)$

The equivalent inductance

$\boxed{L = L1+L2+2M}$

SERIES CONNECTION – OPPOSING

The two coils are connected in series since the current is entering first coil at dotted terminal and leaving other coil at dotted terminal it is called series opposing connection

By applying kirchoff's law

$L1\ di/dt - M\ di/dt + L2\ di/dt - M\ di/dt = V(t)$

$(L1+L2-2M)\ di/dt = V(t)$

The equivalent inductance

$\boxed{L = L1+L2-2M}$

INDUCTORS IN PARALLEL

PARALLEL CONNECTION – AIDING

Consider the two coils connected in parallel (aiding), let I1 and I2 be the currents flowing through coil1 and coil2

Both the currents I1 and I2 enter the coils at the dotted terminal

By applying kirchoff's law

$L1\ di1/dt + M\ di2/dt = V(t)$

$M\ di1/dt\ L2\ di2/dt = V(t)$

Taking the laplace transform for both the equations

$L1_S I1(S) + M_S I2(S) = V(S)$

$M_S I1(S) + L2_S I2(S) = V(S)$

Solving

$I1(S) = V(L2-M)/S(L1L2 – M^2)$

$I2(S) = V(L1-M)/S(L1L2 – M^2)$

Total current $I = I1 + I2$

$I = V(L2-M)/S(L1L2 – M^2)+ V(L1-M)/S(L1L2 – M^2)$

$I = V(L2+L1-2M)/S(L1L2-M^2)$

$V/I = S(L1L2-M^2)/(L2+L1 – 2M)$

But $L= V/I$

Equivalent inductance $L = L1L2-M^2/L1+L2 – 2M$

PARALLEL CONNECTION – OPPOSING

By applying kirchoff's law

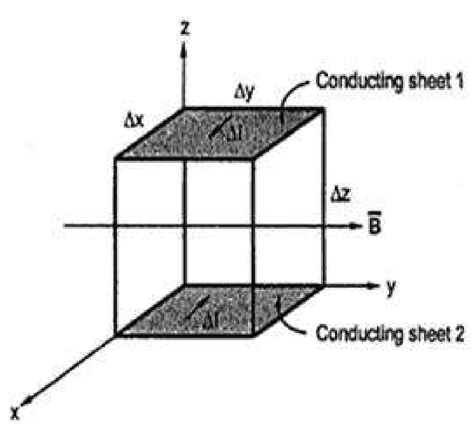
To the above circuit and solving the equations, we get inductance

$L=L1L2 – M^2 /L1+L2 +2M$

ENERGY STORED IN MAGNETIC FIELD

The capacitor when connected to a source of potential would store energy in the electric field.

The inductor when connected to a source of potential would store energy in the magnetic field.



The energy store = work done in establishing the magnetic field.

If the current is increases from 0 to I with the potential difference across the inductor equal to V, then the source is suppyling power equal to VI.

The energy supplied in time dt is

dw = VI dt

W =∫VI dt

=∫L dI / dt.Idt

=L∫IdI / dt.dt (cancelling dt and dt)

$W = ½ LI^2$

A = LI

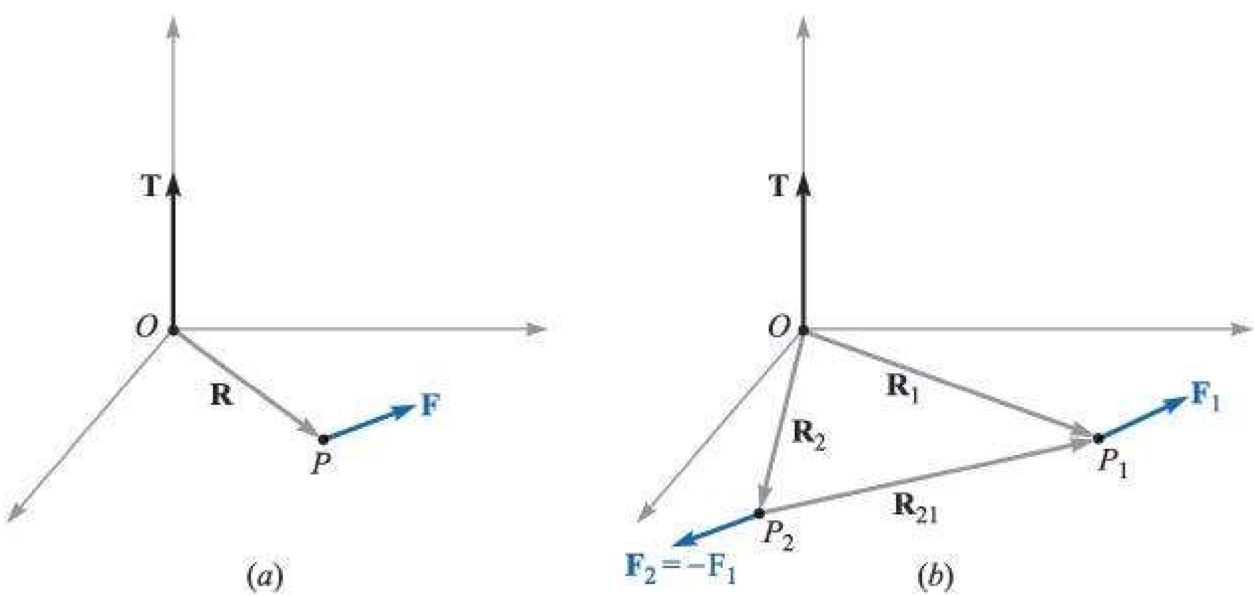
$W = 1/2\ LI^2 = ½\ AI = A^2/2L$

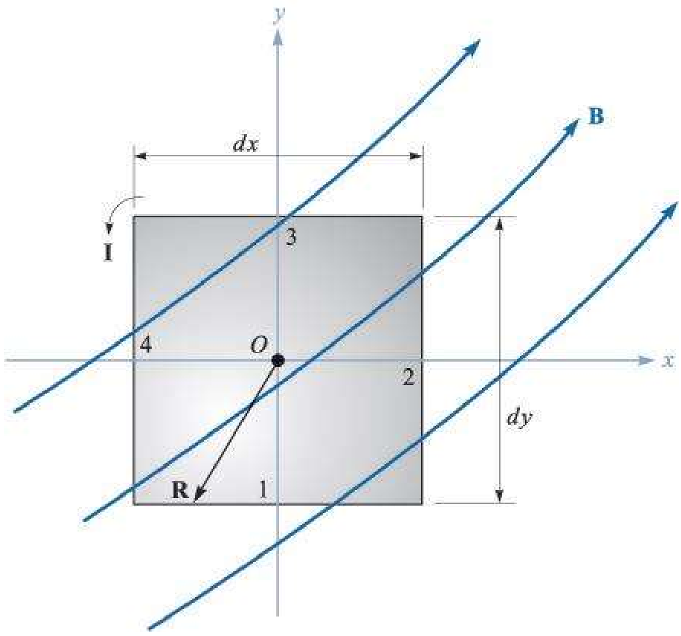
$\boxed{W = ½\ CV2 = ½\ QV = Q^2/2C}$

FORCE AND TORQUE ON CLOSED CIRCUITS (Magnetic torque)

When a current loop is placed parallel to a magnetic field, force act on the loop that tend to rotate it.

The tangential force multiplied by the radial distance at which it acts is called torque or mechanical moment on the loop.





Consider the rectangular loop of length l and breath b carrying a current I in a uniform mag. Field of flux density B as shown in fig b

The loop makes an angle θ with respect to mag. Flux density B

The force acting on the loop

F = BIl sin θ

If the loop plane is parallel to the mag. Field the total torque on the loop

T = 2*Torque on each side

T = 2* force * distance

= 2BIl sin θ b/2

= BIl b sinθ   area = lb

= BIA sinθ

The mag. Moment of loop is IA

The mag. Moment is a vector with the direction given by the unit normal vector(n)to the plane of the loop

m = Ian

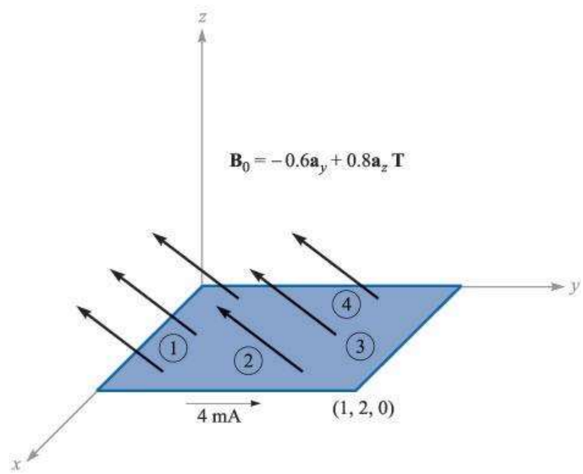
T =m B sinθ n

In vector form the torque can be expressed as

→ → →

T = m * B (vector)

$\vec{m} = \vec{T} / \vec{B}$ (vector)



MAGNETIC BOUNDARY CONDITIONS

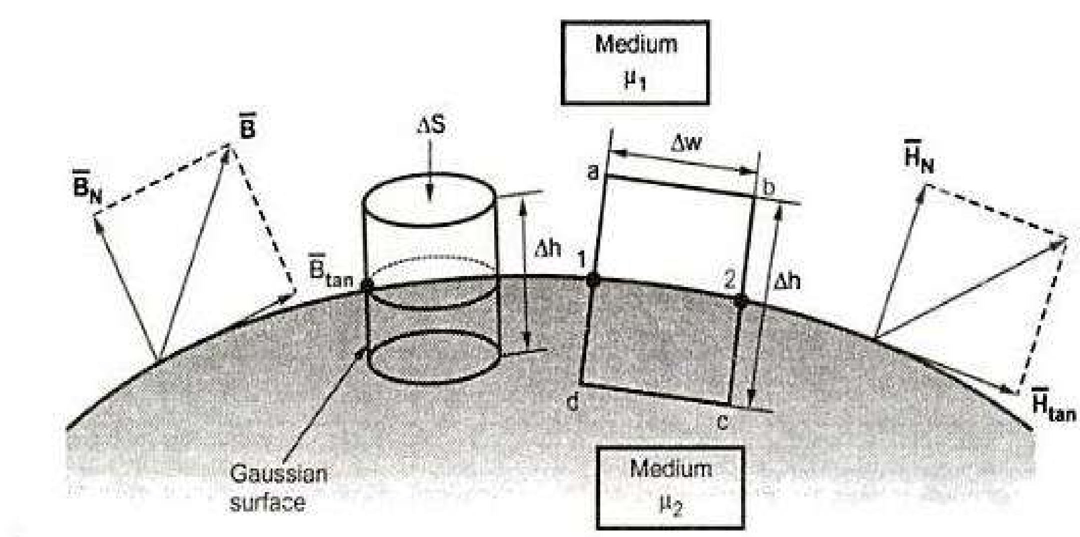
The conditions of magnetic field existing at the boundary of the two media when the magnetic field passes from one medium to other are called boundary conditions or magnetic boundary conditions

In the magnetic boundary condition, the condition of B and H are studied at the boundary

B and H vectors are resolved into two components

Tangential to boundary

Normal (perpendicular) to boundary



Consider a boundary between two isotropic homogeneous linear materials with different permeabilities μ1 and μ2

Boundary conditions for normal component

To find the normal component of E select a closed Gaussian surface in the form of a right circular cylinder

Let the height be Δh and be placed in such a way the Δh / 2 is in medium1 and remaining Δh / 2 is in medium 2.

Also zxis of the cylinder is in the normal direction to the surface

According to gauss's law for the magnetic field

$$\oint_S \overline{B} \cdot d\overline{S} = 0$$

$$\oint_{top} \overline{B} \cdot d\overline{S} + \oint_{bottom} \overline{B} \cdot d\overline{S} + \oint_{lateral} \overline{B} \cdot d\overline{S} = 0$$

For top surfaces :

$$\oint_{Top} \overline{B} \cdot d\overline{S} = B_{N_1} \oint_{Top} d\overline{S} = B_{N_1} \Delta S$$

For bottom surface :

$$\oint_{Bottom} \overline{B} \cdot d\overline{S} = B_{N_2} \oint_{Bottom} d\overline{S} = B_{N_2} \Delta S$$

For lateral surface

$$\oint_{Lateral} \overline{B} \cdot d\overline{S} = 0$$

Sub the above equations

$$B_{N_1} \Delta S - B_{N_2} \Delta S = 0$$

BN1 and BN2 are in opposite direction

$$B_{N_1} \Delta S = B_{N_2} \Delta S$$

$$\boxed{B_{N_1} = B_{N_2}}$$

Thus the normal component of B is continuous at the boundary

As the magnetic flux density and the magnetic field intensity are related by

$$\bar{B} = \mu\bar{H}$$

The above equation becomes

$$\mu_1 H_{N_1} = \mu_2 H_{N_2}$$

$$\boxed{\frac{H_{N_1}}{H_{N_2}} = \frac{\mu_2}{\mu_1} = \frac{\mu_{r2}}{\mu_{r1}}}$$

Hence the normal components of H is not continuous at the boundary. The field strengths in two media are inversely proportional to their relative permeabilities.

Boundary conditions for tangential component

According to Ampere's circuital law

$$\oint \bar{H} \cdot d\bar{L} = I$$

Consider a rectangular loop in clockwise direction as a-b-c-d-a. This closed path is placed in a plane normal to the boundary surface

$$\oint \bar{H} \cdot d\bar{L} = \int_a^b \bar{H} \cdot d\bar{L} + \int_b^1 \bar{H} \cdot d\bar{L} + \int_1^c \bar{H} \cdot d\bar{L} + \int_c^d \bar{H} \cdot d\bar{L} + \int_d^2 \bar{H} \cdot d\bar{L} + \int_2^a \bar{H} \cdot d\bar{L} = I$$

The sides a-b and c-d are parallel to the tangential direction to the surface. While the other two sides are normal to the surface at the boundary

Let the elementary height and elementary width be Δh and Δw

The normal and tangential component in medium1 and in medium2 arein opposite direction

The above equation are written as

To get conditions at boundary, $\Delta h \to 0$. Thus,

$$K \cdot dw = H_{tan1}(\Delta w) - H_{tan2}(\Delta w)$$

$$H_{tan1} - H_{tan2} = K$$

In vector form

$$\boxed{\bar{H}_{tan1} - \bar{H}_{tan2} = \bar{a}_{N_{12}} \times \bar{K}}$$

aN12 – unit vector

the above equ in terms of B

$$\frac{B_{tan1}}{\mu_1} - \frac{B_{tan2}}{\mu_2} = K$$

Special case

The boundary is free of current

So K = 0

$$H_{tan1} - H_{tan2} = 0$$

$$H_{tan1} = H_{tan2}$$

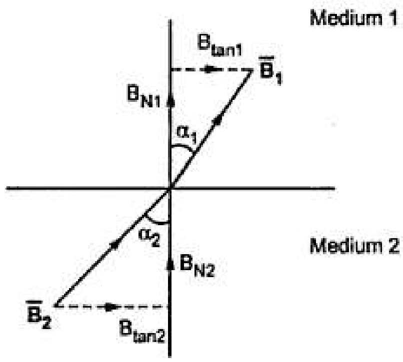
Similarly

$$\frac{B_{tan1}}{\mu_1} = \frac{B_{tan2}}{\mu_2}$$

$$\boxed{\frac{B_{tan1}}{B_{tan2}} = \frac{\mu_1}{\mu_2} = \frac{\mu_{r1}}{\mu_{r2}}}$$

Condition

From the above equ the tangential components of H are continuous while the tangential components of B are discontinuous at the boundary



In medium 1

Tan $\alpha1$ = B tan1 / BN1

In medium 2

Tan $\alpha2$ = B tan2 / BN2

Diving the above equ

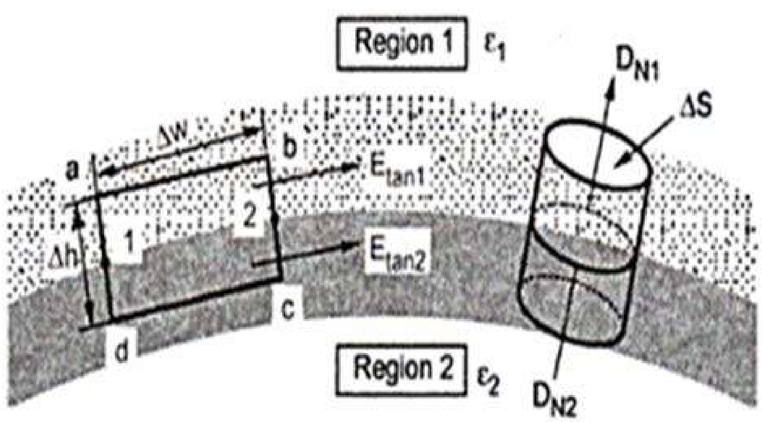
Tan$\alpha1$ / tan $\alpha2$ = Btan1 BN1 . BN2 / B tan2

W.K.T

BN1 = BN2

$\boxed{\text{Tan}\alpha1 / \tan \alpha2 = \text{Btan1} / \text{B tan2} = \mu\text{r1} / \mu2}$

BOUNDARY CONDITIONS BETWEEN CONDUCTOR AND DIELECTRIC



E and D are resolved into two components

Tangential component to the boundary

Normal component to the boundary

Boundary conditions for tangential component

It is known that

$$\oint \bar{E} \cdot d\bar{L} = 0$$

$$\cdot \int_a^b \bar{E} \cdot d\bar{L} + \int_b^c \bar{E} \cdot d\bar{L} + \int_c^d \bar{E} \cdot d\bar{L} + \int_d^a \bar{E} \cdot d\bar{L} = 0$$

The closed contour is placed in such a way that its two sides a-b and c-d is in the conductor

E = 0

$$\int_a^b \bar{E} \cdot d\bar{L} + \int_b^c \bar{E} \cdot d\bar{L} + \int_d^a \bar{E} \cdot d\bar{L} = 0$$

$$\bar{E}_1 = \bar{E}_{1t} + \bar{E}_{1N}$$

$$\bar{E}_2 = \bar{E}_{2t} + \bar{E}_{2N}$$

$$\left|\bar{E}_{1t}\right| = E_{tan1}, \qquad \left|\bar{E}_{2t}\right| = E_{tan2}$$

$$\left|\bar{E}_{1N}\right| = E_{1N}, \qquad \left|\bar{E}_{2N}\right| = E_{2N}$$

$$\int_a^b \bar{E} \cdot d\bar{L} + \int_c^d \bar{E} \cdot d\bar{L} = 0$$

$$\int_a^b \bar{E} \cdot d\bar{L} = E_{tan\,1} \int_a^b d\bar{L} = E_{tan\,1}\,\Delta w$$

$$\int_c^d \overline{E} \cdot d\overline{L} = -E_{\tan 2} \Delta w$$

Thus the tangential component of the electric field intensity is zero at the boundary

$$E_{\tan 1} \Delta w - E_{\tan 2} \Delta w = 0$$

$$\therefore \quad \boxed{E_{\tan 1} = E_{\tan 2}}$$

now

$$\overline{D} = \varepsilon \overline{E}$$

$$D_{\tan 1} = \varepsilon_1 E_{\tan 1} \quad \text{and} \quad D_{\tan 2} = \varepsilon_2 E_{\tan 2}$$

$$\frac{D_{\tan 1}}{\varepsilon_1} = \frac{D_{\tan 2}}{\varepsilon_2}$$

$$\boxed{\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{\varepsilon_1}{\varepsilon_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}}$$

Thus the tangential component of electric flux density is zero at the boundary

Boundary conditions for normal component

According to Gauss' s law

$$\oint \overline{D} \cdot d\overline{S} = Q$$

The surface integral must be evaluated over three surfaces that is top, bottom and lateral

$$\left[\int_{top} + \int_{bottom} + \int_{lateral}\right] \overline{D} \cdot d\overline{S} = Q$$

The bottom surface is in the conductor

Where D = 0

$$\therefore \int_{top} \overline{D} \cdot d\overline{S} + \int_{bottom} \overline{D} \cdot d\overline{S} = Q$$

$\left|\bar{D}\right| = D_{N1}$ for dielectric 1 and D_{N2} for dielectric 2.

$$\int_{top} \bar{D} \cdot d\bar{S} = D_{N1} \int_{top} d\bar{S} = D_{N1} \Delta S$$

From Gauss's law

$D_N \Delta_S = Q$

But at the boundary, the charge exists in the form of surface charge density ρ_s

Since

$D_N \Delta_S = \rho_s \Delta_S$

$D_N = \rho_s$

For the lateral surface $\Delta h = 0$

Since

lateral$\int$ D.ds = 0

considering the above equ

$D_N = \rho_s$

Thus the flux density of normal component is equal to the surface charge density

Similarly

$Q = \rho_s \Delta_S$

$D_{N1} - D_{N2} = \rho_s$

Thus

$$\rho_S = 0$$

$$D_{N1} - D_{N2} = 0$$

$$\boxed{D_{N1} = D_{N2}}$$

$$D_{N1} = \varepsilon_1 E_{N1} \quad \text{and} \quad D_{N2} = \varepsilon_2 E_{N2}$$

$$\frac{D_{N1}}{D_{N2}} = \frac{\varepsilon_1 E_{N1}}{\varepsilon_2 F_{N2}} = 1$$

$$\boxed{\frac{E_{N1}}{E_{N2}} = \frac{\varepsilon_2}{\varepsilon_1} = \frac{\varepsilon_{r2}}{\varepsilon_{r1}}}$$

<u>**Two marks**</u>

1. **State faraday's law of electromagnetic induction.**

   Faraday's law state that electromagnetic force induced in a circuit is equal to the rate of change of magnetic flux linking the circuit.

   Emf=$\frac{d\varphi}{dy}$

2. **Define self-inductance.**

   The self-inductance of a coil is defined as the flux produced by the current flowing through the coil links with the coil itself.

   $L = \frac{\varphi}{i}$

3. **Define mutual inductance.**

   The mutual inductance between two coils is defined as the ratio of induced magnetic flux linkage in one coil to the current through in other coil.

   $M = \frac{N_2\varphi_{12}}{i_1}$

4. **Give the equation for inductance of a solenoid.**

   $$L = \frac{\mu N^2 A}{l}$$

   Where,

   N is number of turns

   A is area of cross section

   l is length of solenoid

   $\mu$is permeability

5. **Give the expression for inductance of a torroid.**

   $L = \frac{\mu N^2 A}{2\pi R}$N is number of turns

   r is radius of the coil

   R is radius of toroid

6. **What is the expression for energy stored in magnetic field**

$$W_E = \frac{1}{2}LI^2$$

Where L is the inductance

I is the current

7. **What is energy density in the magnetic field?**

Energy density = $\frac{1}{2}BH$

8. **State dot rule?**

- If both currents enter dotted ends of coupled coils or if both currents enter undotted ends, then the sign on the M will be same as the sign on the L
- If one current enters a dotted end and the other an undotted end, the sign on the M will be opposite to the sign on the L

9. **What are the difference types of magnetic materials?**

According to their behaviour, magnetic materials are classified as diamagnetic, paramagnetic and ferromagnetic materials

10. **Define boundary condition.**

The conditions existing at the boundary of the two media when field passes from one medium to other are called boundary conditions

UNIT – V

UNIT-V ELECTROMAGNETIC WAVES

Faraday's laws, Induced EMF- Maxwell's Equations: Differential and integral form – Electro Magnetic Wave equations - Wave parameters; velocity, intrinsic impedance, propagation constant – Uniform plane wave and its properties - Waves in free space, dielectrics, conductor- Skin depth-Poynting vector and Poynting Theorem.

MAXWELL'S EQUATIONS – POINT (DIFFERENTIAL) AND INTEGRAL FORM

Maxwell's equations

A set of four expressions derived from Ampere's circuital law, faraday's law, Gauss law for electric field and Gauss law for magnetic field.

These four expressions can be written in the following form:

(i) Point or differential form
(ii) Integral form

MAXWELL'S EQUATION – I

Maxwell's equation – I is derived from ampere's circuital law which states that the line integral of magnetic field intensity H on any closed path is equal to the current enclosed by that path.

Closed ∫ H.dl = I

The above equation can be modified as

Closed ∫ H.dl = surface∫[J +θD / θt].ds

The above equation is derived from ampere's circuital law, this equation is integral form.

Applying stoke's theorem to L.H.S of the above equ

surface∫(ΔXh).ds = surface∫ [J +θD / θt].ds

$$\boxed{\Delta Xh = J + \theta D \ / \ \theta t}$$

The above Maxwell's equation – I in point form or differential form

MAXWELL'S EQUATION – II

Maxwell's equation – II is derived from Faraday's law which states that the emf induced in a circuit is equal to the rate of decrease of the magnetic flux linkage in the circuit.

E=-dΦ/dt Volts

Maxwell's equation – II in integral form

Closed $\int E.dl = -\text{surface}\int \theta B/\theta t.\ ds$

Using stoke's theorem, converting line integral into surface integral in the above equation

Maxwell's equation – II in point form is

$\text{surface}\int(\Delta x E).ds = -\text{surface}\int \theta B/\theta t.\ ds$

$$\boxed{\Delta \text{ x } E = -\theta B / \theta t}$$

The electro motive force around a closed path is equal to the magnetic displacement through that closed path.

MAXWELL'S EQUATION – III

The Maxwell's equation – III is derived from electric Gauss's law which states that the electric flux through any closed surface is equal to the charge enclosed by the surface.

$\Psi = Q$

Maxwell's equation – III in integral form

$\text{surface}\int D.ds = Q$

R.H.S of the above equ can be replaced as

$\text{surface}\int D.ds = \text{volume}\int \rho_v\ dv$

Maxwell's equation – III in point form is

using divergence theorem the above equ can be written as

$\text{volume}\int(\Delta.D)dv = \text{volume}\int \rho_v\ dv$

$$\boxed{\Delta\ .\ D = \rho_v}$$

The total electric displacement through the surface enclosing a volume is equal to the total charge within the volume.

MAXWELL'S EQUATION – IV

Maxwell's equation – IV is derived from magnetic Gauss's law which states that, the magnetic flux through any closed surface is equal to zero.

$\Phi = 0$

Maxwell's equation – IV in integral form surface∫B.ds = 0

Maxwell's equation – IV in point form is

Using divergence theorem

volume∫(Δ . B) dv = 0

$\boxed{\Delta.B = 0}$

The Net Magnetic Flux Emerging through Any Closed Surface is Equal To Zero

Comparative table of Maxwell's equations

Differential form	Integral form	Significanco
$\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t}$	$\oint \bar{E} \bullet d\bar{L} = -\int_S \frac{\partial \bar{B}}{\partial t} \bullet d\bar{S}$	Faraday's law
$\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$	$\oint \bar{H} \bullet d\bar{L} = I + \int_S \frac{\partial \bar{D}}{\partial t} \bullet d\bar{S}$	Ampere's circuital law
$\nabla \bullet \bar{D} = \rho_v$	$\oint_S \bar{D} \bullet d\bar{S} = \int_S \rho_v \, dv$	Gauss's law
$\nabla \bullet \bar{B} = 0$	$\oint_S \bar{B} \bullet d\bar{S} = 0$	No isolated magnetic charges.

The Maxwell's equation, in the free space are as mentioned below.

(A) Point Form :

(i) $\nabla \times \overline{E} = -\frac{\partial \overline{B}}{\partial t}$

(ii) $\nabla \times \overline{H} = \frac{\partial \overline{D}}{\partial t}$

(iii) $\nabla \bullet \overline{D} = 0$

(iv) $\nabla \bullet \overline{B} = 0$

(B) Integral Form :

(i) $\oint \overline{E} \bullet d\overline{L} = -\int_S \frac{\partial \overline{B}}{\partial t} \bullet d\overline{S}$

(ii) $\oint \overline{H} \bullet d\overline{L} = \int_S \frac{\partial \overline{D}}{\partial t} \bullet d\overline{S}$

(iii) $\oint_S \overline{D} \bullet d\overline{S} = 0$

(iv) $\oint \overline{B} \bullet d\overline{S} = 0$

(A) Point Form :

$$
\begin{aligned}
&\text{(i)} & \nabla \times \overline{E} &= -j\omega\,\overline{D} = -j\omega\mu\overline{H} \\
&\text{(ii)} & \nabla \times \overline{H} &= \overline{J} + j\omega\,\overline{D} = \sigma\,\overline{E} + j\omega(\varepsilon\overline{E}) = (\sigma + j\omega\varepsilon)\,\overline{E} \\
&\text{(iii)} & \nabla \bullet \overline{D} &= \rho_v \\
&\text{(iv)} & \nabla \bullet \overline{B} &= 0
\end{aligned}
$$

(B) Integral Form :

$$
\begin{aligned}
&\text{(i)} & \oint \overline{E} \bullet d\overline{L} &= -\int_S j\omega\overline{B} \bullet d\overline{S} = -\int_S j\omega\mu\,\overline{H} \bullet d\overline{S} \\
&\therefore & \oint \overline{E} \bullet d\overline{L} &= -j\omega\mu \int_S \overline{H} \bullet d\overline{S} \\
&\text{(ii)} & \oint \overline{H} \bullet d\overline{L} &= I + \int_S j\omega\overline{D} \bullet d\overline{S} \\
& & &= \int_S \overline{J} \bullet d\overline{S} + \int_S j\omega\varepsilon\,\overline{E} \bullet d\overline{S} \\
& & &= \int_S \sigma\,\overline{E} \bullet d\overline{S} + \int_S j\omega\varepsilon\,\overline{E} \bullet d\overline{S} \\
& & &= (\sigma + j\omega\varepsilon)\int_S \overline{E} \bullet d\overline{S} \\
&\text{(iii)} & \oint_S \overline{D} \bullet d\overline{S} &= \oint_v \rho_v\, dv \\
&\text{(iv)} & \oint \overline{B} \bullet d\overline{S} &= 0
\end{aligned}
$$

POYNTING'S THEOREM

When an electromagnetic wave propagation through space from their source to distant receiving points, there is a transfer of energy from the source to the receivers

There exists a simple and direct relation between the rate of this energy transfer and the amplitudes of electric and magnetic field intensities of the electromagnetic wave.

Statement

The vector product of electric field intensity and magnetic field intensity and magnetic field intensity at any point is a measure of the rate of energy flow per unit area at that point

$$\vec{P} = \vec{E} \times \vec{H}$$

The direction of flow (P) is perpendicular to E & H and P – Poynting vector

4.3 THE ENERGY IN ELECTROMAGNETIC FIELD

The energy flow equation can be obtained from Maxwell's equations.

From Maxwell's I equation

$\Delta Xh = J + \xi\ \theta E/\ \theta t$ -----------1

$J = \Delta Xh - \xi\ \theta E/\ \theta t$-----------2

Multiply eqn 2 by E

$E\ .J = E\ .\ \Delta Xh - \xi\ \theta E/\ \theta t$ ---------------3

By identity

$\Delta .E \text{ x } H = H.\Delta \text{ x } E - E\ .\ \Delta XH$

Or

$\Delta .H \text{ x } E = E.\Delta \text{ x } H - H\ .\ \Delta XE$

Then the equ 3

$$\boxed{E.J = H.\Delta x\ E - \Delta .E \text{ x } H - \xi E\ \ \theta E/\ \theta t}$$

From Maxwell's II equation

$\Delta x\ E = -\mu\ \theta H/\theta t$ --------------5

Sub 5 in 4

$E.J = -\mu H\ \theta H/\theta t - \xi E\ .\ \theta E/\theta t - \Delta\ .\ ExH$

$H\ .\ \theta H/\theta t = ½\ \theta/\theta t\ H^2$ and $E\ .\ \theta E/\theta t = ½\ \theta/\theta t\ E^2$ ----------------7

Sub 7 in 6

Since $E.J = -\ \mu/2\ \ \theta/\theta t\ H^2\ -\ \xi/2\ \theta/\theta t\ E^2\ \ -\ \Delta\ .\ ExH$ --------------8

Integrating equ 8 over a volume

$$\boxed{\text{volume}\int E.J\ dv = -\ \theta/\theta t\,\text{volume}\int(\ \mu/2\ \ \theta/\theta t\ H^2\ -\ \xi/2\ \theta/\theta t\ E^2\ )\ dv\ -\ \text{volume}\int \Delta\ .\ ExH\ dv}$$

Using divergence theorem

volume$\int \Delta\ .\ ExH\ dv$ = closed surface$\int ExH\ ds$

Since volume$\int E.J\ dv = -\ \theta/\theta t$volume$\int(\ \mu/2\ \ \theta/\theta t\ H^2\ -\ \xi/2\ \theta/\theta t\ E^2\ )\ dv$ - surface$\int\ ExH\ ds$

The term volume$\int E.J\ dv$ represents the total power dissipated in a volume V

This term $-\ \theta/\theta t$volume$\int(\ \mu/2\ \ \theta/\theta t\ H^2\ -\ \xi/2\ \theta/\theta t\ E^2\ )\ dv$ represent the rate at which the stored magnetic and electric energy in the volume V is decreasing

This term - surface$\int\ ExH\ ds$ represents the ratio of flow of energy entering the volume from outside

The power flow per unit area is $P = E \times H$

ELECTROMAGNETIC WAVE EQUATION

WAVE EQUATION FOR FREE SPACE

The application of Maxwell's equations is the prediction of existence of electromagnetic wave.

For free space (dielectric medium) the conductivity of the medium is zero (ie $\sigma = 0$) and there is no charge containing in it (ie $\rho = 0$) . the electromagnetic wave equations for free space can be obtained from Maxwell's equations

The Maxwell's equation from faraday's law for free space in point form is

$\Delta \times E = -\ \theta B\ /\ \theta t$

$\Delta \times E = -\ \mu\ \theta H\ /\ \theta t$

Taking curl on both sides

$\Delta x \Delta x E = -\mu\ \Delta \times \theta\ H\ /\ \theta\ t$ ------------------1

But Maxwell's equation from Ampere's law fro free space in point form is

$\Delta XH = \theta D / \theta t = \xi \theta E / \theta t$

Differentiating $\Delta \times \theta H / \theta t = \theta \Delta x H / \theta t$

$\Delta \times \theta H / \theta t = \theta / \theta t \ (\xi \theta E / \theta t)$-------------------------2

Sub 2 in equ 1

$\Delta x \Delta x E = -\mu \xi \theta^2 E / \theta t^2$--------------------------3

But the identity is given by

$\Delta x \Delta x E = \Delta (\Delta . E) - \Delta^2 E$

But $\Delta . E = 1/\xi \Delta . D = \rho / \xi = 0$

Then $\Delta x \Delta x E = -\Delta^2 E$ -----------------------4

Comparing 3 and 4

$\Delta^2 E - \mu \xi \theta^2 E / \theta t^2 = 0$ ---------------------5

This is the wave equation in free space in terms of electric field

The wave equation for free space in terms of magnetic field H is obtained in a similar manner as follows

The Maxwell's equation from Ampere's law for free space in point form is given by

$\Delta \times H = \xi \theta E / \theta t$

Taking curl on both sides

$\Delta x \Delta x H = -\xi \Delta \times \theta E / \theta t$ ------------------6

But Maxwell's equation from Ampere's law fro free space in point form is

$\Delta XE = - \mu \theta H / \theta t$

Differentiating $\Delta \times \theta E / \theta t = - \mu \theta^2 H / \theta t^2$-----------------------7

Sub 7 in equ 6

$\Delta x \Delta x H = -\mu \xi \theta^2 H / \theta t^2$--------------------------8

But the identity is given by

$\Delta x \Delta x H = \Delta (\Delta . H) - \Delta^2 H$---------------------9

But $\Delta . H = 1/\mu \Delta . B = 0$

Then $\Delta x \Delta x H = -\Delta^2 H$ -----------------------10

Comparing 8 and 10

$\Delta^2 H = \mu \xi \theta^2 H / \theta t^2$

$\Delta^2 H - \mu \xi \theta^2 H / \theta t^2 = 0$ ------------------11

This is the wave equation in free space in terms of Magnetic field

4.5 WAVE EQUATION FOR CONDUCTING MEDIUM

Electromagnetic wave equation can be obtained from Maxwell's equation

The Maxwell's equation from faraday's law in point form is given by

$\Delta X E = - \theta B / \theta t = - \mu \theta H / \theta t$

Taking curl on both sides

$\Delta x \Delta X E = - \mu \Delta x \theta H / \theta t$ ------------------1

But Maxwell's equation from Ampere's law in point form is

$\Delta X H = J + \theta D / \theta t$

$\Delta X H = \sigma E + \xi \theta E / \theta t$

Differentiating

$\Delta X \theta H / \theta t = \sigma \theta E / \theta t + \xi \theta^2 E / \theta t^2$ --------------------2

Sub 2 in 1

$\Delta x \Delta X E = - \mu[\sigma \theta E / \theta t + \xi \theta^2 E / \theta t^2]$

$\Delta x \Delta X E = - \mu\sigma \theta E / \theta t - \mu \xi \theta^2 E / \theta t^2$-----------------------3

But according to identity

$\Delta x \Delta X E = \Delta (\Delta . E) - \Delta^2 E$ ----------------------------4

w.k.t $\Delta . E = 1/\xi \Delta . D$

since there is no net charge within the conductor the charge density $\rho = 0$

$\Delta . D = 0$

$\Delta . E = 0$

Then equ 4 becomes

$\Delta x \Delta x E = -\Delta^2 E$ -----------------------5

Comparing equ 3 and 5

$\Delta^2 E = = - \mu\sigma\, \theta E \;/\; \theta t - \mu\, \xi\, \theta^2 E \;/\; \theta t^2 = 0$ --------------------6

This is the wave equation for electric field

The wave equation for magnetic field H is obtained in a similar manner as follows

The Maxwell's equation from Ampere's law in point form is given by

$\Delta X H = \sigma E + \xi\, \theta E \;/\; \theta t$

Taking curl on both sides

$\Delta x \Delta X h = \sigma \Delta \text{ x } E + \xi\, \Delta \text{ x } \theta E \;/\; \theta t$ ----------------------7

But Maxwell's equation from faraday's law

$\Delta x E = - \mu\, \theta H \;/\; \theta t$

Differentiating w.r.t 't'

$\Delta \theta E / \theta t = - \mu\, \theta^2 H \;/\; \theta t^2$

Sub $\Delta x E$ and $\Delta x\, \theta E \;/\; \theta t$ in equ 7

$\Delta x \Delta X h = - \mu\, \sigma\, \theta H \;/\; \theta t - \mu\, \xi\, \theta^2 H \;/\; \theta t^2$------------------------8

But the identity is

$\Delta x \Delta X H = \Delta\, (\Delta . H) - \Delta^2 H$

But $\Delta . B = \mu\, \Delta . H = 0$

Then $\Delta x \Delta x H = -\Delta^2 H$-----------------------------9

Comparing equ 8 and 9

$\Delta^2 H - \mu\sigma\, \theta H \;/\; \theta t - \mu\, \xi\, \theta^2 H \;/\; \theta t^2 = 0$ ---------------------10

This is the wave equation for magnetic field H

UNIFORM PLANE WAVE

If the phase of a wave is the same for all points on a plane surface it is called plane wave. If the amplitude is also constant in a plane wave, it is called uniform plane wave.

<u>Properties of uniform plane waves</u>

1. At every point in space, electric field (E) and magneticfield (H) are perpendicular to each other and to the direction of travel.
2. The field vary harmonically with time and at the same frequency everywhere in space.
3. Each field has the same direction, magnitude and phase at every point in any plane perpendicular to the direction of wave travel.

If the electric field is in x direction and the magnetic field in y direction, then the wave is travelling is S direction.

The wave equation for free space is given by

$$\nabla^2 E = \mu\varepsilon \frac{\partial^2 E}{\partial t^2}$$

$$\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2} + \frac{\partial^2 E}{\partial z^2} = \mu\varepsilon \frac{\partial^2 E}{\partial t^2}$$

Consider electric field E various in x direction and E is independent of direction y and z

Then the wave equation becomes

$$\frac{\partial^2 E}{\partial x^2} = \mu\varepsilon \frac{\partial^2 E}{\partial t^2} \qquad \left[\frac{\partial^2 E}{\partial y^2} = \frac{\partial^2 E}{\partial z^2} = 0\right]$$

It can be written as in term of the components of E

$$\frac{\partial^2 Ex}{\partial x^2} = \mu\varepsilon \frac{\partial^2 Ex}{\partial t^2}$$

$$\frac{\partial^2 Ey}{\partial x^2} = \mu\varepsilon \frac{\partial^2 Ey}{\partial t^2}$$

$$\frac{\partial^2 Ez}{\partial x^2} = \mu\varepsilon \frac{\partial^2 Ez}{\partial t^2}$$

For free space there is no charge density

$$\nabla . D = \varepsilon \nabla . E = 0$$

$$\nabla . E = 0$$

$$\frac{\partial Ex}{\partial x} + \frac{\partial Ey}{\partial y} + \frac{\partial Ez}{\partial z} = 0$$

For uniform plane wave, E is independent of y and z , then

$$\frac{\partial Ex}{\partial x} = 0$$

This equation shows that there is no variation of Ex in the x direction

Diff. this equation w.r.t 't'

$$\frac{\partial^2 Ex}{\partial x^2} = 0$$

- It requires that either Ex be zero or constant
- If Ex is constant, it would not part of wave motion and so Ex must be zero
- Therefore a uniform plane wave propagating in the x direction has no x component of E. A similar analysis would show that there is no x component of H.

$$\nabla . B = \mu \nabla . H = 0$$

$$\nabla . H = 0$$

$$\frac{\partial Hx}{\partial x} + \frac{\partial Hy}{\partial y} + \frac{\partial Hz}{\partial z} = 0$$

Since H is propagating in x direction, it is independent of y and z.

Then

$$\frac{\partial Hx}{\partial x} = 0 \ \& \frac{\partial^2 Hx}{\partial x^2} = 0$$

Since Hx is not a constant, Hx must be zero

∴ Hx=0 for uniform plane wave.

5.2 CHARACTERISTIC IMPEDANCE OR INTRINSIC IMPEDANCE (η_0)

- Consider the plane wave propagating in x-direction. The wave equation for free space is

$$\frac{\partial^2 E}{\partial x^2} = \mu\varepsilon \frac{\partial^2 E}{\partial t^2}$$

The general solution of this differential equation is in the form.

$$E = f_1(x - V_0 t) + f_2(x + V_0 t)$$

Where $V_0 = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

$f_1 \& f_2$ are any functions of $(x - V_0 t) \& (x + V_0 t)$ respectively.

- The solution of wave equation consists of two waves, one travelling in positive direction and other travelling in negative direction.
- Consider the wave travel in positive direction alone.

$$f_2(x + V_0 t) = 0 \qquad \text{(Negative direction)}$$

The general solution of wave equation becomes

$$E = f(x - V_0 t)$$

$$\nabla \times E = \begin{vmatrix} \vec{x} & \vec{y} & \vec{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ Ex & Ey & Ez \end{vmatrix}$$

$$= \vec{x}\left[\frac{\partial Ez}{\partial y} - \frac{\partial Ey}{\partial z}\right] + \vec{y}\left[\frac{\partial Ex}{\partial z} - \frac{\partial Ez}{\partial x}\right] + \vec{z}\left[\frac{\partial Ey}{\partial x} - \frac{\partial Ex}{\partial y}\right]$$

Since wave travelling in x direction, E & H are both independent of y and z i.e E_x=H_x=0 &$\frac{\partial E}{\partial y} = \frac{\partial E}{\partial z} = 0$

$$\nabla \times E = -\frac{\partial Ez}{\partial x}\vec{y} + \frac{\partial Ey}{\partial x}\vec{z}$$

Similarly

$$\nabla \times H = -\frac{\partial Hz}{\partial x}\vec{y} + \frac{\partial Hy}{\partial x}\vec{z}$$

But

$$\nabla \times H = \mu\varepsilon\frac{\partial E}{\partial t}$$

Comparing these two equation

$$-\frac{\partial Hz}{\partial x}\vec{y} + \frac{\partial Hy}{\partial x}\vec{z} = \varepsilon\left[\frac{\partial Ez}{\partial x}\vec{y} + \frac{\partial Ey}{\partial x}\vec{z}\right] \qquad [\therefore\ Ex = 0]$$

Equating $\vec{y}$ and $\vec{z}$ terms

$$-\frac{\partial Hz}{\partial x} = \varepsilon\frac{\partial Ey}{\partial t}$$

$$\frac{\partial Hy}{\partial x} = \varepsilon\frac{\partial Ez}{\partial t}$$

From second maxwell's equation for free space

$$\nabla \times E = -\mu\frac{\partial H}{\partial t}$$

But $\nabla \times E = -\frac{\partial Ez}{\partial x}\vec{y} + \frac{\partial Ey}{\partial x}\vec{z}$

Equating these two equation

$$-\frac{\partial Ez}{\partial x}\vec{y} + \frac{\partial Ey}{\partial x}\vec{z} = -\mu\left[\frac{\partial Hy}{\partial t}\vec{y} + \frac{\partial Hz}{\partial t}\vec{z}\right] \qquad [H_x=0]$$

Equating $\vec{y}$and $\vec{z}$ terms

$$\frac{\partial Ez}{\partial x} = \mu\frac{\partial Hy}{\partial t}$$

$$\frac{\partial Ey}{\partial x} = -\mu\frac{\partial Hz}{\partial t}$$

Let the solution of that equation is given by

$$E_y = f(x - V_0 t)$$

Diff

$$\frac{\partial Ey}{\partial x} = \frac{\partial f}{\partial (x - V_0 t)} = \frac{\partial (x - V_0 t)}{\partial t}$$

$$\frac{\partial Ey}{\partial x} = f'(x - V_0 t)(-V_0)$$

Simply $f'(x - V_0 t)$can be written as f'

$\frac{\partial Ey}{\partial x} = -V_0 f'$

But $-\frac{\partial Hz}{\partial x} = \varepsilon \frac{\partial Ey}{\partial t}$

$\frac{\partial Hz}{\partial x} = -\varepsilon(V_0 f') = -\varepsilon V_0 f'$

$= \frac{1}{\sqrt{\mu\varepsilon}} \varepsilon f'$

$$\frac{\partial Hz}{\partial x} = \sqrt{\frac{\varepsilon}{\mu}} f'$$

Integrating

$Hz = \sqrt{\frac{\varepsilon}{\mu}} \int f' dx$

$$Hz = \sqrt{\frac{\varepsilon}{\mu}} \; f$$

$= \sqrt{\frac{\varepsilon}{\mu}} Ey$

$\frac{Ey}{Hz} = \sqrt{\frac{\varepsilon}{\mu}}$

Similarly ,

$$\frac{Ey}{Hy} = -\sqrt{\frac{\mu}{\varepsilon}}$$

If E is the total electric field

$$E = \sqrt{{E_y}^2 + {E_z}^2}$$

And H is the total magnetic field,

$$H = \sqrt{{H_y}^2 + {H_z}^2}$$

Then $\frac{E}{H} = \sqrt{\frac{\mu}{\varepsilon}}$

- It is referred to as characteristics impedance or intrinsic impedance of the medium.
- It is the radio of square root of permeability to the dielectric constant of the medium and it is denoted by n.

$$\eta = \frac{E}{H} = \sqrt{\frac{\mu}{\varepsilon}}$$

For free space $\varepsilon_r = \mu_r = 1(air)$, the characteristics impedance or intrinsic imoedance for free space is given by

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

$$= \frac{4\pi \times 10^{-7}}{\frac{1}{36\pi \times 10^9}}$$

$$= \sqrt{4\pi \times 36\pi \times 10^2}$$

$$\eta_0 = 120\pi$$

$$\boxed{\eta_0 \approx 377\Omega}$$

Problems

1. Find the characteristics impedance of the medium whose relative permittivity is 3 and relative permeability is 1

$$\varepsilon_r = 3$$

$$\mu_r = 1$$

$$Characteristics\ impedance\ \eta = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_r \mu_0}{\varepsilon_r\ \varepsilon_0}}$$

$$= \eta_0 \sqrt{\frac{\mu_r}{\varepsilon_r}}$$

Where $\eta_0 = \frac{\mu_0}{\varepsilon_0} = 120\pi$

$$\eta = 120\pi\sqrt{\frac{1}{3}}$$

$$\eta = 217.66\ ohms$$

2. Uniform plane eave in free space is described by E=100 $e^{-(\pi z/3)}\overrightarrow{ax}$. Determine the frequency and wave length.

E=100$e^{-(\pi z/3)}\overrightarrow{ax}$

$$\beta = \frac{2\pi}{\lambda} = \frac{\pi}{3}$$

$$\lambda = 6m$$

$$f = \frac{e}{\lambda} = \frac{3 \times 10^8}{60} = 50MHz$$

3. The velocity of uniform plane wave in a loss-less dielectric is $1*10^8$m/sec. Find the dielectric constant.

For loss-less dielectric $\mu_r = 1$

V= $\frac{1}{\sqrt{\mu_r \mu_o\ \varepsilon_0\ \varepsilon_r}}$

= $\frac{1}{\sqrt{\mu_o\ \varepsilon_0}\sqrt{\mu_r\ \varepsilon_r}}$

$1\times 10^8 = \frac{3\times10^8}{\sqrt{\varepsilon_r}}$

$$\sqrt{\varepsilon_r} = 3$$

$$\varepsilon_r = 9$$

5.3 WAVE PROPAGATION IN A LOSSLESS MEDIUM

The wave equation for free space (loss less medium) is

$$\nabla^2 E = \mu E \frac{\partial^2 E}{\partial t^2}$$

The phasor value of E is

E(x,t)=Re[E(x)$e^{i\omega t}$]

Applying to the above equation

$$\nabla^2 \mathrm{Re}\left[\mathrm{E}e^{i\omega t}\right] = \mu\varepsilon \frac{\partial^2}{\partial t^2} \mathrm{Re}[\mathrm{E}e^{i\omega t}]$$

$$\nabla^2 \mathrm{Re}\left[\mathrm{E}e^{i\omega t}\right] = \mu\varepsilon Re[-\omega^2 \mathrm{E}e^{i\omega t}]$$

$$\mathrm{Re}\left[(\nabla^2 \mathrm{E} + \mu\varepsilon\omega^2 E)e^{i\omega t}\right] = 0$$

$$\nabla^2 E + \mu\varepsilon\omega^2 E = 0$$

This is the wave equation for lossless medium in phasor form and it is called vector Helmholtz equation.

$$\nabla^2 E + \beta^2 E = 0$$

Where, $\beta^2 = \mu\varepsilon\omega^2$

$\beta = \sqrt{\mu\varepsilon}\omega$ $\qquad\qquad$ $\beta \rightarrow Phase\ shift\ constant$

The velocity of propagation is

$$\mathrm{V}=\frac{\omega}{\beta}=\frac{1}{\sqrt{\mu\varepsilon}}$$

The wave propagates in x direction i.e no variation in y and z

$$\frac{\partial^2 E}{\partial x^2} + \beta^2 E = 0$$

The solution of the wave equation is

$$\mathrm{E} = C_1 e^{-j\beta x} + C_2 e^{j\beta x}$$

5.4 WAVE PROPAGATION IN A CONDUCTING MEDIUM

The wave equation for conducting medium is

$$\nabla^2 E - \mu E\frac{\partial^2 E}{\partial t^2} - \mu\sigma\frac{\partial E}{\partial t} = 0$$

The phasor form of wave equation is

$$\nabla^2 E - j^2\mu\varepsilon\omega^2 E - j\mu\omega\sigma E = 0$$

$$\nabla^2 E - jE[\mu\omega\sigma + j\mu\varepsilon\omega^2) = 0$$

$$\nabla^2 E - jE\mu\omega(\sigma + j\omega\varepsilon) = 0$$

$$\nabla^2 E - \gamma^2 E = 0$$

Where $\gamma^2 = j\mu\omega(\sigma + j\omega\varepsilon)$

$\gamma$is called propagation constant, which has both real and imaginary parts.

$$\gamma = \alpha + j\beta$$

Where,

α *is attenuation constant*

β *is phase shift*

$\gamma = \alpha + j\beta$

$= \sqrt{j\mu\omega(\sigma + j\omega\varepsilon)}$

Squaring on both sides

$\alpha^2 - \beta^2 + 2j\alpha\beta = j\mu\omega\sigma - \omega^2\mu\varepsilon$

Equating real and imaginary parts

$\alpha^2 - \beta^2 = -\omega^2\mu\varepsilon$

$$2\alpha\beta = \mu\omega\sigma$$

To solve these two equation

w.k.t

$\alpha^2 + \beta^2 = \sqrt{(\alpha^2 - \beta^2)^2 + 4\alpha^2\beta^2}$

But

$$(\alpha^2 - \beta^2)^2 = (-\omega^2\mu\varepsilon)^2$$

$$(2\alpha\beta)^2 = (\mu\omega\sigma)^2$$

$$\alpha^2 + \beta^2 = \sqrt{\omega^4\mu^2\varepsilon^2 + \omega^2\mu^2\sigma^2}$$

$\alpha^2 - \beta^2 = -\omega^2\mu\varepsilon$

Adding these two equations

$2\alpha^2 = -\omega^2\mu\varepsilon + \sqrt{\omega^4\mu^2\varepsilon^2 + \omega^2\mu^2\sigma^2}$

$$\alpha^2 = -\frac{\omega^2\mu\varepsilon}{2} + \frac{\omega^2\mu\varepsilon}{2}\sqrt{1 + \frac{\alpha^2}{\omega^2\varepsilon^2}}$$

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1 + \frac{\alpha^2}{\omega^2\varepsilon^2}} - 1\right]}$$

Attenuation factor is given by

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\frac{\alpha^2}{\omega^2\varepsilon^2}}-1\right]}$$

By subtracting $\alpha^2-\beta^2$from $\alpha^2+\beta^2$, the value of β becomes

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\frac{\alpha^2}{\omega^2\varepsilon^2}}-1\right]}$$

5.5 WAVE PROPAGATION IN A GOOD DIELECTRIC

- The ratio of conductor current density to displacement current density in the medium is $\frac{\sigma}{\omega\varepsilon}$. Hence $\frac{\sigma}{\omega\varepsilon}$=1 can be consider to mash the dividing line between conductor and dielectrics
- For good conductor $\frac{\sigma}{\omega\varepsilon}$ is much greater than unity
- For good conductor $\frac{\sigma}{\omega\varepsilon}$ is very much less than unity
- For dielectric

$$\frac{\sigma}{\omega\varepsilon} \ll 1$$

$$\sqrt{1+\frac{\sigma^2}{\omega^2\varepsilon^2}} = \left(1+\frac{\sigma^2}{\omega^2\varepsilon^2}\right)^{\frac{1}{2}}$$

$$\approx \left(1+\frac{\sigma^2}{2\omega^2\varepsilon^2}\right)$$

The attenuation factor is

$$\alpha \cong \omega\sqrt{\frac{\mu\varepsilon}{2}\left[1+\frac{\sigma^2}{\omega^2\varepsilon^2}-1\right]}$$

$$\cong \omega\sqrt{\frac{\mu\sigma^2}{4\omega^2\varepsilon}}$$

$$\sigma \cong \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$$

The phase shift is

$$\beta \cong \omega\sqrt{\frac{\mu\varepsilon}{2}\left[1+\frac{\sigma^2}{\omega^2\varepsilon^2}+1\right]}$$

$$\cong \omega\sqrt{\frac{\mu\varepsilon}{2}\left[2+\frac{\sigma^2}{2\omega^2\varepsilon^2}\right]}$$

$$\cong \omega\sqrt{\mu\varepsilon\left[1+\frac{\sigma^2}{4\omega^2\varepsilon^2}\right]}$$

$$\cong \omega\sqrt{\mu\varepsilon}\left[1+\frac{\sigma^2}{4\omega^2\varepsilon^2}\right]^{1/2}$$

$$\cong \omega\sqrt{\mu\varepsilon}\left[1+\frac{\sigma^2}{8\omega^2\varepsilon^2}\right]$$

The velocity of the wave in the dielectric is

$$v = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{\mu\varepsilon}\left[1+\frac{\sigma^2}{8\omega^2\varepsilon^2}\right]}$$

$$\cong \frac{1}{\sqrt{\mu\varepsilon}}\left[1-\frac{\sigma^2}{8\omega^2\varepsilon^2}\right] \qquad \left[\, V_0 = \frac{1}{\sqrt{\mu\varepsilon}}\right]$$

$$v = V_0\left[1-\frac{\sigma^2}{8\omega^2\varepsilon^2}\right]$$

The intrinsic or characteristic impedance of medium is given by

$$\eta = \sqrt{\frac{j\mu\varepsilon}{\sigma+j\omega\varepsilon}}$$

$$\eta = \sqrt{\frac{j\mu\varepsilon}{j\omega\varepsilon\left(1+\frac{\sigma}{j\omega\varepsilon}\right)}} = \sqrt{\frac{\mu}{\varepsilon}\left(1+\frac{\sigma}{j\omega\varepsilon}\right)^{-1}}$$

$$= \sqrt{\frac{\mu}{\varepsilon}\left(1-\frac{\sigma}{j\omega\varepsilon}\right)}$$

$$= \sqrt{\frac{\mu}{\varepsilon}}\left(1-\frac{\sigma}{\omega\varepsilon}\right)^{1/2}$$

$$\eta == \sqrt{\frac{\mu}{\varepsilon}}\left(1+\frac{j\sigma}{2\omega\varepsilon}\right)$$

5.6 WAVE PROPAGATION IN GOOD CONDUCTOR

For good conductor $\frac{\sigma}{\omega\varepsilon} \gg 1$

$$\gamma == \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$$

$$= \sqrt{j\omega\mu(1 + \frac{j\omega\varepsilon}{\sigma})}$$

$$\approx \sqrt{j\omega\mu\sigma}$$

$$\gamma = \sqrt{\omega\mu\sigma}\,\angle 45°$$

$$\therefore \alpha = \beta = \frac{\sqrt{\omega\mu\sigma}}{2}$$

The velocity of the wave in conductor

$$v = \frac{\omega}{\beta} = \frac{\omega}{\frac{\sqrt{\omega\mu\sigma}}{2}}$$

$$v = \sqrt{\frac{2\omega}{\mu\sigma}}$$

The intrinsic impedance of the conductor

$$\eta \approx \sqrt{\frac{j\omega\mu}{j\omega\varepsilon(1+\frac{\sigma}{j\omega\mu})}}$$

$$\approx \sqrt{\frac{j\omega\mu}{j\omega\varepsilon.\frac{\sigma}{j\omega\varepsilon}}}$$

$$\approx \sqrt{\frac{j\omega\mu}{\sigma}}$$

$$\eta \approx \sqrt{\frac{\omega\mu}{\sigma}}\,\angle 45°$$

- It is found that in good conductor, α & β are large since σ is large., (i.e) the wave is attenuated greatly as it progress through the conductor.
- But the velocity & characteristics impedance are reduced considerably.

Problems

1. A uniform plane wave propagating in semi-infinite region z>0 is expressed (at zero) as 100 cos $10^9\pi$t $\overrightarrow{ax}\,V/m$. The regin has $\sigma = 0.25$ ε_r =9 $\mu_r = 1000$. Determine the propagation and attenuation constants.

Sol.

E=100 cos 10^9 πt $\overrightarrow{ax}\ V/m$

$$\omega = \pi \times 10^9$$
$$\sigma = 0.25$$

ε_r=9

$$\mu_r = 1000$$

Propagation constant $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$

$$\gamma = \sqrt{j\omega\mu\sigma(1 + \frac{j\omega\varepsilon}{\sigma})}$$

$$\frac{\omega\varepsilon}{\sigma} = \frac{\pi \times 10^9 \times 9}{0.25 \times 36\pi \times 10^9} = 1$$

$$\gamma = \sqrt{j\omega\mu\sigma(1 + j)}$$

$$= \sqrt{j\pi \times 10^9 \times 4\pi \times 10^{-7} \times 1000 \times 0.25 \times (1 + j)}$$

$= \sqrt{j986960(1 + j)}$ $= 1181.428\ \underline{|67.5°}$

Attenuation constant $\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left(1 + \frac{\sigma^2}{2\omega^2\varepsilon^2} - 1\right)}$

$$\frac{\sigma}{\omega\varepsilon} = \frac{0.25}{\pi \times 10^9 \times \frac{9}{36\pi \times 10^9}} = 1$$

$$\alpha = \pi \times 10^9\sqrt{\frac{4\pi \times 10^{-7} \times 1000 \times \frac{9}{36\pi \times 10^9}}{2}}\left[1 + \frac{1}{2} - 1\right]$$

$$\alpha = 496.73$$

2. A lossy dielectric is characterized by ε_r=2.5, $\mu_r = 4$ & $\sigma = \frac{10^{-3}mho}{m}$ aat 10MHz. Let E=10$e^{-Vz}\overrightarrow{ax}\frac{V}{m}$. Find i)$\alpha$, $(ii)\beta$, $(iii)\lambda$, $(iv)Vp$, $(V)\eta$

Given

$\sigma = 10^{-3}mho/m$

ε_r=2.5

$$\mu_r = 4$$

E=10$e^{-Vz}\overrightarrow{ax}\frac{V}{m}$

$$f = 10MHz$$

$$\frac{\sigma}{\omega\varepsilon} = \frac{10^{-3}}{2\pi \times 10 \times 10^6 \times \frac{2.5}{36\pi \times 10^9}} = 0.72 \approx 1$$

(i) $\alpha = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}}$

$= \frac{10^{-3}}{2}\sqrt{\frac{4}{2.5}} \times 120\pi$ $= 238.43 \times 10^{-3}$

(ii) $\beta = \omega\sqrt{\mu\varepsilon}[1 + \frac{1}{2} + (\frac{\sigma}{2\omega\varepsilon})^2]$

$= 2\pi \times 10 \times 10^6 \sqrt{4\pi \times 10^{-7} \times 4 \times \frac{2.5}{36\pi \times 10^9}}[1 + \frac{1}{2}(\frac{0.72}{2})2]$

$=0.6623 \times 1.0648$

$=0.7052$

(iii) $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{0.7052} = 8.91m$

(iv) $Vp = f\lambda$

$= 10 \times 10^6 \times 8.91 = 8.91 \times 10^6 m/sec$

(v) $\eta = \sqrt{\frac{\mu}{\varepsilon}}$

$$= \sqrt{\frac{\mu_r}{\varepsilon_r}} \times 120\pi$$

$$= \sqrt{\frac{4}{2.5}} \times 120\pi$$

$$= 476.86\ ohms$$

3. Determine the propagation constant for material with $\mu_r = 1\ and\ \varepsilon_r = 8$ and conductivity $\sigma = 0.25\ Ps/m$ and wave at frequency 1.63MHz.

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega)\varepsilon}$$

$= \sqrt{j\omega\mu(0.25 + j(1.6 \times 10^{-6})\ \varepsilon_0\ \varepsilon_r}$

$= \sqrt{j(1.6 \times 10^{-6})(4\pi \times 10^{-7})(0.25 + j(1.6 \times 10^{-6})8(8.854 \times 10^{-12})}$

$= \sqrt{j \times 2 \times 10^{-12}[0.25 + j(1.133)10^{-6}]}$

$= \sqrt{j5 \times 10^{-13} - [2.66 \times 10^{-28}]}$

$$\gamma = 7.07 \times 10^{-7} m^{-1}$$

5.7 PHASE VELOCITY AND GROUP VELOCITY

The relation between Up and phase constant β is

$U_p = \omega / \beta$ (m/s)

Consider a simplest case of a leaving point that consist of a two travelling waves having equal amplitude and slightly different angular frequency ω and phase constant β

The relationship between U_g and U_p

$d\beta / d\omega = d / d\omega (\omega / \mu_p)$

it is subdivided as

1. No dispersion
2. Normal
3. Avalanche

5.8 DEPTH OF PENETRATION

- The depth of penetration (δ) id defined as that depth in which the wave has been attenuated to 1/e or approximately 37 percent of its original value. It is also know as skin effect.
- The alternating current flows more easily through outer portion of a conductor than through inner portion. This effect is called skin effect.
- The amplitude of the wave decrease by the factor $e^{-\alpha x}$ as it propagates through a distance x.

  By definition

$$e^{-\alpha x} = \frac{1}{e}$$

$$e^{-\alpha x} = e^{-1}$$

$$\alpha\delta = 1$$

$$\boxed{\delta = \frac{1}{\alpha}}$$

$\rightarrow$ *This is the depth of peneration (or) skin effect*

$$\delta = \frac{1}{\alpha}$$

$$= \frac{1}{\omega\sqrt{\frac{\mu\varepsilon}{2}(1 + \frac{\sigma^2}{\omega^2\varepsilon^2} - 1)}}$$

For a good conductor the depth of penetration is

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega\mu\sigma}}$$

Problem

Find the depth of penetration of a plane wave in copper at a power frequency of 60Hz and at microwave frequency $10^{10} Hz$. Given$\sigma = 5.8 \times 10^7 \frac{mho}{m}$

Sol.

Depth of penetration $\delta = \frac{1}{\sigma} = \sqrt{\frac{2}{\omega\mu\sigma}}$

f=60Hz ,$\sigma = 5.8 \times 10^7 \frac{mho}{m}, \mu_r = 1$

$$\delta = \sqrt{\frac{2}{2\pi \times 60 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7}}$$

$$\delta = 8.53 \times 10^{-3} m$$

<u>**Two marks**</u>

1. **List out the properties of a uniform plane wave**

If the plane of wave is the same for all points on a plane surface, it is called plane wave. If the amplitude is also constant in a plane wave, it is called uniform plane wave. The properties of uniform plane waves are:

At every point in space E and H are perpendicular to each other and to the direction of travel

The fields vary with time at the same frequency, everywhere in space

Each field has the same direction, magnitudes and phase at every point in any plane perpendicular to the direction of wave travel.

2. **Give the expression for the characteristic impedance of the wave**

The characteristic impedance or intrinsic impedance is the ratio of the electric field intensity to the magnetic field intensity

E / H = $\sqrt{\mu/\xi}$

Where μ is the permeability of the medium and ξ is the permittivity of the medium.

3. **Give the wave equation for a conducting medium**

The wave equation for a conducting medium in phasor form is given as

$\Delta^2 E - j(\omega\mu\sigma + j\mu\varepsilon\omega^2) E = 0$

4. **What is skin effect and skin depth**

In a good conductor the wave is attenuated as it progresses. At higher frequencies the rate of attenuation is very large, and the wave may penetrate only at very short distance before being reduced to a small value. This effect is called skin effect.

The skin depth (δ) is defined as that depth in which the wave has been attenuated to 1/e or app 37% of its original value. It is also known as depth of penetration

$\Delta = 1/\alpha$

Where α = attenuation constant

$\alpha = \omega\sqrt{\mu\varepsilon}(\sqrt{1+\sigma^2/\omega^2\varepsilon} - 1)$

5. **Give the expression for attenuation constant and phase shift constant for a wave propagating in a conducting medium**

The attenuation constant for a wave propagating in a conductin medium is

$\alpha = \omega\sqrt{\mu\varepsilon}(\sqrt{1+\sigma^2/\omega^2\varepsilon} - 1)$

The phase shift constant for a wave propagating in a conducting medium

$\beta = \omega\sqrt{\mu\varepsilon}(\sqrt{1+\sigma^2/\omega^2\varepsilon} + 1)$

6. **Give the expression for the velocity of propagation of a wave in any medium**

The velocity of propagation of a wave in any medium is

$\upsilon = \omega / \beta = 1/\sqrt{\mu\varepsilon}$

Where ω is the angular velocity and β is the phase shift

7. **Define a wave**

If a physical phenomenon that occurs at one place at a given time is reproduced at other places at later times, that time delay being proportional to the space separation from the first location then the group of phenomena constitutes a wave.

8. **Mention the properties of uniform plane wave**

The properties of uniform plane waves are:

At every point in space E and H are perpendicular to each other and to the direction of travel

The fields vary with time at the same frequency, everywhere in space

Each field has the same direction, magnitudes and phase at every point in any plane perpendicular to the direction of wave travel.

9. Define intrinsic impedance or characteristic impedance

It is the ratio of electric field to magnetic field or it is the ration of square root of permeability to permittivity of medium

10. Define propagation constant

Propagation constant is a complex number

A = a + jb

Where A = attenuation constant

b is phase constant

a = juμ (o +juc)

11. What is meant by displacement current

Displacement current is the current that flows through capacitor

J = D / t

12. Define depth of penetration

It is defined as the depth in which the wave has been attenuated to 1/e or app 37% of its original value

13. State the principle of superposition of fields

The total electric field at a point is the algebraic sum of the individual electric field at that point

14. Define surface impedance?

Surface impedance is defined as the ratio of the tangential component of the electric field at the surface of the conductor to the linear current density

15. Define Polarization

The polarization of a uniform plane wave refers to the time varying behaviour of the electric field strength vector at some fixed point in space.

16. What are the classification of polarization

Linear Polarization

Circular Polarization

Elliptical Polarization

17. Write the Mathematical expressions for polarization

$$\boxed{\bar{p} = Q\,\bar{d}}$$

Q = Magnitude of one of the two charges

$\bar{d}$ = Distance vector from negative to positive charge

n = Number of dipoles per unit volume

Δv = Total volume of the dielectric

N = Total dipoles = $n\,\Delta v$

18. Define Linear Polarization

$E = \sqrt{E_X^2 + E_Y^2}$

$\theta = \tan^{-1}(E_Y / E_X)$

if both E_X and E_Y are present and are in phase with the resultant electric field has a direction at an angle of $\tan^{-1}(E_Y / E_X)$

if the direction of the resultant vector is constant with time, the wave is said to linearly polarized.

19. Define Circular Polarization

If E_X and E_Y have equal magnitudes and a $\pi/2$ phase difference, the locus of the resultant E is a circle and the wave is said to be circularly polarized.

E_X and E_Y have the same magnitude , E_a differ 90 degree in phase.

The resultant electric field in vector form is

$E = a_x E_a + j\, a_y E_a$

The corresponding time varying field is

$E = a_x E_a \cos \omega t + j\, a_y E_a \sin \omega t$

The components are

$E_{X} = E_{a} \cos \omega t$

$E_{Y} = E_{a} \sin \omega t$

$E_X^2 + E_Y^2 = E_a^2$

This equation shows that the locus of the resultant E is a circle whose radius is E_a

20. Define Elliptical Polarization

E_X and E_Y have different amplitude and $\pi/2$ phase difference, the locus of the resultant E is an ellipse and the wave is said to be Elliptically Polarized.

Let E_X the magnitude A and E_Y has the magnitude B and differ 90 degree in phase.

The resultant electric field in vector form is

$E = a_x A + j a_y B$

The corresponding time varying field is

$E = a_x A \cos \omega t + j B E_a \sin \omega t$

The components are

$E_{X} = A \cos \omega t$

$E_{Y} = B \sin \omega t$

Then $E_X/A = \cos \omega t$

$E_Y/A = - \sin \omega t$

$E_X^2/A^2 + E_Y^2/B^2 = 1$ [since $\cos^2 \omega t + \sin^2 \omega t = 1$]

21. How are Perfect Conductor in refraction and reflection classified

(a) wave incident normally on a perfect conductor
(b) wave incident obliquely on a perfect conductor

22. how are Perfect Dielectric in refraction and reflection classified

(a) wave incident normally on a perfect dielectric
(b) wave incident obliquely on a perfect dielectric

Two marks

4. Write the maxwell's equation from ampere's law both in integral and point forms

maxwell's equation from ampere's law

$\oint \vec{H}dl = \int \left(\sigma E + \varepsilon \frac{d\vec{E}}{dt}\right) ds$ (integral form)

$\nabla \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{d\vec{E}}{dt}$ (Point form)

5. **Write the maxwell's equation from faraday's law both in integral and point forms.**

Maxwell's equation from faraday's law

$\oint Edl = -\int \frac{\partial B}{\partial t} ds$ (integral form)

$\nabla \times \vec{E} = -\frac{\overrightarrow{\partial B}}{\partial t}$ (Point form)

6. **Write down the maxwell's equation from electric gauss's law in integral and point form.**

Maxwell's equation from electric Gauss's law

$\oint_S \vec{D}.ds = \int_V \rho_v dv$ (integral form)

$\nabla \cdot \vec{D} = \rho_v$ (Point form)

7. **Write down the maxwell's equations from magnetic Gauss's law in integral and point form.**

$\oint_S \vec{B}.ds = 0$ (integral form)

$\nabla.\vec{B} = 0$ (Point form)

8. **Write down the wave equation from E and H in a non-dissipative (free space) medium.**

$$\nabla^2 E - \mu_0 \varepsilon_0 \frac{\partial^2 E}{\partial t^2} = 0$$
$$\nabla^2 H - \mu_0 \varepsilon_0 \frac{\partial^2 H}{\partial t^2} = 0$$

9. **Write down the wave equations for E and H in a conducting medium.**

$$\nabla^2 E - \mu\varepsilon \frac{\partial^2 E}{\partial t^2} - \mu\sigma \frac{\partial E}{\partial t} = 0$$
$$\nabla^2 E - \mu\varepsilon \frac{\partial^2 H}{\partial t^2} - \mu\sigma \frac{\partial H}{\partial t} = 0$$

10. **State poynting theorem.**

The vector product of electric field intensity and magnetic field intensity at any point is a measure of the rate of energy flow per unit area that point.

$\vec{P} = \vec{E} \times \vec{H}$

11. **What is an emf?**

An electro-motive force is a voltage that arises from conductors moving in a magnetic field or from changing magnetic fields.

12. State Faraday's law

Faraday's law states that, the total emf induced in a closed circuit is equal to the time rate of decrease of the total magnetic flux linking the circuit.

$E=-d\Phi/dt$ Volts

13. State Lenz's law

The Lenz's law states that , the induced current in the loop is always in such a direction as to produce flux opposing the change in flux density.

14. Explain briefly the different types of emf's produced in a conductor placed in a magnetic field

There are two ways in which we can induce emf in a conductor. If a moving conductor is placed in a static magnetic field then the emf produced in the conductor is called dynamically induced emf. If the stationary conductor is placed in a time varying magnetic field, then the emf produced is called statically induced emf.

15. Distinguish between the conduction current and displacement current

Conduction current I_c is flowing through a conductor having resistance R, when potential V is applied across the conductor

$I_c = V / R$ Ampere

Displacement current Id is flowing through a capacitor when ac voltage is applied across the capacitor.

$I_d = C\ Dv / dt$

16. What is Eddy current and Eddy current loss

In electrical machines, the alternating magnetic fields induce emf in the cores also apart from the coil. This small amount of emf induced in the core circulates current in the core. This current is called eddy current and the power loss. Which appears in the form of heat, due to these eddy current is called eddy current loss.

17. Give the wave equation in terms of electric field and magnetic field

The electromagnetic wave equation in terms of electric field is,

$\Delta^2 E - \mu\sigma\ \theta E / \theta t - \mu\varepsilon\ \theta^2 E / \theta^2 t = 0$

The electromagnetic wave equation in terms of magnetic field is,

$\Delta^2H – \mu\sigma\ \theta H / \theta t – \mu\varepsilon\ \theta^2H / \theta^2t= 0$

18. Give the wave equation in terms of free space

The wave equation in free space in terms of electric field is

$\Delta^2E - \mu\varepsilon\ \theta^2E / \theta^2t= 0$

The wave equation in free space in terms of magnetic field is

$\Delta^2H - \mu\varepsilon\ \theta^2H / \theta^2t= 0$